Network Theory

Workbook

Electronics & Communication Engineering
Electrical Engineering
Electrical & Electronics Engineering

GATE & ESE

unacademy

Network Theory

Workbook

Electronics & Communication Engineering

Electrical Engineering

Electrical & Electronics Engineering

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GATE Syllabus

Network Theory (EC) : Network solution methods: nodal and mesh analysis; Network theorems: superposition, Thevenin's and Norton's, maximum power transfer; Wye-Delta transformation; Steady state sinusoidal analysis using phasors; Time domain analysis of simple linear circuits; Solution of network equations using Laplace transform; Frequency domain analysis of RLC circuits; Linear 2-port network parameters: driving point and transfer functions; State equations for networks.

Electric Circuits (EE) : Network graph, KCL, KVL, Node and Mesh analysis, Transient response of dc and ac networks, Sinusoidal steady-state analysis, Resonance, Passive filters, Ideal current and voltage sources, Thevenin's theorem, Norton's theorem, Superposition theorem, Maximum power transfer theorem, Two -port networks, Three phase circuits, Power and power factor in ac circuits.

ESE Syllabus

Network Theory (EC) : Network graphs & matrices; Wye - Delta transformation; Linear constant coefficient differential equations- time domain analysis of RLC circuits; Solution of network equations using Laplace transforms- frequency domain analysis of RLC circuits; 2-port network parameters driving point & transfer functions; State equations for networks; Steady state sinusoidal analysis.

Electrical Circuits (EE) : Circuit elements, network graph, KCL, KVL, Node and Mesh analysis, ideal current and voltage sources, Thevenin's, Norton's, Superposition and Maximum Power Transfer theorems, transient response of DC and AC networks, Sinusoidal steady state analysis, basic filter concepts, two port networks, three phase circuits, Magnetically coupled circuits.

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1 Basics Of Network Elements

Practice Questions

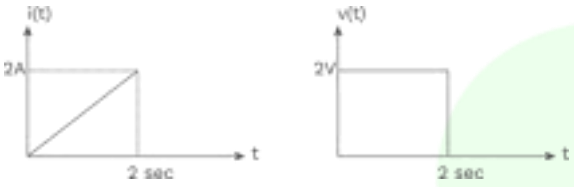
Q.1 A current of 0.5A is drawn from a filament of an electric bulb in 10 minutes. Find the amount of electric charge that flows in the circuit.

Q.2 A conductor has a constant current of 5 amperes. How many electrons passes from a fixed point on the conductor in one minute?

(A) 1H (B) 4H

(C) 2H (D) 5H

Q.3 The voltage and current waveforms for an element are shown in the figure below-



The circuit element and its value are:

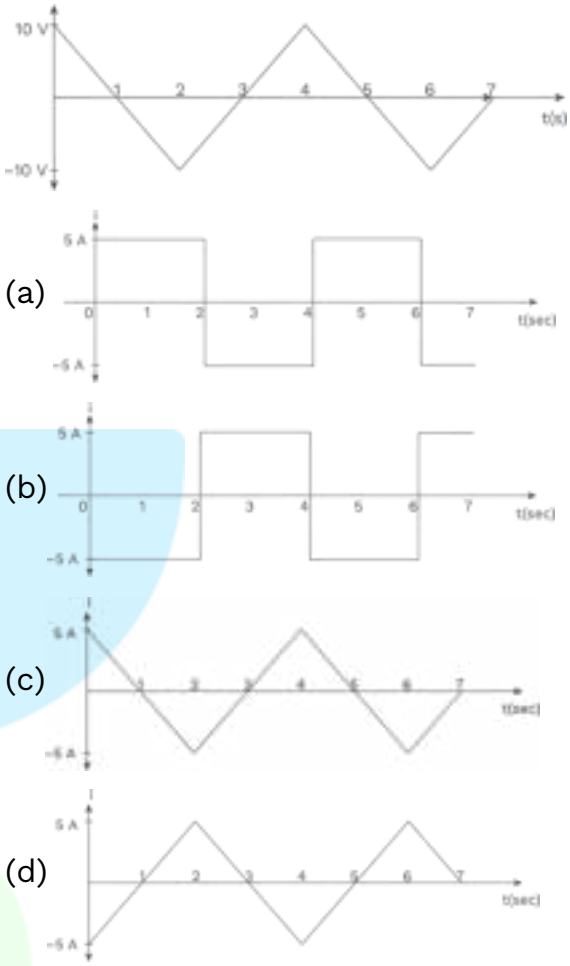
(A) Capacitor of 2F

(B) Inductor of 1 H

(C) Inductor of 2 H

(D) Resistor of 1 Ω

Q.4 A triangular wave voltage, as shown in figure, is applied across the terminals of a 0.5F pure capacitor at time t = 0. The corresponding current wave is:



Q.5 If an ideal voltage source and ideal current source are connected in series, the combination-

(A) has the same properties as a current source alone.

(B) has the same properties as a voltage source alone.

(C) has the same properties as the source which has a higher value.

(D) results in the branch being redundant.

Q.6 Consider the following statements:

Any element connected in

(A) Series with an ideal current source is redundant.

(B) Parallel with an ideal current source is redundant.

(C) Series with an ideal voltage source is redundant.

(D) Parallel with an ideal voltage source is redundant.

Q.7 Which of the following are satisfied in a non-linear network? MSQ

(A) Associative (B) Superposition

(C) Homogeneity (D) Bilaterality

Q.8 The V-I characteristics of an element is shown in the figure given below. The element is:



(A) Non-linear, active, non-bilateral

(B) Linear, active, non-bilateral

(C) Non-linear, passive, non-bilateral

(D) Non-linear, active, bilateral

Q.9 A network contains linear resistors and ideal voltage sources. If value of all resistors are doubled, then the voltage across each resistor is:

(A) halved

(B) doubled

(C) increased by four times

(D) not changed

Q.10 The circuit shown in the figure represent a

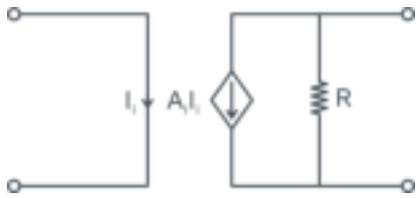


diagram i_i not same in diagram

(A) Voltage controlled voltage source

(B) Voltage controlled current source

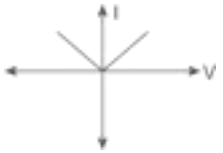
(C) Current controlled current source

(D) Current controlled voltage source

Q.11 Which of the following graphs, represents an active element?

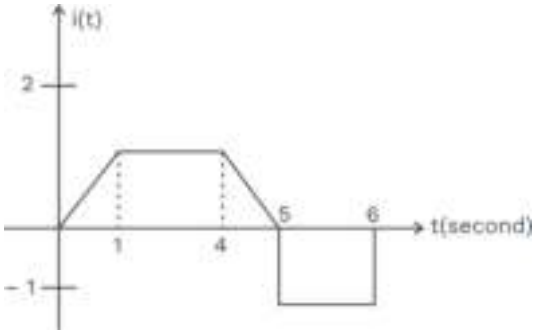
(A)


(B)


(C)


(D)


Q.12 The current i(t) flowing through a capacitor as shown below –

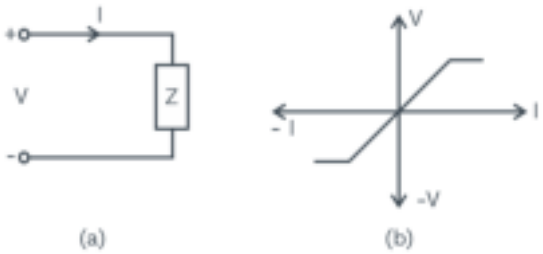


The charge acquired by the capacitor up to 4 second.

(A) 6 C (B) 9 C

(C) 7 C (D) 10 C

Q.13 V – I characteristic of an element shown in fig(a) is plotted in fig (b). Determine the type of element –



(A) Linear, active, bilateral

(B) Non linear, passive, unilateral

(C) Non linear, passive, bilateral

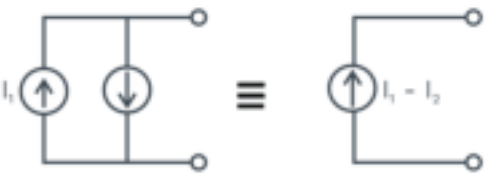
(D) Non linear, active, bilateral

Q.14 Which one of the is not correct equivalence regarding voltage & current sources –

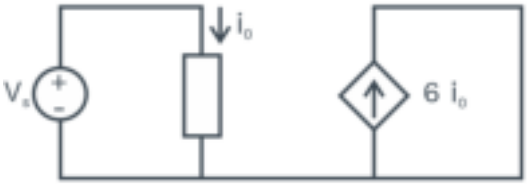
(A) 

(B) 

(C) 

(D) 

Q.15. The dependent source in the figure is –



(A) Voltage – controlled current source

(B) Voltage controlled voltage source

(C) Current controlled voltage source

(D) Current controlled current source

Q.16. Which one of the statements is correct?

(A) All passive networks are always symmetrical.

(B) All the linear networks are always reciprocal.

(C) All the active networks are always non reciprocal.

(D) All bidirectional networks are linear also.

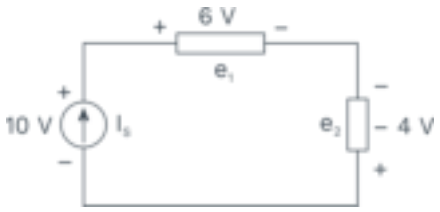
Q.17. Find the power delivered to an element at t = 3 ms if the current entering its positive terminal is

$i = 5 \cos 60 \pi t$ A

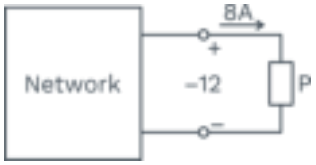
and the voltage is: $v = 3\frac{di}{dt}$

Power (P)=_____kW

Q.18 In the following circuit, what is the value of I_S, such that net power absorbed by two elements in 4W?



Q.19 In the circuit shown below, power P is



(A) 96 W, absorbed

(B) 36 W, supplied

(C) 96 W, supplied

(D) None of these

Q.20 A current of 0.5 A is drawn from a filament of an electric bulb in 10 minutes. Find the amount of electric charge that flows in the circuit.

Q.21 Is the circuit BIBO stable?



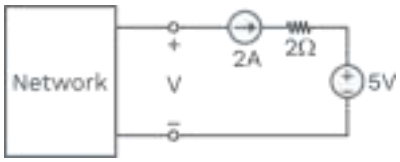
(A) Stable for low current

(B) Stable for high values of C

(C) Not stable

(D) Stable for large current

Q.22 The voltage V in figure is always equal to



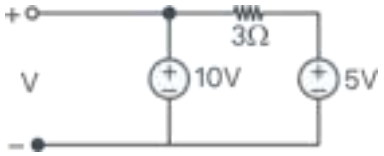
(A) 9V

(B) 5V

(C) 1V

(D) None of these

Q.23 The voltage V in figure is –



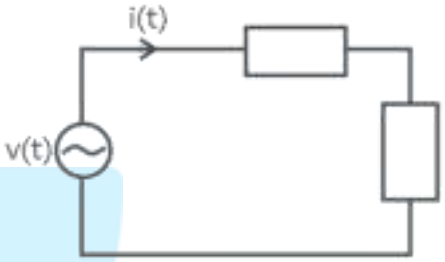
(A) 10V

(B) 15V

(C) 5V

(D) None of these

Q.24 The two elements in the circuit given below are –



Where $v(t) = 60\cos\left(5000t + 45^\circ\right)$

$i(t) = 10\cos\left(5000t - 30^\circ\right)$

(A) $R = 1.75\Omega,\ L = 1.89mH$

(B) $R = 1.55\Omega,\ L = 1.15mH$

(C) $R = 1.89\Omega, L_1 = 1.75mH$

(D) $R = 1.43\Omega, L = 2mH$

Q.25 Which one of the statement is correct?

(A) All passive networks are always symmetrical.

(B) All the linear networks are always reciprocal.

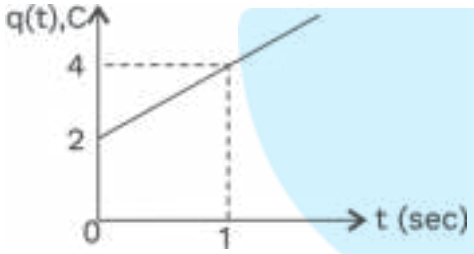
(C) All the active networks are always non reciprocal.

(D) All bidirectional networks are linear also.

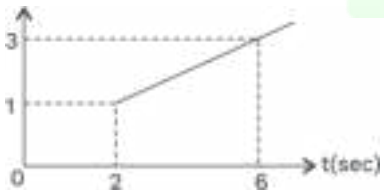
Q.26 The expression for the charge entering to the upper terminal of the given figure is –

$q = \frac{1}{\alpha} - \left(\frac{t}{\alpha} + \frac{1}{\alpha^2}\right)e^{-\alpha t}$ coulomb. Find the maximum value of the current entering the terminal (= 0.03679 s^{-1})

Q.27 The charge entering a circuit element is shown in the figure below. What is the current in the element in the time interval $0 \le t \le 1$ sec. ?



Q.28 The current entering the circuit element is shown in figure below. The charge that passes through element from t = 0 to t = 6 sec will be :



(A) 5 C (B) 20 C

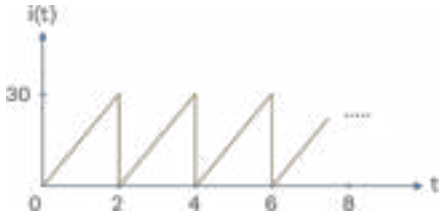
(C) 40 C (D) 10 C

Q.29 The wires A and B of the same material but of different lengths L and 2L have the radius r and 2r respectively. The ratio of specific resistance will be -

(A) 1 : 4 (B) 1 : 1

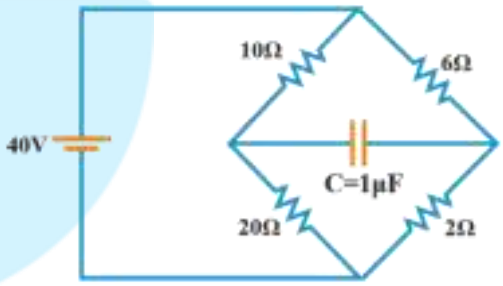
(C) 1 : 8 (D) 1 : 2

Q.30 The current characteristics is shown below. The given current passes through a 10 resistor.



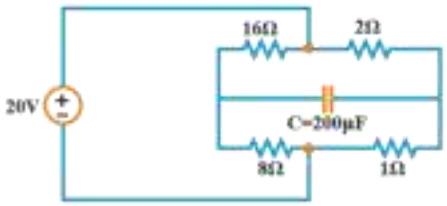
The average power dissipated in the load resistor is _____ W. (Rounded off to the nearest integer)

Q.31 A circuit is shown in the figure below.



The energy stored in the capacitor is ____ μJ. (Up to two decimal places)

Q.32. For the bridge circuit shown below, the charge accumulated in the capacitor in steady state is –

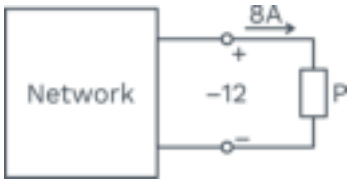


(A) 0μC (B) 4μC

(C) 200μC (D) 400μC

Q.33 A current of 0.5A is drawn from a filament of an electric bulb in 10 minutes. Find the amount of electric charge that flows in the circuit.

Q.34 In the circuit shown below, power P is



diagram

8A current not BA

(A) 96 W, absorbed

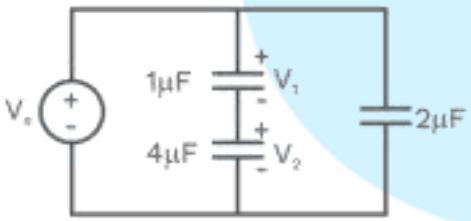
(B) 36 W, supplied

(C) 96 W, supplied

(D) None of these

Q.35 For the given circuit the value of $V_1(t)$ for $t \geq 0$ is _____.

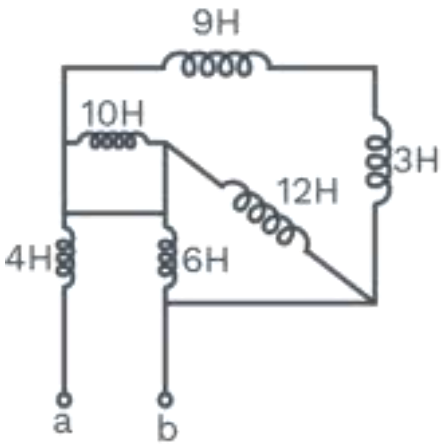
If $V_s = 100e^{-80t}$ and $V_1(0) = 20$



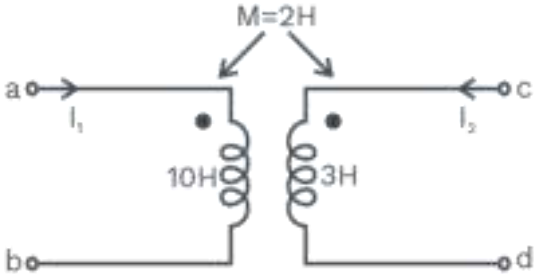
(A) $\left(100e^{-80t} - 80\right)V$ (B) $\left(60e^{-80t} - 80\right)V$

(C) $\left(80e^{-80t} - 80\right)V$ (D) $\left(80e^{-80t} - 60\right)V$

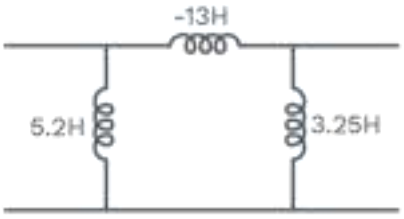
Q.36 The equivalent inductance seen across the terminals a & b is _ ___ H.



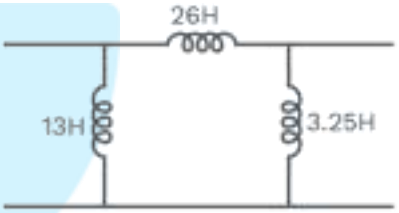
Q.37 Determine the π – equivalent circuit of linear transformer which is given below –



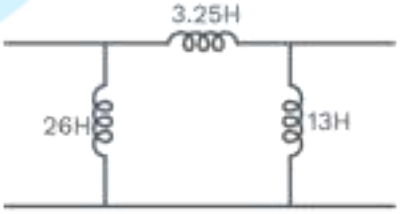
(A)



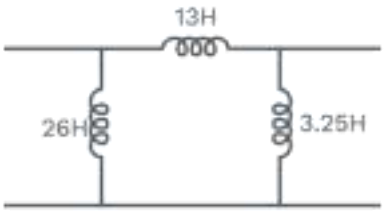
(B)



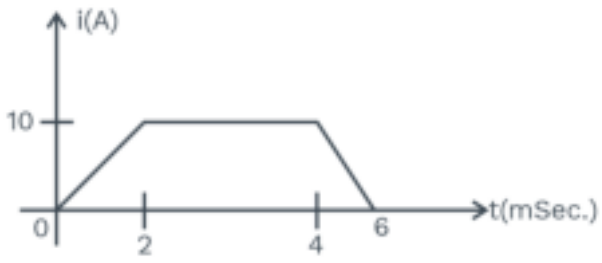
(C)



(D)



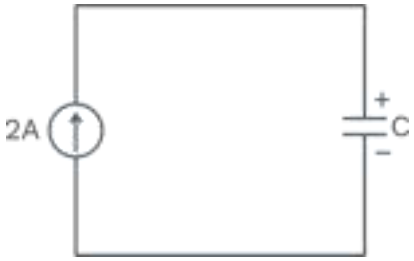
Q.38 The current through a 5mH inductor is shown in fig. determine the voltage across the inductor at t = 5 msec. in mV –



(A) 25 (B) 0

(C) -25 (D) 50

Q.39 Is the circuit BIBO stable?



(A) Stable for low current

(B) Stable for high values of C

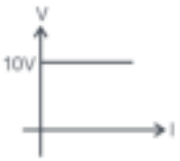
(C) Not stable

(D) Stable for large current

Q.40 Consider the ideal voltage source shown given below figure.



The correct V-I characteristic for the given ideal voltage source is -

(A) 

(B) 

(C) 

(D) Both (a) and (c)

Q.41 A resistance is connected across supply of 20V and dissipates 80 W at 25°C temperature. The power dissipated at 35°C temperature across same supply will be ..... W. (Take α = 0.015/°C)

Q.42 Consider a capacitor having charge q and voltage (sin t) volts across it. Charge q and voltage V relation is also given by $q = V + \frac{V^3}{3}$. Current through capacitor is

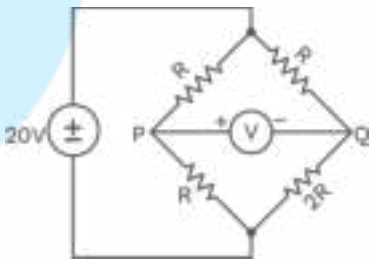
(A) $(1 + \sin^2 t) \cos t$ A

(B) sin t A

(C) $(1 + \sin^2 t) \cos t$ A

(D) None

Q.43 Consider the circuit shown in the figure. The reading of the ideal voltmeter between P and Q.



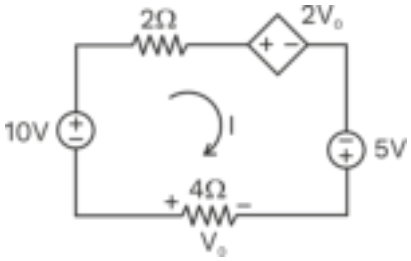
(A) $-\frac{20}{3}$ V

(B) $-\frac{10}{3}$ V

(C) $-\frac{40}{3}$ V

(D) None

Q.44 Determine the value of V_0 and I in the circuit shown below.



(A) 15V, – 7.5 A (B) 30V, – 7.5 A

(C) 30V, – 8.5 A (D) 15V, – 8.5 A

Q.45 The voltage and current through circuit element are

V = 250 sin (314t + 30°)

I = 20 sin (314t - 75°)

Identify the element and find its value.

(A) Inductor of value 0.0397 H

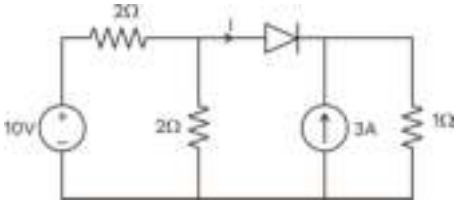
(B) Inductor of value 63.7 H

(C) Capacitor of value 63.7 F

(D) Capacitor of value 0.0397 F

2 Circuit Analysis

Q.1 Find the value of current I?



(A) 0 Amp (B) 1 Amp

(C) 1.5 Amp (D) 0.5 Amp

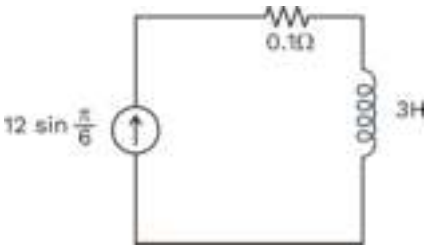
Q.2 When a complex network of N is connected to load resistance R_L. The power dissipation in the load resistance is P. When two identical complex network connected to load R_L. Find the power dissipation in the load.



(A) P (B) 4P

(C) P to 4P (D) 2P

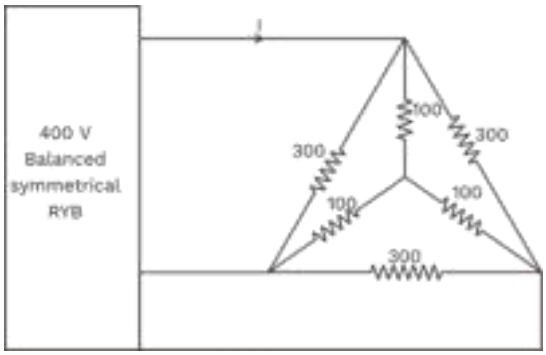
Q.3 Find the maximum energy stored in inductor and then calculate how much energy is dissipated in the resistor in time during which the energy is being stored and recovered in the inductor



(A) 432J (B) 43.2 J

(C) 216 J (D) 21.6 J

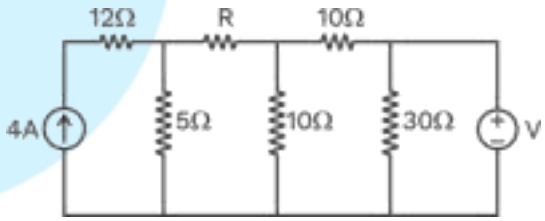
Q.4 In the circuit shown find the value of current (I) in ampere –



(A) 4.2 A (B) 4.3 A

(C) 4.6 A (D) 4.5 A

Q.5

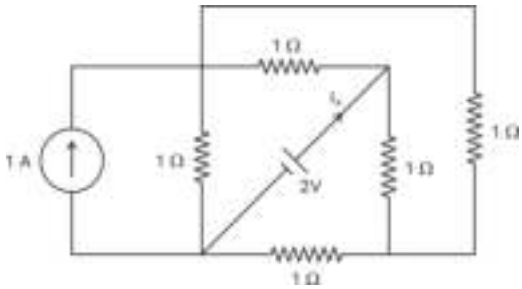


When the resistance R value is 20Ω the power consumed buy resistance is '0' watt if the resistance value changes for 10Ω then find the value of V?

(A) 45 V (B) 35 V

(C) 30 V (D) 40 V

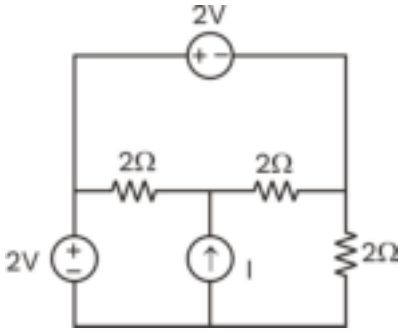
Q.6 Find I_x ?



(A) 6.5 A (B) 5.5 A

(C) 2 A (D) 1.5 A

Q.7 The power delivered by the current source in the figure is –



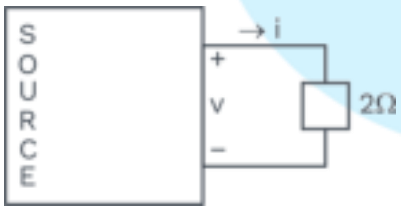
(A) – 2 Watt

(B) 1 Watt

(C) 2 Watt

(D) – 1 Wat

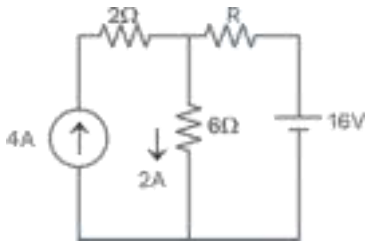
Q.8 As shown in figure, a 1Ω resistance is connected across a source that has a load line V + I = 90 the current through resistance is –



(A) 25 (B) 30

(C) 50 (D) 100

Q.9 If the 6Ω resistor draw a current of 2A as shown in figure, the value of R is .... (Ω)



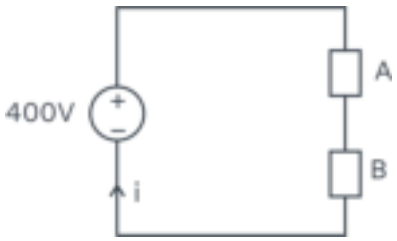
(A) 3Ω (B) 9Ω

(C) 6Ω (D) 8Ω

Q.10 A and B are identical circuit element connected as shown in figure. The VI relationship are either element is –

$V = 2I^2$

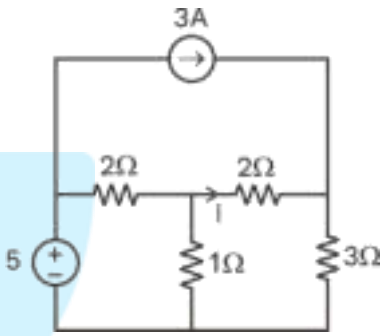
Find power absorbed by each element?



(A) – 2 KW (B) 2 KW

(C) 4 KW (D) – 4 KW

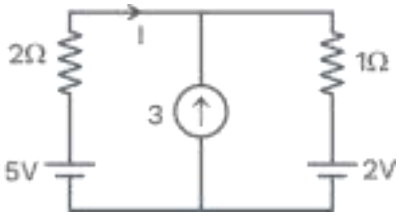
Q.11 Find the value of current I?



(A) 1.29 A (B) 2.59 A

(C) – 2.59 A (D) – 1.29 A

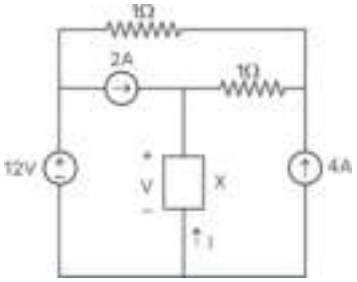
Q.12 Find I in the given network?



(A) 3 A (B) 0 A

(C) 1 A (D) 2 A

Q.13 The V-I relation of the unknown element 'X' in the given network is V = AI + B. The Value of A (in Ω) and B (in volt) respectively –



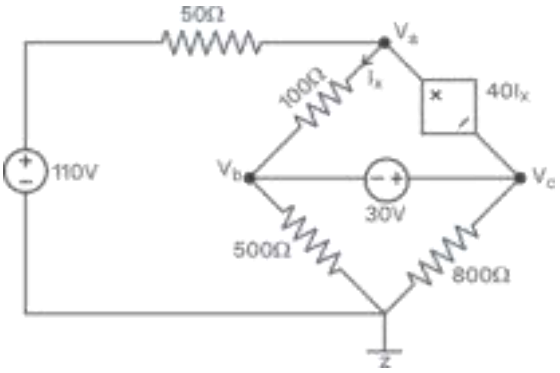
(A) A = 2Ω, B = 20 V

(B) A = 4Ω, B = 20 V

(C) A = 2Ω, B = 40 V

(D) A = 4Ω, B = 25 V

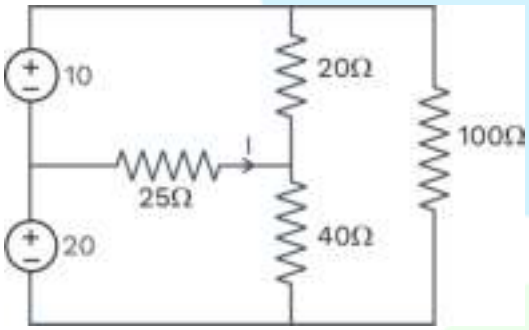
Q.14 Determine V_a, V_b, V_c and find $|V_b - V_a|$ =?



(A) 0 volt (B) 50 volt

(C) 100 volt (D) 30 volt

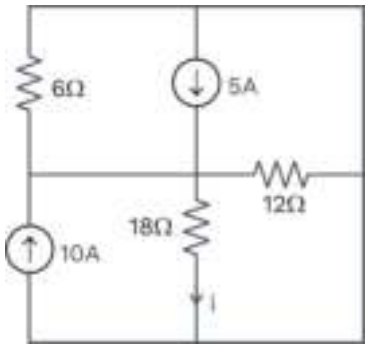
Q.15 Find the value of current I?



(A) 0 A (B) 8 A

(C) 5 A (D) 4 A

Q.16 Consider the circuit shown in figure shown -

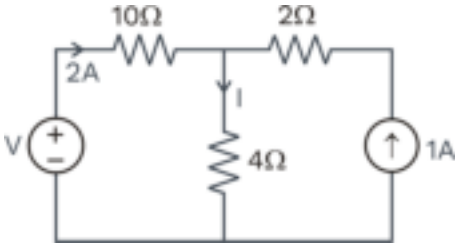


The current i flowing through 18Ω resistor is -

(A) 3.2 A (B) 2.5 A

(C) 3 A (D) 2.72 A

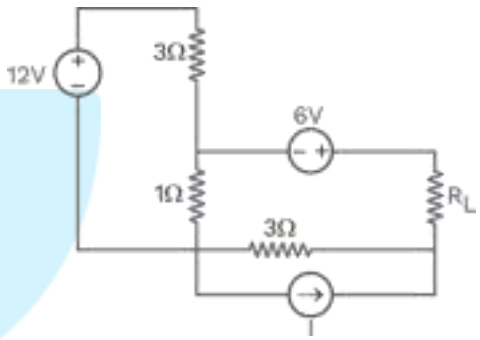
Q.17 In the figure, the value of source voltage is –



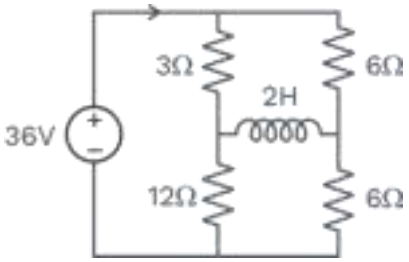
(A) 8 V (B) 24 V

(C) 32 V (D) 16 V

Q.18 Find the current I so that the current flowing through R_L is 0 Amp.



Q.19 Find the value of I?



(A) 6 (B) 12

(C) 15 (D) 9

Q.20 If P_1 = 36 W, P_2 = 100 W and P_3 = 144 W then find the value of P_{min}=?

(A) 0 watt

(B) 64 watt

(C) 16 watt

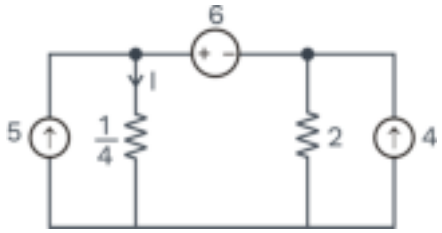
(D) 256 watt

Q.21 If P_1 = 36 W, P_2 = 100 W and P_3 = 144 W then find the value of P_{min}= ?

(A) 0 watt (B) 64 watt

(C) 16 watt (D) 256 watt

Q.22

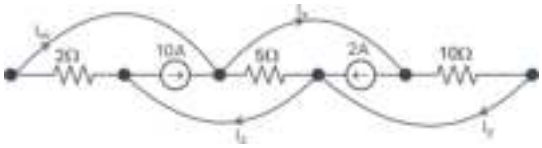


For the given circuit find the value of I _____ in Amp

(A) 10 A (B) 10.5 A

(C) 10.67 A (D) 11 A

Q.23

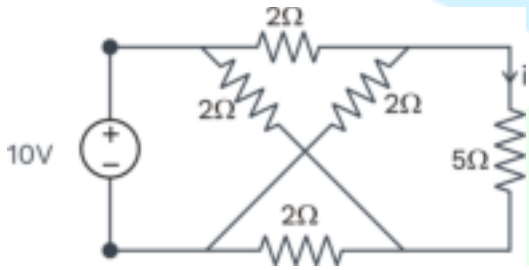


Find the value of I_x and I_z? MSQ

(A) 1 A (B) 3 A

(C) 5 A (D) –5 A

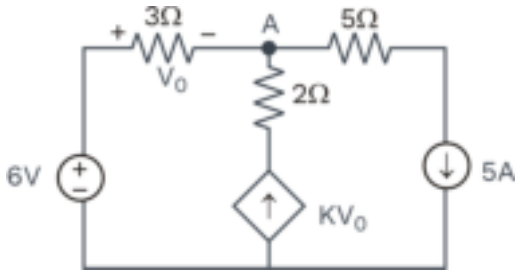
Q.24 The circuit I (in ampere) in 5Ω resistor of the given network –



(A) 1.42 A (B) 0 A

(C) 2 A (D) 7 A

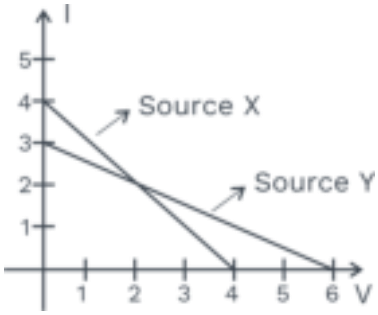
Q.25 In the given circuit the parameter K is positive and the power dissipated in the 3Ω resistor is 12 W. The value of K is

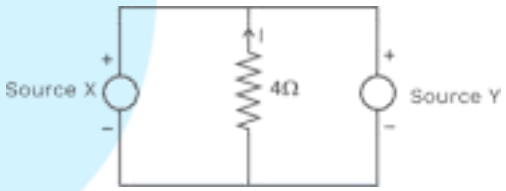


(A) A (B) 1

(C) 1/2 (D) 0

Q.26 The linear V-I Characteristic of 2-terminal non-ideal DC sources X and Y are shown in figure. If the sources are connected to 4Ω resistor as shown, the current through the resistor in ampere is ____ Amp

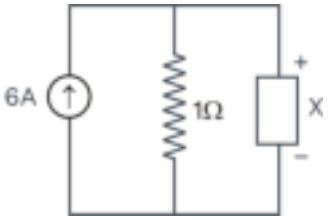




(A) $\frac{-5}{11}$ Amp (B) $\frac{1}{7}$ Amp

(C) $\frac{5}{11}$ Amp (D) $\frac{-1}{7}$ Amp

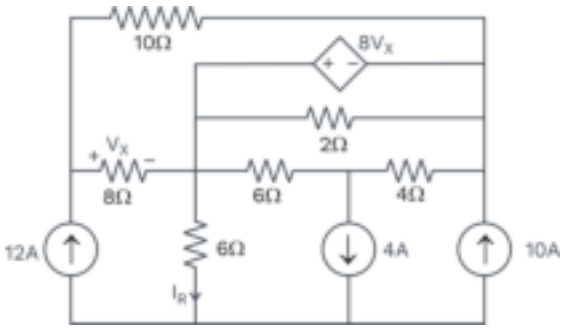
Q.27 In the given circuit X is an element which absorbs power. During particular operation, it sets up a voltage of 5 V as polarities as shown and absorbs a power P watts. It is possible that X can absorb same power P watt for another value of voltage V. find the value of V?



(A) 5 V (B) 1 V

(C) 3 V (D) 2 V

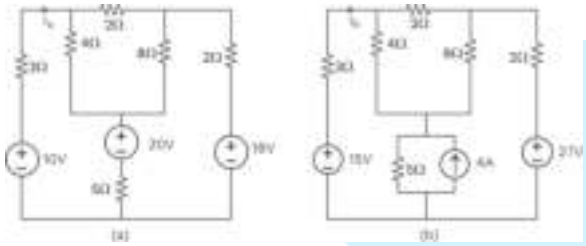
Q.28 Find the current I_R in the given network –



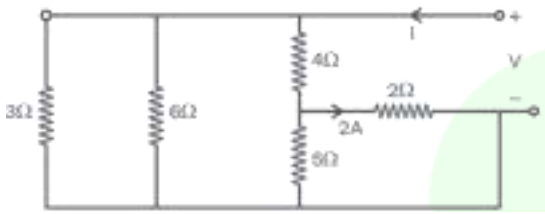
(A) 26 (B) 25

(C) 18 (D) 22

Q.29 If I_a = 3 Amp, then find I_b (in Amp)



Q.30 Consider the circuit shown below figure.

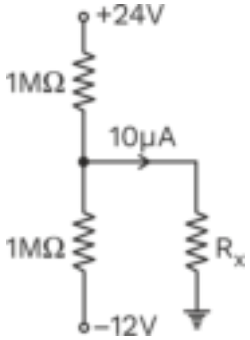


The value of V and I is respectively,

(A) 11.2V, 4.9A (B) 12.2V, 9.4A

(C) 13.3V, 5.4A (D) 21.1V, 4.7A

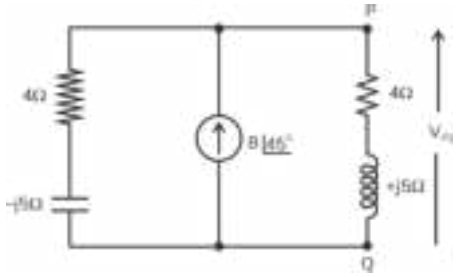
Q.31 For the circuit shown in figure, the value of 'R_x' is



(A) 1 kΩ (B) 10 kΩ

(C) 100 kΩ (D) 1 MΩ

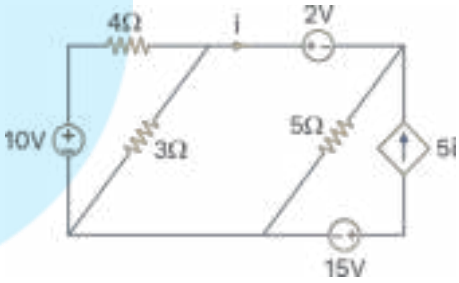
Q.32 Determine the phase voltage V_{PQ} in the ac network shown in figure



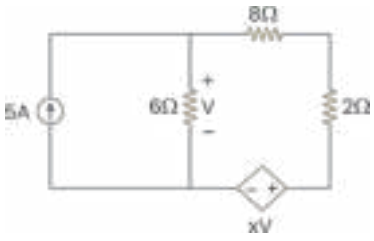
(A) $-41\angle 45^0$ (B) $40\angle 45^0$

(C) $41\angle 45^0$ (D) $40\angle 135^0$

Q.33 For the circuit shown below, the current I is _____ mA.(Up to one decimal place)



Q.34 For the given circuit, the value of x is- MSQ



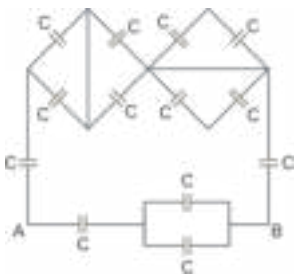
(A) For V = 30Volts; x = 1

(B) For V = 25Volts; x = 0.66

(C) For V = 45 Volts; x = 1.55

(D) For V = 60Volts;x = 1.83

Q.35 Find the equivalent capacitance between the terminals A and B for C = 2F.

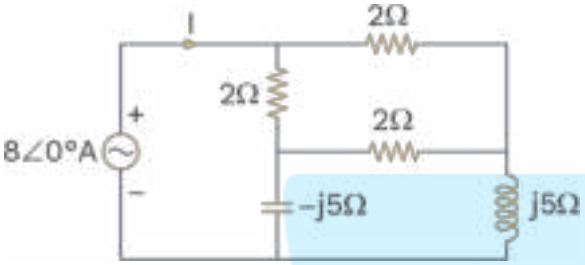


(A) 6F (B) 0.6F

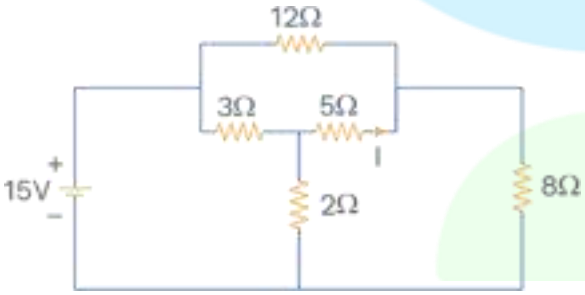
(C) 2F (D) 3F

Q.36 The average power delivered to an impedance (5-6j) by a current source 4cos(20t+50) is ____ Watts.(Rounded off to the nearest integer)

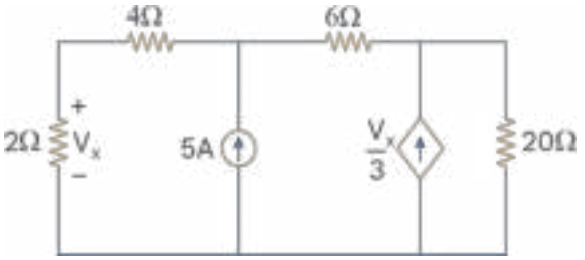
Q.37 In the circuit shown below, the current I is _____ Amp. (upto 3 decimal places)



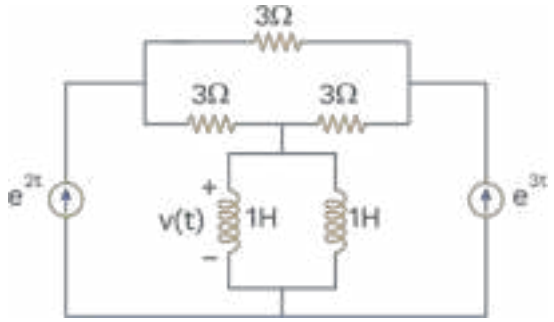
Q.38 For the circuit shown in the figure below, the current flowing through the 5Ω resistance is ____ (in Amp).



Q.39 The power (in Watts) of the given circuit at $\frac{V_x}{3}$ source is_____ (Upto two decimal places)



Q.40 The voltage v(t) in the circuit shown below is-



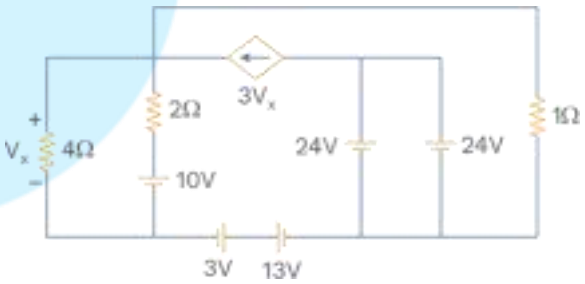
(A) $e^{2t} + e^{3t}$

(B) $2e^{2t} + 3e^{3t}$

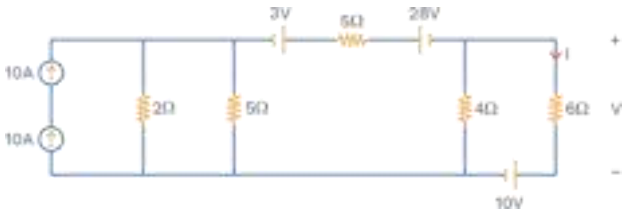
(C) $\frac{1}{2}\left(2e^{2t} + 3e^{3t}\right)$

(D) $2(2e^{2t} + 3e^{3t})$

Q.41 For the circuit shown below, the value of V_x is _____ (in volts) (Rounded off to the nearest integer)



Q.42 For the circuit shown below, which of the statement(s) given in the options is/are correct? MSQ



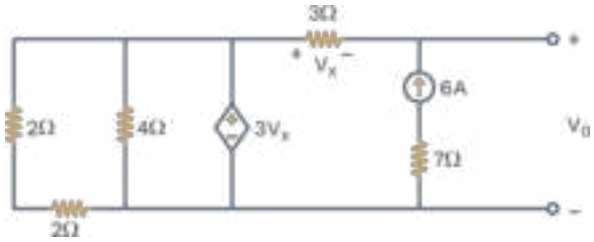
(A) The voltage across 2Ω resistor is 16.666V

(B) The magnitude of voltage V = 0V.

(C) The current I is –1.666 A.

(D) The polarity of I and V are opposite to that considered in the circuit

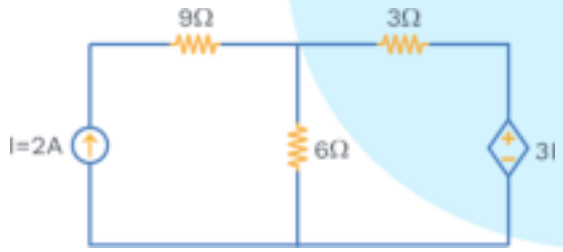
Q.43 Consider the network shown below.



The open circuited voltage is _____ (in Volts).

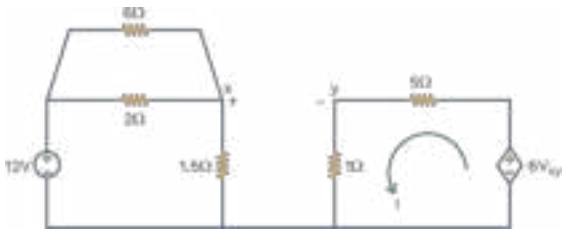
(Rounded off to the nearest integer)

Q.44 In the circuit shown below, the power absorbed by the dependent source is _____ (in Watts). (Up to two decimal places)

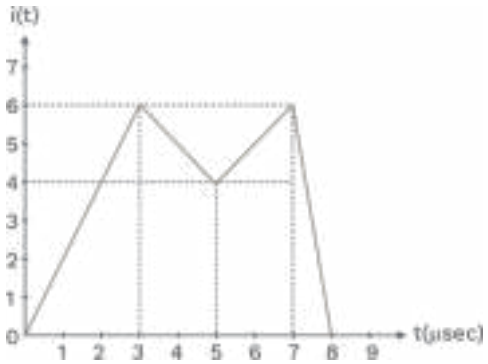


Q.45 In the circuit shown below, the current I is _____ (in Amp).

(Rounded off to the nearest integer)



Q.46 The current characteristics is shown in the figure below.

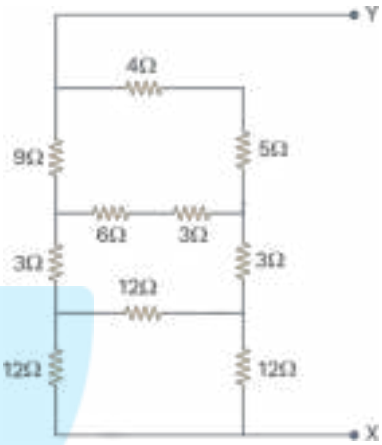


The current passes through a capacitor. The charge (□C) stored in the capacitor after 6 μsec is-

(A) 15 (B) 19.5

(C) 23.5 (D) 32

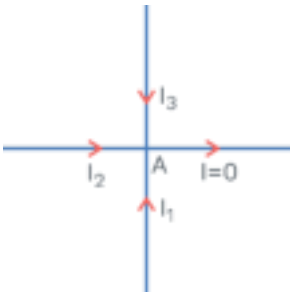
Q.47 In the circuit shown below, the value of the resistance across X and Y i.e., R_{XY} is-



(A) 10Ω (B) 12Ω

(C) 13.5Ω (D) 15Ω

Q.48 The currents entering at node-Aare given as I_1, I_2, I_3 and resultant current I = 0, as shown in the figure below. If $I_1 = -3\sin(\square t)$ mA and $I_2 = 4\cos(\square t)$ mA, thenthe current I_3 is-



(A) 7 sin(ωt + 90°) mA

(B) $\frac{5}{\sqrt{2}}\cos(\omega t + 90°)$ mA

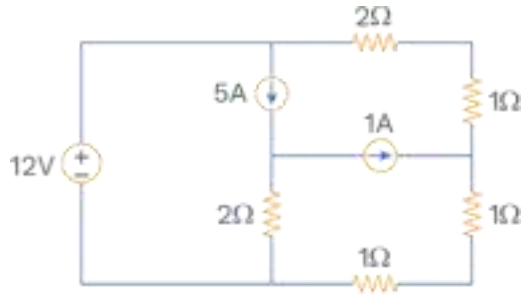
(C) −5cos (ωt + 36.87) mA

(D) 7 cos (9t + 36.87) mA

Q.49 The voltage and current phasors of a circuit are 5□47° and 3□17°

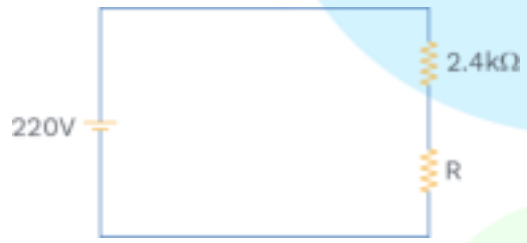
respectively. The ratio of active power toreactive power is _____. (Upto 3 decimal places)

Q.50 A circuit is shown in the figure below.



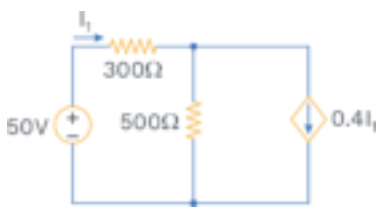
The power supplied by the 12V voltage source is _____ W. (Rounded off to the nearest integer)

Q.51 A circuit is shown in the figure below.



A voltmeter with internal resistance 1.8kΩ is connected across the 2.4kΩ resistance and reads 120V across the resistance. The value of resistance 'R' is _____ Ω. (Upto three decimal places)

Q.52 For the given circuit, the current I_1 and the power absorbed by the 500Ω resistor are respectively-



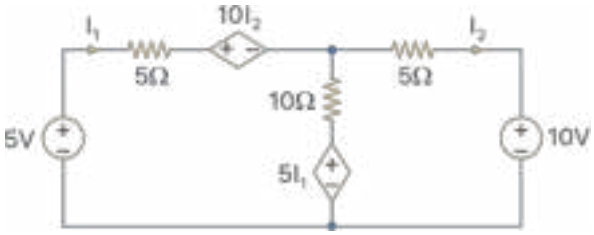
(A) 96.6 mA and 6.77W

(B) 39.9 mA and 2.29W

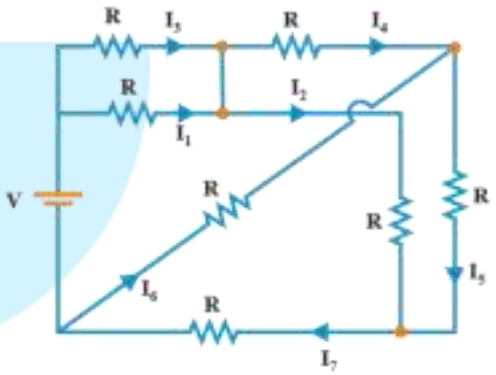
(C) 47.7 mA and 5.25W

(D) 83.3 mA and 3.47W

Q.53 In the given circuit, the current I_2 is _____Amp.(Up to two decimal places)



Q.54 For the circuit shown in the figure below, the only current equation valid for the circuit is-



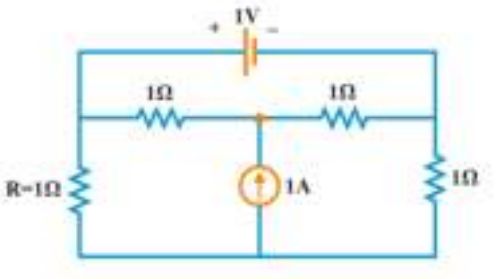
(A) $I_2 + I_5 + I_6 = I_7$

(B) $I_1 + I_2 + I_6 = I_7$

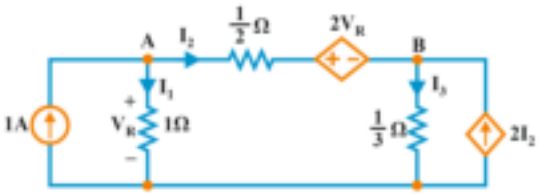
(C) $I_2 + I_4 + I_6 - I_7 = 0$

(D) $I_3 + I_6 + I_4 + I_5 = 0$

Q.55 The current through the resistance 'R' shown in the figure is _____ A. (Rounded off to the near estinteger)

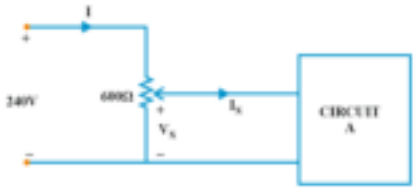


Q.56 A circuit is shown in the figure below.



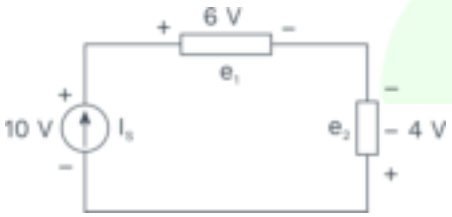
If voltage at A is–1V, then voltage at B will be_____ V. (Up to two decimal places)

Q.57 A DC voltage of 240V is applied to a tapped-out resistor of 600Ω as shown in the circuit below.

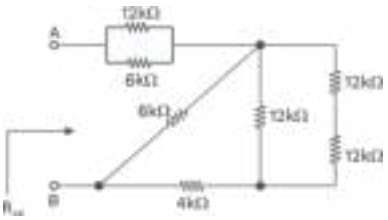


If the current $I_X = 0.2A$ and $V_X = 30V$, then the power supplied by the 240V voltage source is _____ W. (Upto 2 decimal places)

Q.58 In the following circuit, what is the value of I_S, such that net power absorbed by two elements in 4W?



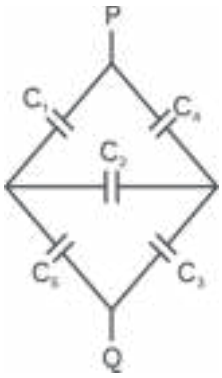
Q.59 Find the equivalent resistance of the circuit shown below at the terminals A – B.



(A) 15 kΩ (B) 5 kΩ

(C) 10 kΩ (D) 8 kΩ

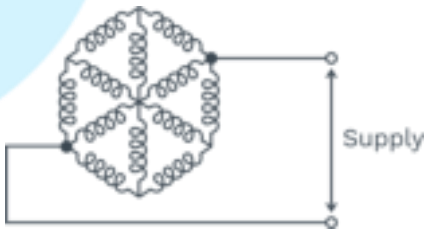
Q.60 Calculate the capacitance between P and Q in μF when C_1, C_3, C_4, C_5 = 8 μF each and C_2 = 20 μF is



(A) 2 μF (B) 4 μF

(C) 6 μF (D) 8 μF

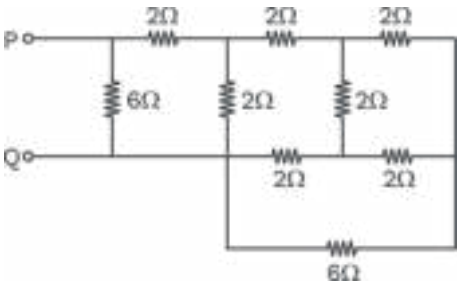
Q.61 Find the equivalent in the circuit shown below, if each inductance is $\frac{1}{2}$H.



(A) $\frac{2}{5}$H (B) $\frac{3}{5}$H

(C) $\frac{1}{5}$H (D) $\frac{4}{5}$H

Q.62 The equivalent resistance between PQ is



(A) $\frac{5}{8}\Omega$ (B) $\frac{15}{8}\Omega$

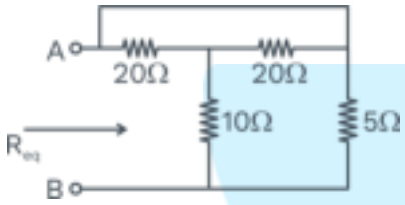
(C) 20Ω (D) $\frac{24}{11}\Omega$

Q.63 Twelve 6Ω resistors are used as edges to from a cube, the resistance between two diagonally opposite corners of the cube is -

(A) $\frac{5}{6}$ (B) 6Ω

(C) $\frac{6}{5}\Omega$ (D) 5Ω

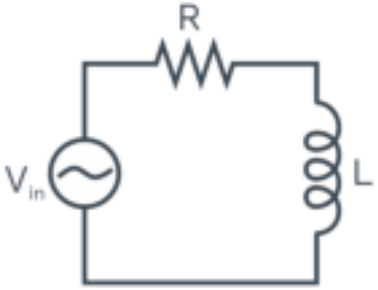
Q.64 For the circuit shown below R_{eq} will be



(A) 4Ω (B) 5Ω

(C) $\frac{10}{3}\Omega$ (D) $\frac{5}{2}\Omega$

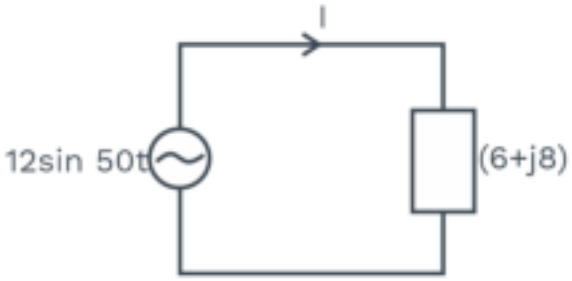
Q.65 The RL circuit of the figure is fed from a constant magnitude, variable frequency sinusoidal voltage source V_{in} at 100Hz, the R & L elements each have a voltage drop U_{rms}. If the frequency of the source is changed to 50Hz, then new voltage drop across resistance R is _____.



(A) $\sqrt{\frac{5}{8}}U_{rms}$ (B) $\sqrt{\frac{8}{5}}U_{rms}$

(C) $\sqrt{\frac{2}{3}}U_{rms}$ (D) $\sqrt{\frac{3}{2}}U_{rms}$

Q.66 The energy (in joules) consumed by the impedance (6+j8)Ω in 5 cycles for the circuit shown below is _____.



Q.67 The input voltage given to a converter is –

$V_i = 50\sqrt{2}\sin(100\pi t)V$

the current drawn by the converter is –

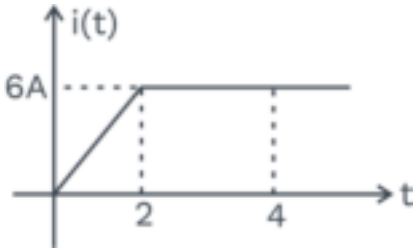
$$i_i = 2\sqrt{2}\cos\left(100\pi t + \frac{\pi}{3}\right) + 3\sqrt{2}\sin\left(300\pi t + \frac{\pi}{4}\right) + 5\sqrt{2}\sin\left(500\pi t + \frac{\pi}{6}\right)A$$

the active power drawn by the converter is –

Q.68 The current flowing through a coil if resistance 1Ω and inductor of 2 H Determine MSQ

(A) The energy absorbed by coil up to 4 second

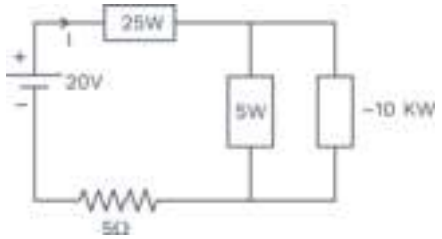
(B) Energy stored in the inductor up to 4 second



(A) 48 J (B) 132 J

(C) 36 J (D) 144 J

Q.69 Consider the shown figure below, in which the power absorbed buy each unknown element is marked on it -



The value of the current I in the above DC circuit will be -

(A) –2A (B) 1A

(C) – 1A (D) 2A

Q.70 Current through a resistor at –

0° C — 1A

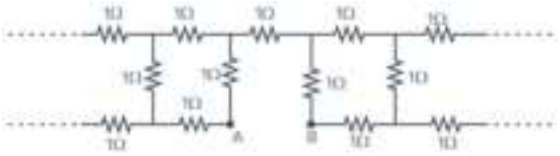
100°C — 0.7A

Find the current at 1200°C temperature:

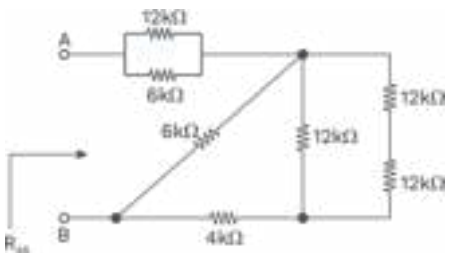
(A) 0.15A (B) 0.1225A

(C) 0.5A (D) 0.45A

Q.71 Each resistance shown in figure is an infinite network of resistance 1Ω. The effective resistance between point A & B.



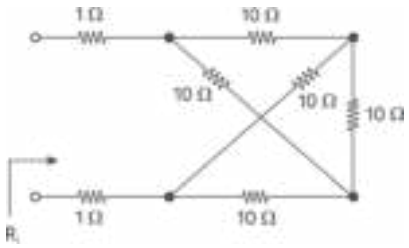
Q.72 Find the equivalent resistance of the circuit shown at the terminals A – B.



(A) 15 kΩ (B) 5 kΩ

(C) 10 kΩ (D) 8 kΩ

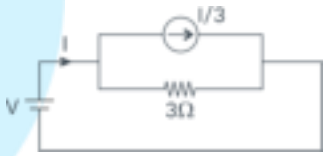
Q.73 The value of input resistance R_i in the circuit shown in figure is-



(A) 2 Ω (B) 12 Ω

(C) 22 Ω (D) None of these

Q.74 For the circuit shown in below, the effective resistance faced by the voltage source.



(A) 4 (B) 3

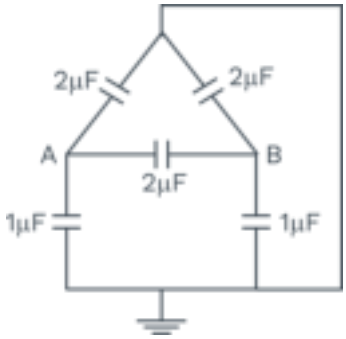
(C) 2 (D) 1

Q.75 R_1 = 36Ω and R_2 = 75Ω, each having to tolerance of ± 5% are connected in series. The value of resultant resistance is –

(A) 111 ± 0Ω (B) 111 ± 2.778Ω

(C) 111 ± 5.55Ω (D) 111 ± 7.23Ω

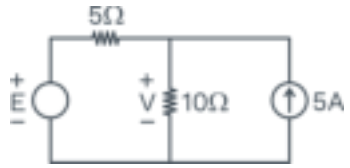
Q.76 For the circuit shown in figure, the capacitance measured between terminals A and B will be –



(A) 1.5 μF (B) 12 μF

(C) 10 μF (D) 3.5 μF

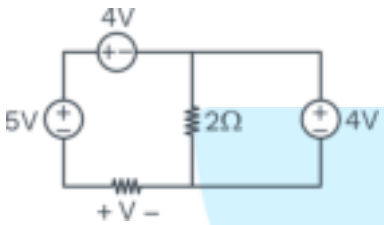
Q.77 If the voltage V across 10Ω resistance is 10V, what is the voltage E of the voltage source in the circuit below?



(A) –50V (B) –10V

(C) +10V (D) +50V

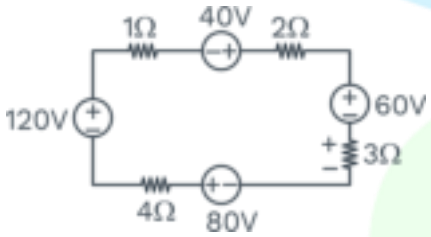
Q.78 The voltage V in figure is equal to



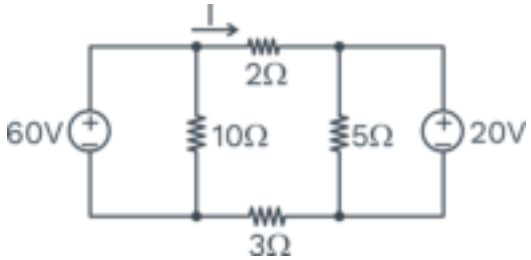
(A) 3V (B) –3V

(C) 5V (D) None of these

Q.79 Find the current in the given circuit.



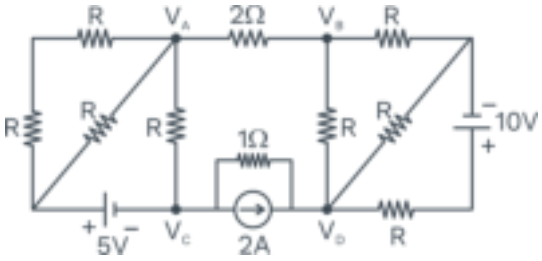
Q.80 The circuit shown in figure. Find the current in 2Ω resistor.



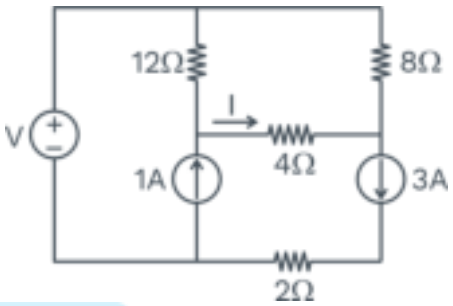
(A) 8A (B) 2A

(C) 4A (D) 16A

Q.81 If $V_A - V_B = 6V$ then $V_C - V_D$ is –



Q.82 In the circuit shown in below, find current I and voltage V. MSQ



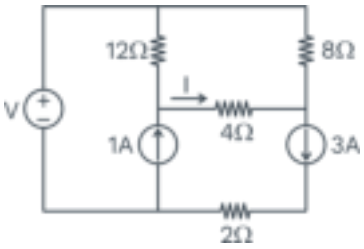
(A) 3A

(B) 1.5A

(C) 6V

(D) Voltage V cannot be obtained

Q.83 In the circuit shown in below, find current I and voltage V. MSQ



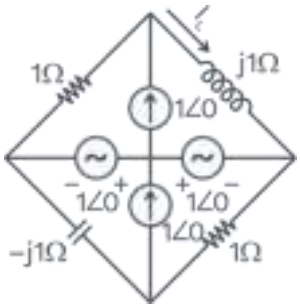
(A) 3A

(B) 1.5A

(C) 6V

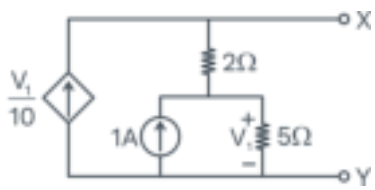
(D) Voltage V cannot be obtained

Q.84 The circuit shown in figure. Find the magnitude current I_L (current in inductor)



Q.85 The circuit shown in figure. Find open circuit voltage and short circuit current at X and Y terminals.

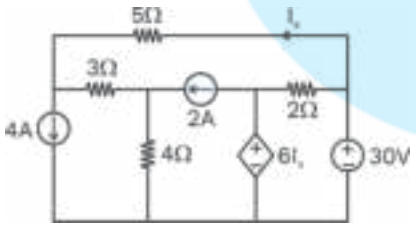
MSQ



(A) 10 V (B) 12 V

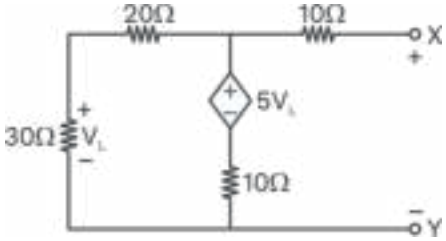
(C) 0.8571 A (D) None of these

Q.86 The circuit shown in figure.



Find power consumed by 2Ω resistance

Q.87 The equivalent resistance of the network given below while looking through terminals xy is



Q.88 The capacitance of linear time varying capacitor is given by –

$$C(t) = C_0\left(1 - e^{-\alpha t}\right)$$

If voltage across capacitor is

$$V(t) = V_0\, t^2$$

Determine capacitor current.

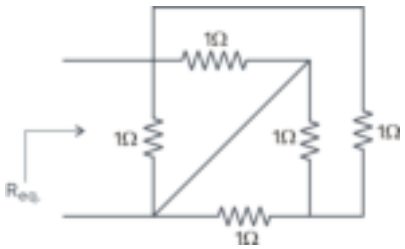
(A) $2V_0\, C_0\, t(1-e^{-\alpha t}) + V_0\, C_0\, t^2\, \alpha e^{-\alpha t}$

(B) $2\,\alpha e^{-t}\,(1-e^{-\alpha t}) + V_0\, C_0\, t^2\, e^{-\alpha t}$

(C) $2\,\alpha e^{-t}\,(1-e^{-\alpha t}) - V_0\, C_0\, t^2\, e^{-\alpha t}$

(D) $2V_0\, C_0\, t^2\, e^{-\alpha t}$

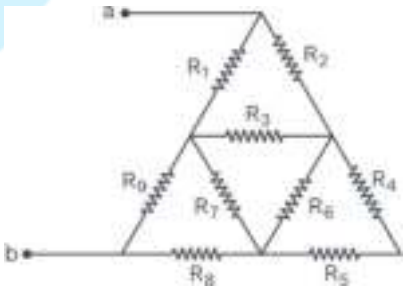
Q.89 Find R_{eq}?



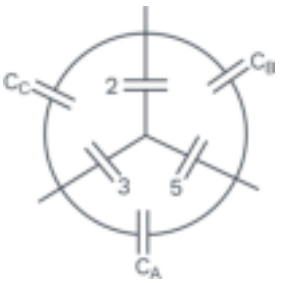
(A) $\frac{2}{8}$ (B) $\frac{5}{8}$

(C) $\frac{1}{8}$ (D) $\frac{3}{8}$

Q.90 Find R_{ab} if value of each resistance 30 ohm



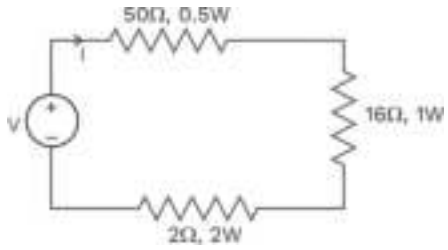
Q.91 Determine the value of C_A, C_B, C_C?



(A) 15.5, 10.34, 6.2 (B) 1, 0.5, 2

(C) 1.5, 1, 0.6 (D) 6, 10, 15

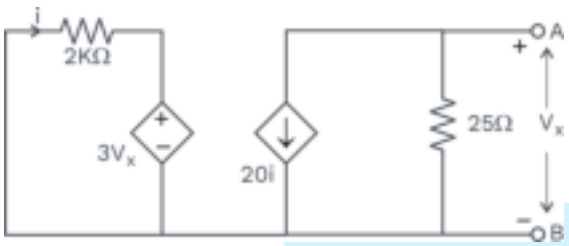
Q.92 In the circuit shown below the safe maximum value of the current I is –



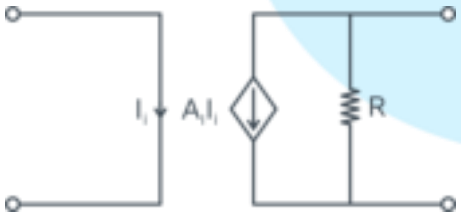
(A) 0.25 A (B) 0.5 A

(C) 0.1 A (D) 1 A

Q.93 Find equivalent resistance between AB terminals of circuit given below –



Q.94 The circuit shown in the figure represent a



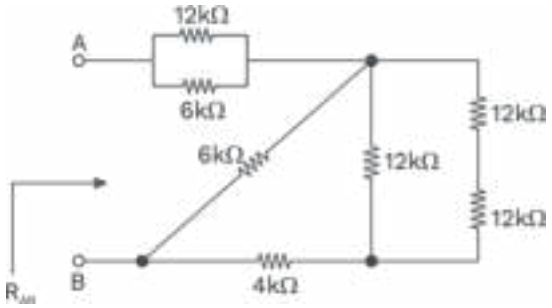
(A) Voltage controlled voltage source

(B) Voltage controlled current source

(C) Current controlled current source

(D) Current controlled voltage source

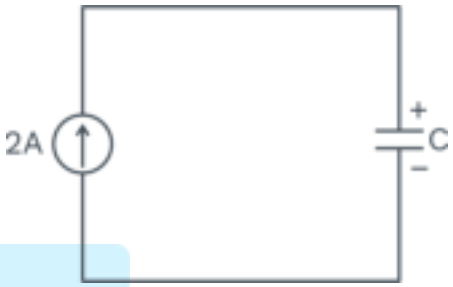
Q.95 Find the equivalent resistance of the circuit shown below, at the terminals A – B.



(A) 15 kΩ (B) 5 kΩ

(C) 10 kΩ (D) 8 kΩ

Q.96 Is the circuit BIBO stable?



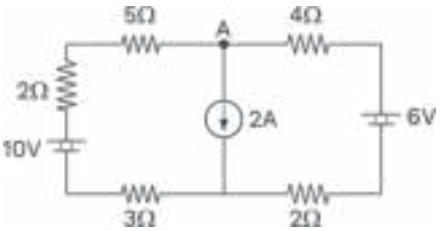
(A) Stable for low current

(B) Stable for high values of C

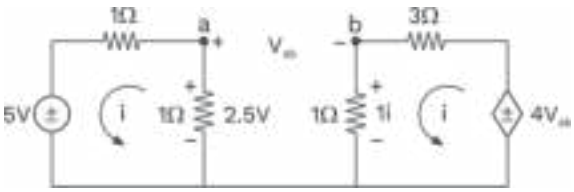
(C) Not stable

(D) Stable for large current

Q.97 Consider below circuit, voltage at node A is _____ V.



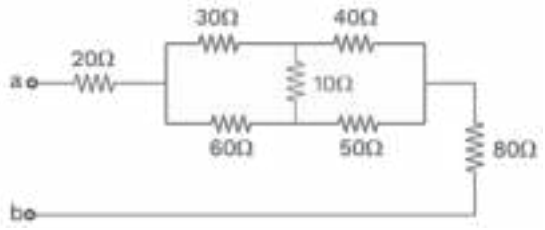
Q.98 In the circuit shown in figure, the value of the current i will be given by-



(A) 0.31 A (B) 1.25 A

(C) 1.75 A (D) 2.5A

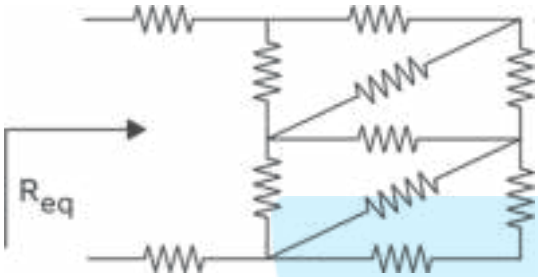
Q.99 Obtain the equivalent resistance R_{ab} in the circuits of figure.



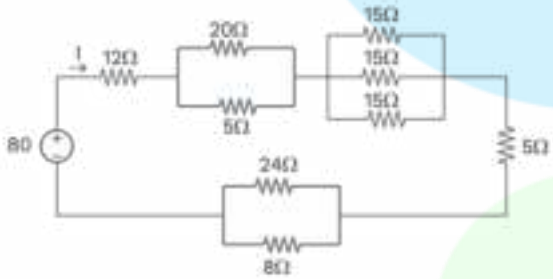
(A) 52 Ω (B) 30 Ω

(C) 142.22 Ω (D) 140 Ω

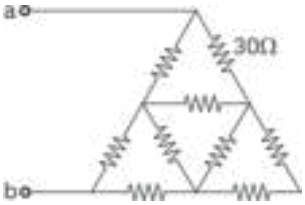
Q.100 For the circuit shown in figure. The equivalent resistance is _____Ω. All resistor are 1Ω.



Q.101 For the Circuit shown in figure. The value of I _____ in amperes.



Q.102 Obtain the equivalent resistance R_{ab} in the circuits of the figure. All resistors have a value of 30 Ω.



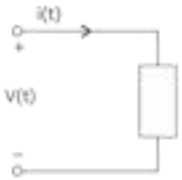
(A) 33.93 kΩ (B) 33.33 Ω

(C) 330 Ω (D) 110 Ω

Q.103 The current through and voltage across the element shown in figure are given as MSQ

V(t) = 4 cos 3t V,

i(t) = 5 sin 3t A



Consider the following statements:

(A) The maximum value of power absorbed is 10W.

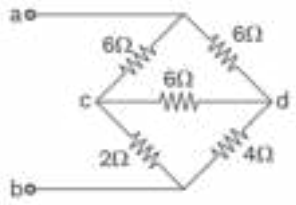
(B) At t = 0.5 sec, element absorbs power.

(C) At t = 1 sec, element absorbs power.

(D) None of these

Which of the above statements are true

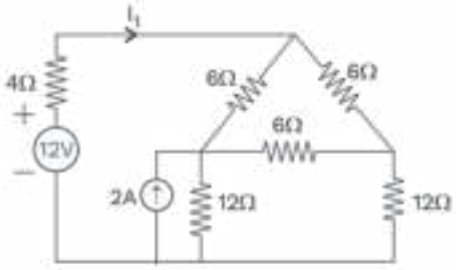
Q.104 Determine the resistance (ohm) across the terminals 'ab' of the interconnected resistors of figure.



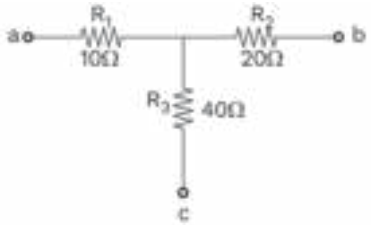
(A) 2.2 (B) 1.1

(C) 4.4 (D) 0.1

Q.105 Find the current I_s(A) shown in the network.



Q.106 Transform the wye network in figure to a delta network.



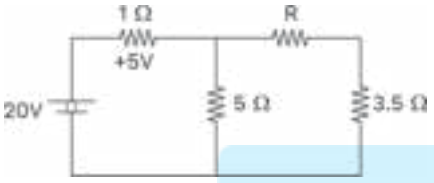
(A) R_a = 140 Ω, R_b = 70 Ω, R_c = 35 Ω

(B) R_a = 140 Ω, R_b = 140 Ω, R_c = 70 Ω

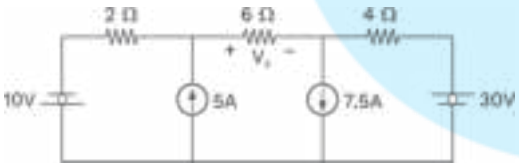
(C) R_a = R_b = 140 Ω, R_c = 30 Ω

(D) R_a = 30 Ω, R_b = 140 Ω, R_c = 70 Ω

Q.107 In the circuit shown below, the value of R is _____ Ω.



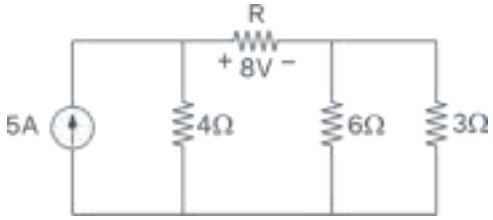
Q.108 In the circuit shown below, determine the value of V_x = ?



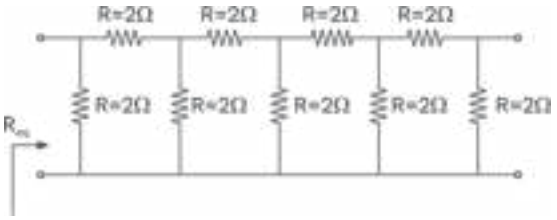
(A) 60 V (B) 20 V

(C) 0 V (D) 40 V

Q.109 The power absorbed by 4 Ω resistor in the circuit shown below is _____ (W).

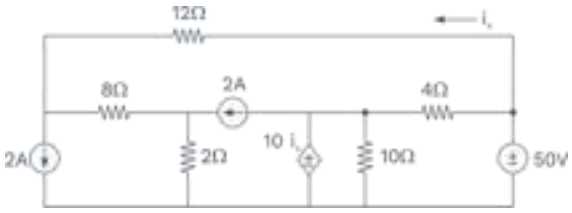


Q.110 The equivalent resistance in the infinite ladder network shown in the figure, is R_{eq}.



The value of R_{eq} _____.

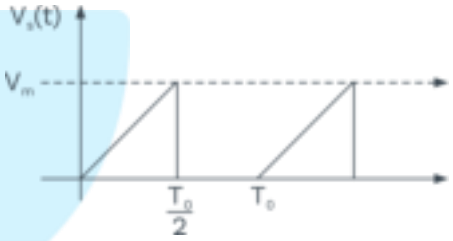
Q.111 Consider the circuit given below power dissipated in 4 Ω resistor is-



(A) 0 W (B) 25 W

(C) 50 W (D) 100 W

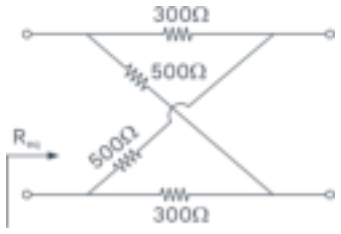
Q.112 The average power absorbed by the resistor R connected to an independent voltage source, with excitation shown below is _____



(A) $\frac{2V_m}{R}$ (B) $\frac{V_m^2}{6R}$

(C) $\frac{V_m^2}{R}$ (D) $\frac{2V_m^2}{R}$

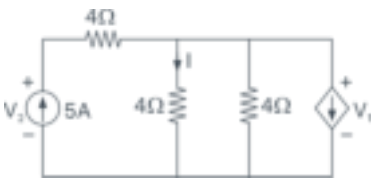
Q.113 For the circuit shown in figure, the value of R_{eq} is



(A) 200 Ω (B) 600 Ω

(C) 400 Ω (D) 800 Ω

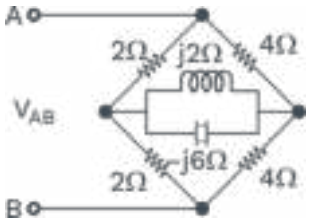
Q.114 In the given circuit, the values of V_1 and V_2 respectively are MSQ



(A) 5 V (B) 10 V

(C) 50 V (D) 25 V

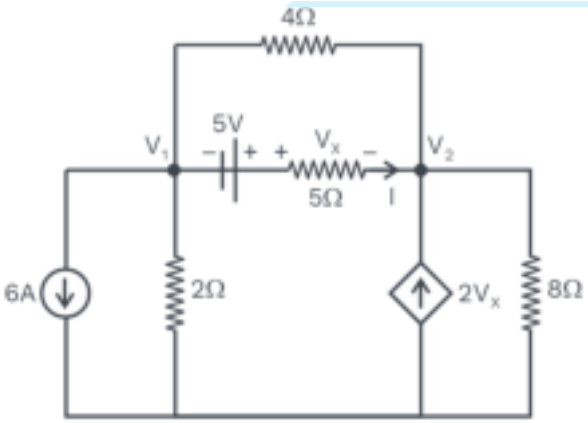
Q.115 Find the Z_{AB}-



(A) $\frac{3}{8}\Omega$ (B) 0 Ω

(C) 1 Ω (D) $\frac{8}{3}\Omega$

Q.116 Consider the circuit shown below figure:



The voltage across 5Ω resistor is____ V (up to two decimal places)

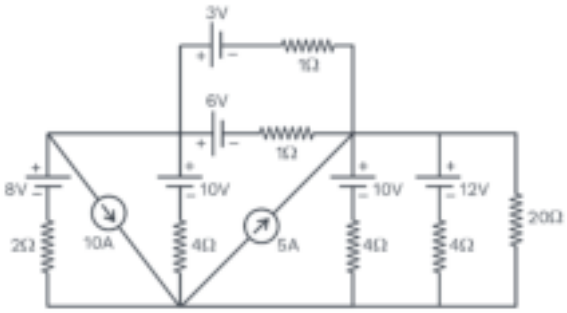
Q.117



The value of 'R such that the current through 60V voltage source is zero is _____Ω

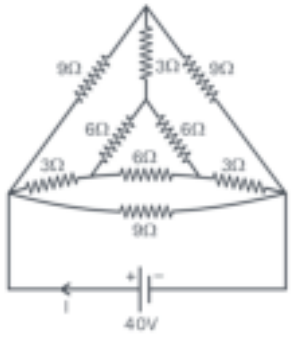
Q.118 5 resistors of equal value when connected in series across a supply dissipates 30W across each resistor. If the same resistors are now connected in parallel across the same supply, the power dissipated will be ____ W.

Q.119 Consider the circuit shown given below figure.



The power absorbed by 20Ω resistor is _____ W

Q.120 Consider the circuit shown given below figure.

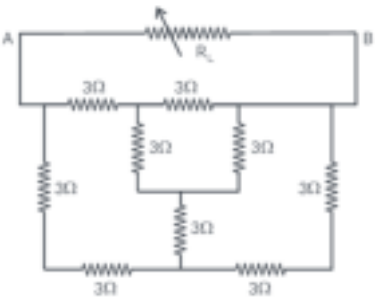


The value of current I will be –

(A) 9.23 A (B) 10.67 A

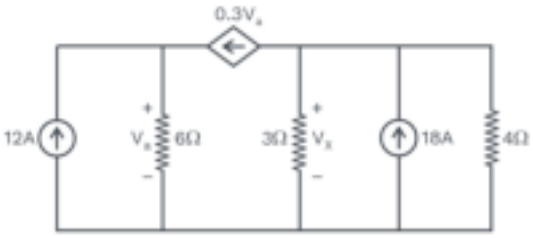
(C) 11.99 A (D) 3.99 A

Q.121 Consider the circuit shown given below figure.



If the open circuit voltage across the load resistance A and B, is 60V then the maximum power absorbed by the load resistor is _____ W.

Q.122 Consider the circuit shown given below figure.



Power absorbed by 3Ω is:

(A) 44.45 W (B) 56.34 W

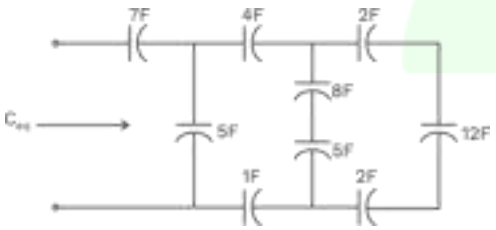
(C) 82.33 W (D) 79.26 W

Q.123 A single phase 240V, 2 kW heater is connected across single phase 250V, 50 Hz supply through a diode calculate the power delivered to the heater element.

(A) 1085 W (B) 868 W

(C) 2000W (D) 1414W

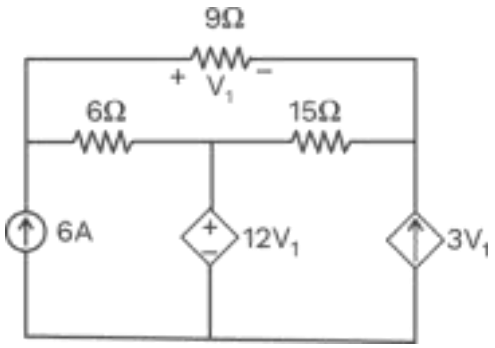
Q.124 Calculate C_{eq} of the circuit given below?



(A) $\frac{17}{3}$ F (B) $\frac{119}{38}$ F

(C) 5 F (D) $\frac{114}{19}$ F

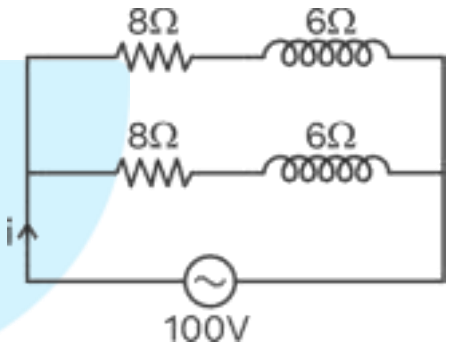
Q.125 Calculate V_1 for the circuit given below



(A) $\frac{128}{15}$ V (B) $\frac{148}{35}$ V

(C) $\frac{125}{36}$ V (D) $\frac{108}{145}$ V

Q.126 For the circuit shown, the supply current 'i' is



(A) 10 A (B) Zero

(C) 20 A (D) 14.28 A

Q.127 The average power delivered to an impedance $(4 - j3)\ \Omega$ by a current $5\cos(100\pi t + 100)$ A is ..... in W

Q.128 If the voltage across it is,

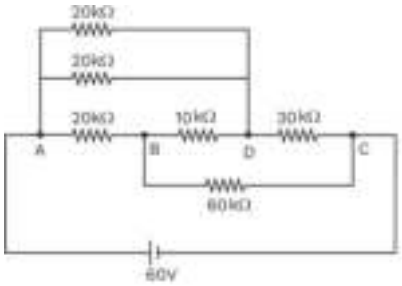
$$V(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t < 0 \end{cases}$$

Then find the energy stored at t = 5s by using the expression of current?

(A) 180.42kJ (B) 156.25kJ

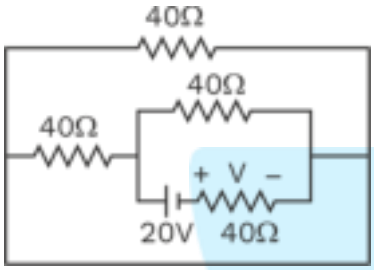
(C) 132.28kJ (D) 164.32kJ

Q.129

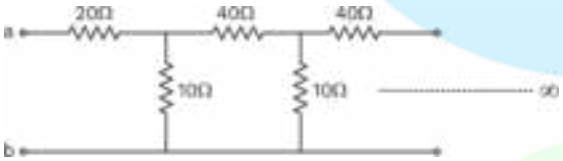


Find the current through 10 ohm resistor in the circuit shown in figure.

Q.130 The value of 'V' from the given below circuits is _____V.



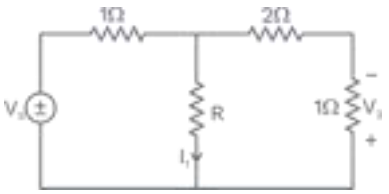
Q.131 Find equivalent resistance across terminal ab



(A) 26.5 Ω (B) 30 Ω

(C) 28.28 Ω (D) None of these

Q.132 For the given circuit, I_1 = 1 A, V_2 = 2 V as shown in the figure then find the value of V_S.



(A) 14 V (B) 7 V

(C) –14V (D) –7V

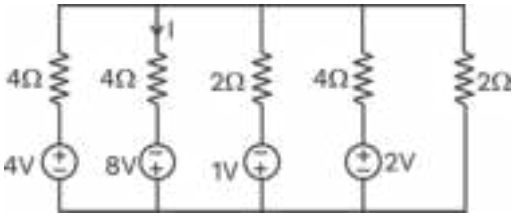
Q.133 Calculate the average power dissipated by current i(t) for a 2Ω resistor?

$$i(t) = 2 + 2\sqrt{2}\cos(2t + 45°) + 2\sin 2t$$

(A) 24W (B) 12W

(C) 32W (D) 28W

Q.134 Consider the given network below:-



Find the value of current 'I' in the circuit?

(A) 3.56 A (B) 2.95 A

(C) 1.86 A (D) 0.98 A

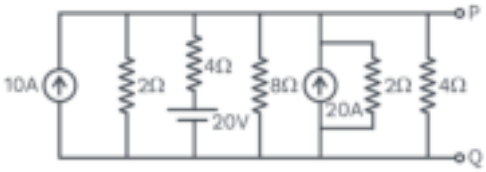
Q.135 If the voltage across the 10 H inductor is, $V(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t < 0 \end{cases}$

then find the energy stored at t = 2 sec by using the expression of current ?

(A) 640 J (B) 240 J

(C) 320 J (D) 480 J

Q.136 The circuit left to terminal P and Q can be replaced by a voltage source in series with resistor. The value of voltage source and resistor are:



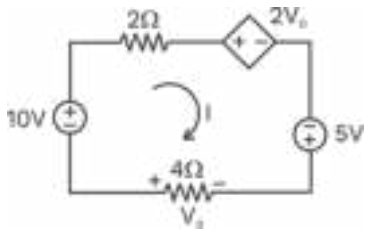
(A) 35 V, 0.615 W

(B) 21.525 V, 0.615 W

(C) 35 V, 0.725 W

(D) 21.525 V, 0.725 W

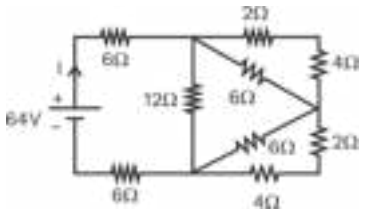
Q.137 Determine the value of V_0 and I in the circuit shown below.



(A) 15V, – 7.5 A (B) 30V, – 7.5 A

(C) 30V, – 8.5 A (D) 15V, – 8.5 A

Q.138 Find 'I' in the network shown below:



(A) 4A (B) 3A

(C) 5A (D) 8A

Q.139 Math the duality of electrical network of Group-1 with Group-2.

Group-1

P. Link

Q. Mesh or loop

R. Loop current

S. Number of loops

Group-2

1. Node pair voltage
2. Number of node pairs
3. Node or node pairs
4. Tree branch

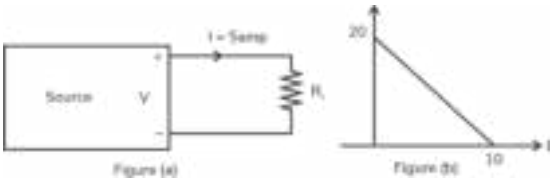
(A) P-4, Q-3, R-2, S-1

(B) P-4, Q-1, R-2, S-3

(C) P-4, Q-2, R-3, S-1

(D) P-4, Q-3, R-1, S-2

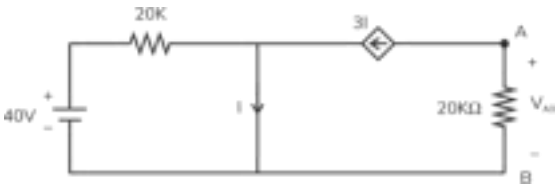
Q.140 The load line of source in the figure (a) is represented in figure (b). The value of load resistance R_L so that the current across it will be of 5 amp.



(A) 5Ω (B) 4Ω

(C) 2Ω (D) 10Ω

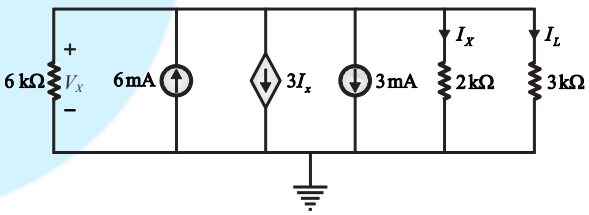
Q.141 The voltage across the 2kΩ resister V_{AB} in the given network will be



(A) 0 V (B) 6 V

(C) -5 V (D) 5 V

Q.142 Consider the following circuit shown below.



The value of V_X and I_L are respectively-

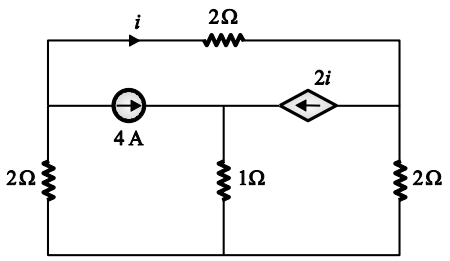
(A) 15 V, 0.3 mA

(B) 18/15 V, 0.4 mA

(C) 0.4 V, 15/18 mA

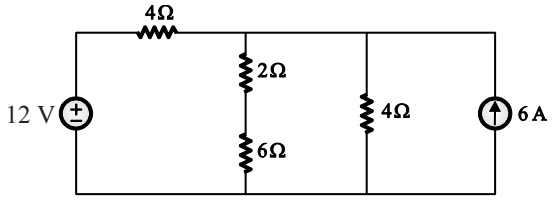
(D) 0.3 V, 15 mA

Q.143 Consider the circuit shown in figure below



The value of current i is _____ A. (in integer

Q.144 Consider the circuit shown in figure below



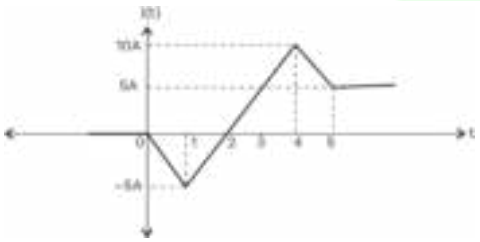
The power dissipated in 2Ω resistor is _____ W. (correct upto two decimal places)

Q.145 An inductor of inductance ' L' have 'N' number of turns and relative permeability (μ_r) as 1. The area of cross-section and length of the inductor is 'A' and 'L' respectively. Now the inductor is immersed in liquid with $\mu_r = 0.32$, then the value of new inductance will be-

(A) $\frac{L}{4}$ (B) $\frac{8}{25}L$

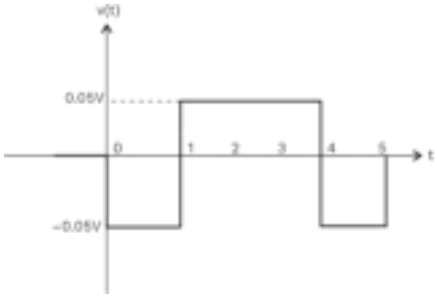
(C) $\frac{L}{8}$ (D) $\frac{3}{16}L$

Q.146 Current waveform is shown in the figure below.

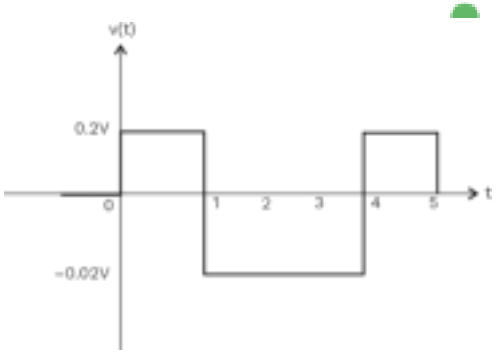


The current i(t) is applied to an inductor of 10mH. The corresponding waveform will be-

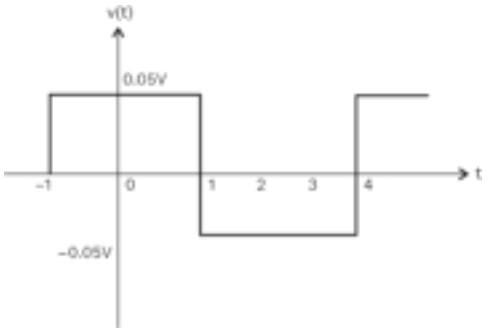
(A)



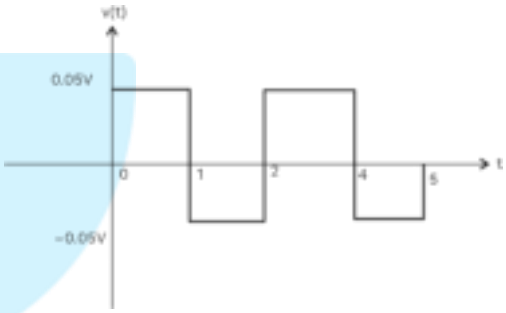
(B)



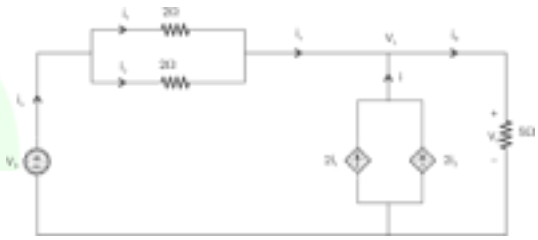
(C)



(D)



Q.147 A circuit is shown in the figure below.



Which of the following statements is/are true? MSQ

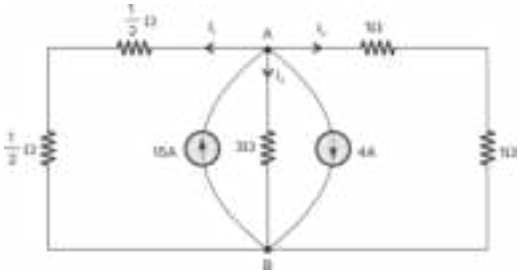
(A) The supply voltage $V_S = \frac{15}{16}V_R$

(B) The current $i_R = \frac{1}{5}V_R$

(C) The current $i_s = \frac{1}{5}V_R$

(D) The current $i = \frac{2}{15}V_R$

Q.148 Consider a circuit shown below.

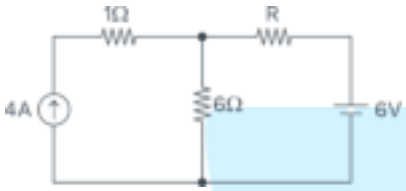


The power dissipated across 3Ω is

(A) 4Ω (B) 12Ω

(C) 18Ω (D) 36Ω

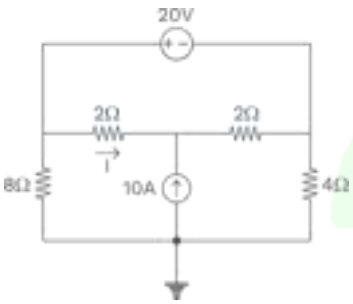
Q.149 If the 6Ω resistor draws a current of 2A as shown in the figure, the value of resistance R is-



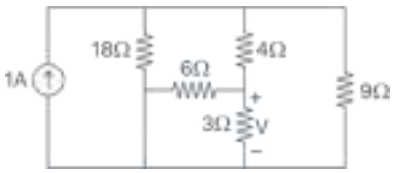
(A) 3Ω (B) 6Ω

(C) 9Ω (D) 12Ω

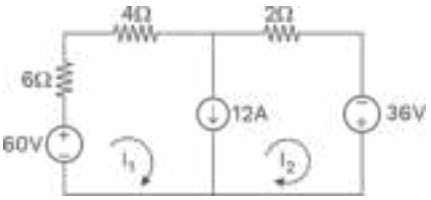
Q.150 In the circuit shown below the current (I) is equal to _____ A.



Q.151 In the resistive network shown below the value of voltage V is-



Q.152 The value of I_1 and I_2 in the network of figure are-



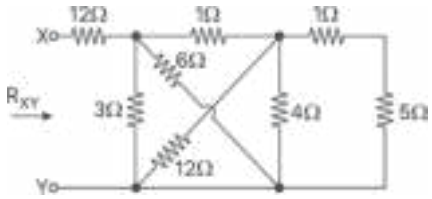
(A) I_1 = 2A, I_2 = –10A

(B) I_1 = –4A, I_2 = 16A

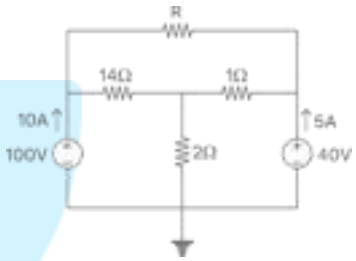
(C) I_1 = –10A, I_2 = 2A

(D) I_1 = 10A, I_2 = –2A

Q.153 Calculate the equivalent resistance R_{XY} in the circuit given below-



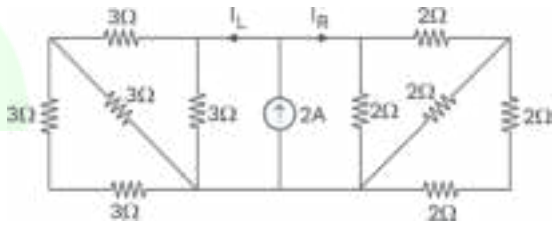
Q.154 The circuit shown in below. Find the value of R is _____.



(A) 3Ω (B) 6Ω

(C) 9Ω (D) 12Ω

Q.155 For the circuit shown below, Find the value of (I_R – I_L) in ampere-



Q.156 The current $i(t)$ flowing in a series combination of a resistor of 20Ω and an inductor of 4mH is given as $i(t) = 8 + 6\sin(200\pi t + 30^0) + 6\sin(400\pi t + 45^0)$ The rms value of current and power dissipated in the circuit are respectively:

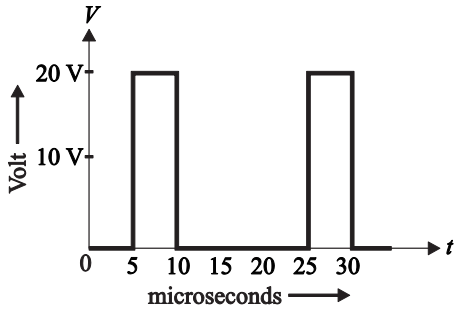
(A) 14.13 A, 3993.14 W

(B) 20 A, 8000 W

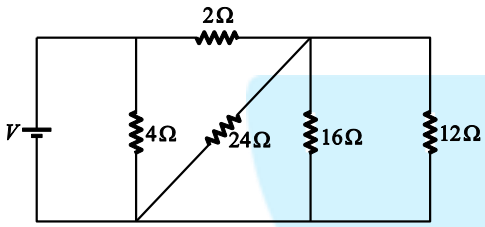
(C) 7.34 A, 1078.75 W

(D)10 A, 2000 W

Q.157 The value indicated by a DC voltmeter for the waveform given below is _____ V. (in integer)

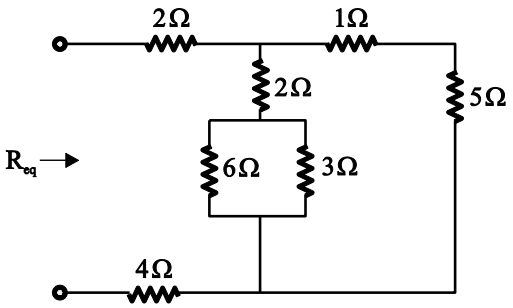


Q.158 Consider the circuit shown in figure below



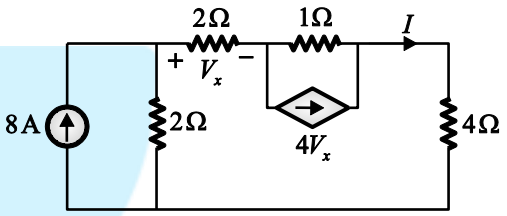
If the power dissipated by $12\,\Omega$ resistance is 144W, then the value of voltage source (V) is _____V. (rounded upto two decimal places)

Q.159 Consider the resistive network shown in figure below,



The value of R_{eq} is _____Ω. (rounded upto one decimal place)

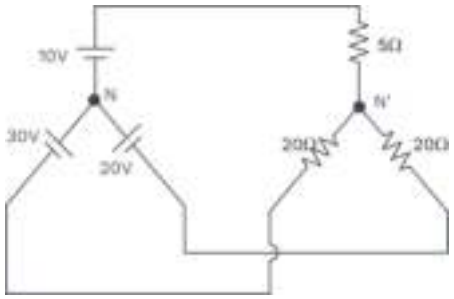
Q.160 Consider the circuit shown in figure below



The value of current I is _____A. (in integer)

3 Network Theorem

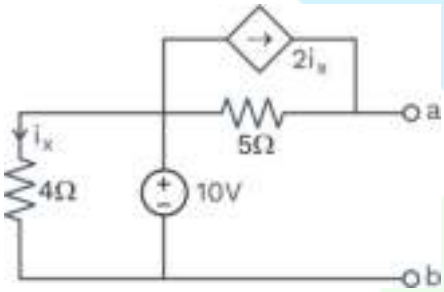
Q.1 Using Norton's theorem, find the current flowing through 10Ω resistance connected between terminals N and N'



(A) 1.5 A (B) 1.25 A

(C) 1.124 A (D) 1 A

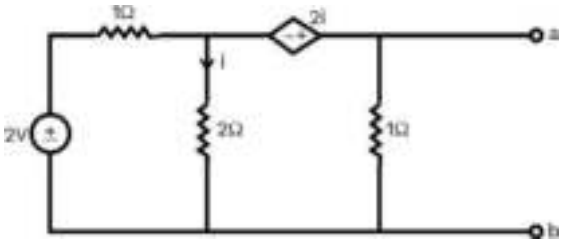
Q.2 Obtain the Norton equivalent I_N and R_N?



(A) 5 Amp (B) 7 Amp

(C) 5Ω (D) 2.5Ω

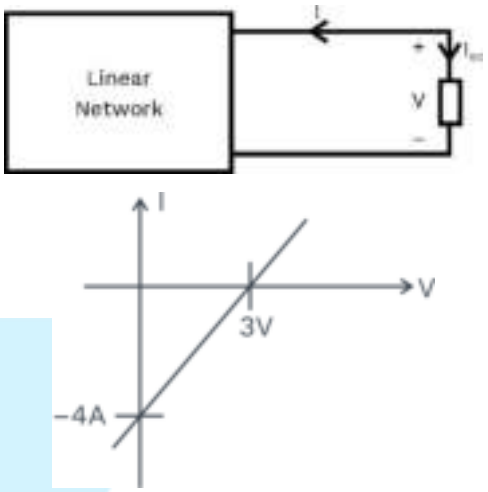
Q.3 Calculate the value of V_{th} and R_{th}?



(A) $\frac{4}{7}\Omega$ (B) $\frac{2}{7}\Omega$

(C) $\frac{8}{7}$V (D) $\frac{8}{14}$V

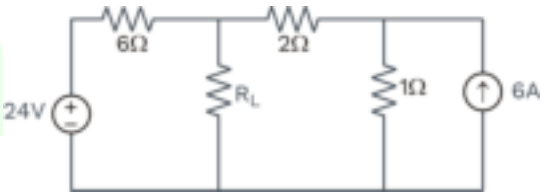
Q.4 For the linear network shown in the figure voltage current characteristic are shown. Find Norton equivalent resistance in□□.



(A) 0.5Ω (B) 0.75Ω

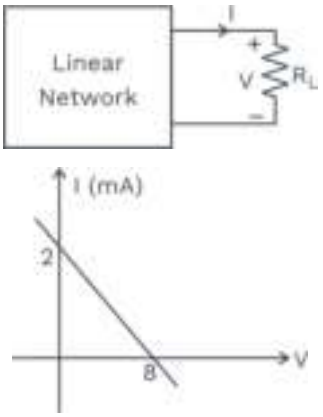
(C) – 0.5Ω (D) – 0.75Ω

Q.5 Find the maximum power trans furred to RL ..... (W)

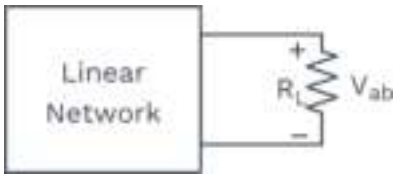


The maximum power transferred to R_L -

Q.6 The V I relation for the circuit below is plotted in figure the maximum power that can be transferred to load R_L will ..... (inmW)



Q.7 Consider the network shown below –



The power absorbed by load resistance R_L is shown in table –

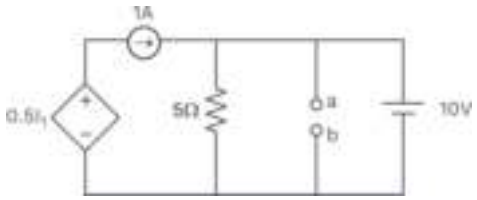
R_L	10KΩ	30KΩ
P	3.6MW	4.8MW

What is the value of R_L (in Ω) that would absorb maximum power -

(A) 30 Ω (B) 20 Ω

(C) 24 Ω (D) 25 Ω

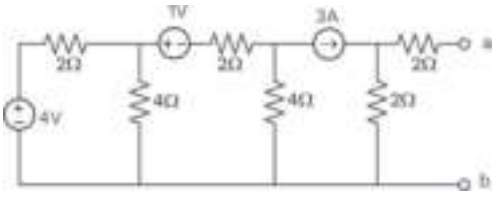
Q.8 Find the venin's equivalent resistance at terminal ab –



(A) 7.5 Ω (B) 6 Ω

(C) 5.5 Ω (D) 2.5 Ω

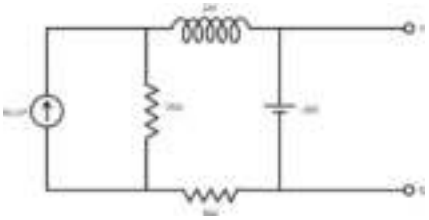
Q.9 Find the Thevenin's voltage and resistance across terminal ab –



(A) 6V (B) 6Ω

(C) 4V (D) 3Ω

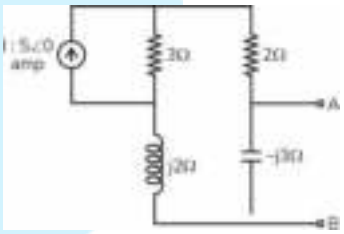
Q.10 Find the Norton equivalent current with respect to the terminal P and Q is ..... (A)



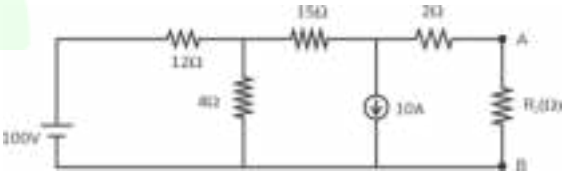
(A) 4Ω (B) 6V

(C) 3Ω (D) 4V

Q.11 In the given network shown in figure, the Nortons's current across the terminal AB is_____



Q.12 For the maximum power transfer to the network shown in the figure the value of R_L will be_____.



Q.13 Consider the following statements:-

1. Tellegen's theorem is applicable to any network linear or nonlinear, active or passive, time varying or time invariant.
2. Superposition theorem is not applicable to ac circuit
3. Maximum power transfer from source having complex internal

impedance to a complex load take place only if source impedance is same as load impedance

4. Superposition theorem is applicable to networks having more than one independent source.

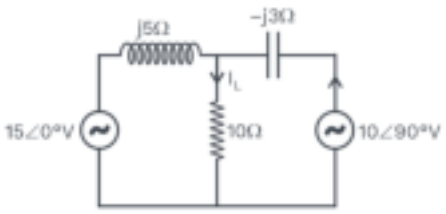
Which of these statements are correct?

(A) 1 and 4 (B) 2 and 3

(C) 1 and 3 (D) All of the above

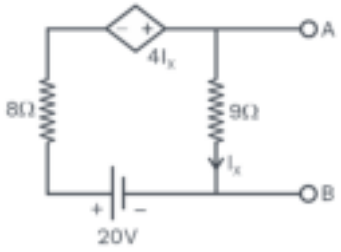
Q.14 A load is connected to an active network. At the terminals to which load is connected, the value of Thevenin's resistance is given by R_{th} = 15Ω and the venin's voltage is V_{th} = 30V. For maximum power that can be delivered to this load, the energy delivered to the load in one hour is_____ kJ

Q.15 Consider the circuit shown given below figure.



By using Norton's theorem calculate current across 10Ω resistor is _____ A

Q.16 Consider the circuit shown below figure.

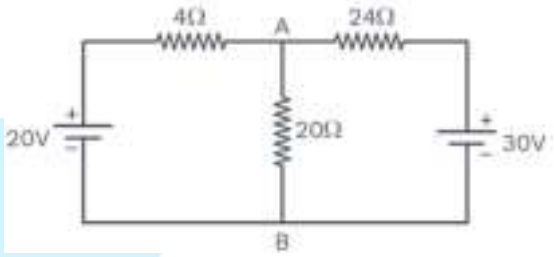


The Norton's equivalent circuit across the terminals A–B is –

(A) 7.62 A, 3.54Ω (B) 3.4 A, 4.45 Ω

(C) 6.32 A, 3.45 Ω (D) 2.5 A, 5.54 Ω

Q.17 Consider the circuit shown below figure:



The Thevenin's equivalent across 20Ω resistance is –

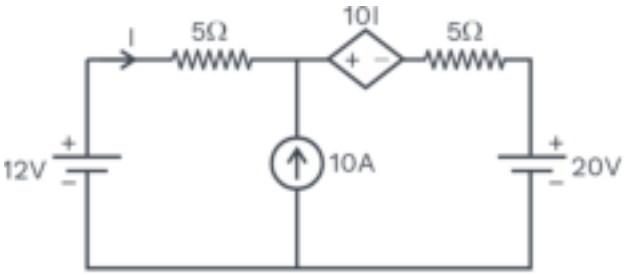
(A) 21.43 V, 0.91 A

(B) 25.32 V, 0.87 A

(C) 33.24 V, 0.78 A

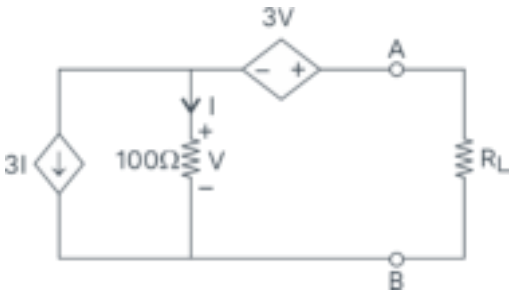
(D) None of these

Q.18 Consider the circuit shown below figure:



Power absorbed by 5Ω resistor is left side of the dependent voltage source_ W

Q.19 The maximum power delivered to resistance R_L is _____ W.



Q.20 Which of the following statements is incorrect about reciprocity theorem-

(A) Initial conditions are assumed to be zero.

(B) There should not be extra dependent or independent sources in the network.

(C) It is not applicable for magnetically coupled circuits.

(D) The network should be bilateral

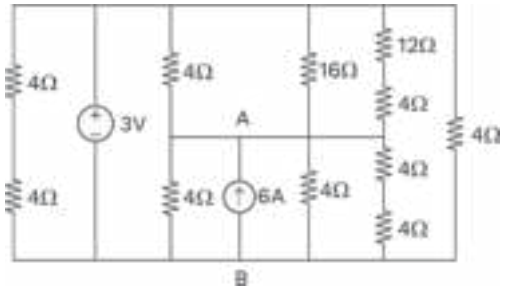
Q.21.



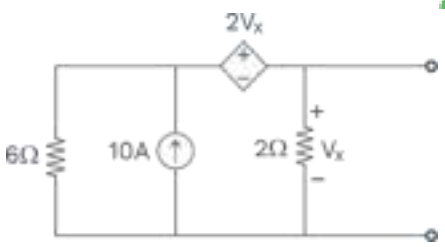
(A) 5 Ω (B) 2.5 Ω

(C) 1.25 Ω (D) 10 Ω

Q.22 In the circuit shown the value of R_{Th} across AB is _____ (Ω).



Q.23 The value of R_{Norton} and I_{Norton} across AB for the circuit is-



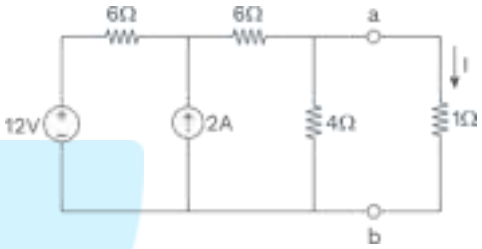
(A) $R_{Norton} = 2\Omega$, $I_{Norton} = 10A$

(B) $R_{Norton} = 1\Omega$, $I_{Norton} = 10A$

(C) $R_{Norton} = 4\Omega$, $I_{Norton} = 5A$

(D) $R_{Norton} = 2\Omega$, $I_{Norton} = 5A$

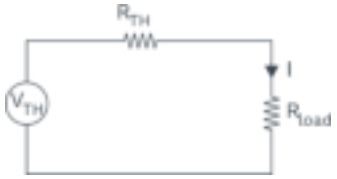
Q.24 The value of R_{Th} across ab and I in the circuit given below is-



(A) 3Ω, 2A (B) 2Ω, 3A

(C) 4Ω, 1.5A (D) 3Ω, 1.5A

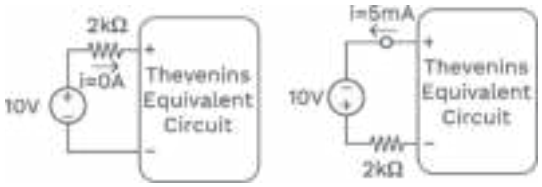
Q.25 Given circuit :



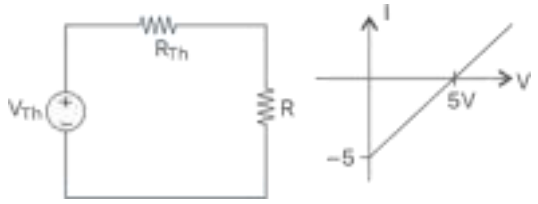
At normal load resistance = R_{load} = 20 Ω and I = 50A

At 10% higher load I = 48A. The value of R_{TH} is _____ Ω.

Q.26 The current I is measured in two experiments which are performed on Thevenin's equivalent circuit as shown. The Thevenin's equivalent resistance is _____ (kΩ).



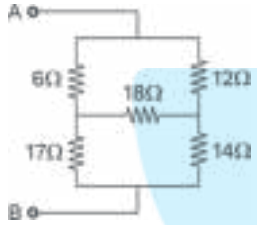
Q.27 The VI graph for the below circuit is shown 2nd figure. V and I are voltage and current through R. The value of R_{TH} is-



(A) 1 Ω (B) 10 Ω

(C) 5 Ω (D) 2 Ω

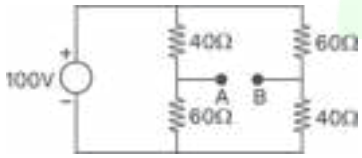
Q.28 The equivalent resistance between AB is-



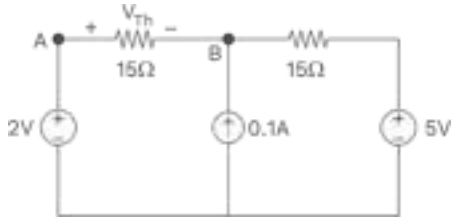
(A) 6 Ω (B) 12 Ω

(C) 10 Ω (D) 18 Ω

Q.29 In the circuit shown below, the Thevenin's resistance looking from the terminals A and B is _____ (Ω)



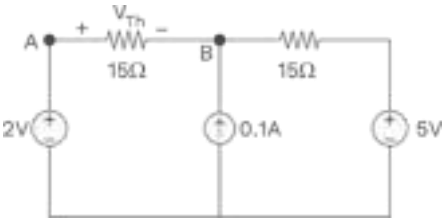
Q.30 The Norton's current I_N is _____ (A).



(A) 0.3 (B) −0.3

(C) 0.5 (D) −0.5

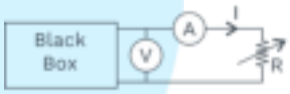
Q.31 The Thevenin's voltage V_{Th} across the terminals A and B is _____ (V).



(A) −4.5 V (B) 4.5 V

(C) 9 V (D) −9 V

Q.32 A black box a circuit in it is connected to variable resistor. An ideal ammeter and an ideal voltmeter are use to measure current and voltage as shown in figure. The result are shown in the table.



R Ω	V(v)	i(A)
2	3	1.5
8	8	1
14	10.5	0.75

Find the maximum power from the box.

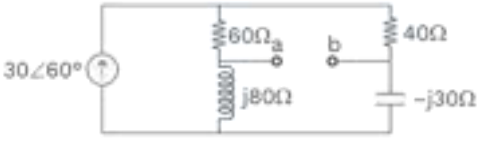
(A) 40 W (B) 5 W

(C) 3 W (D) 8.1 W

Q.33 Find the value of current in amperes when the network satisfies the reciprocity theorem-



Q.34 For the circuit shown in figure, find the Norton at terminals a – b



(A) $44.72\angle 63.43°\Omega$ (B) $(30 + j80)\Omega$

(C) $40\angle -40°\Omega$ (D) $(1 + j3)\Omega$

Q.35 The reciprocity theorem is applicable for-

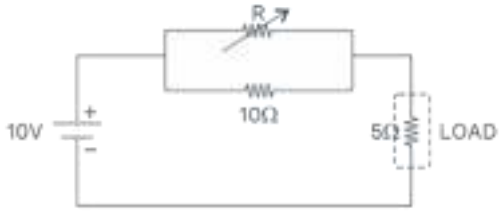
(A) Multi-source networks

(B) Single-source network

(C) Both single and multi-source network

(D) None of these

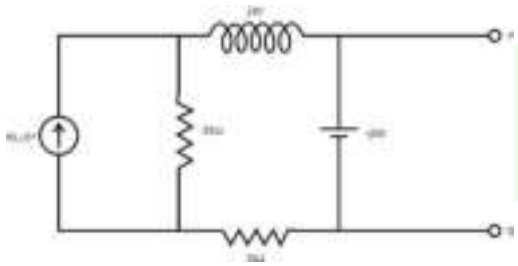
Q.36 For the circuit given below, the value of R required for the transfer maximum power to the load having a resistance of '5Ω' is:



(A) 5 Ω (B) 10 Ω

(C) Infinity (D) zero

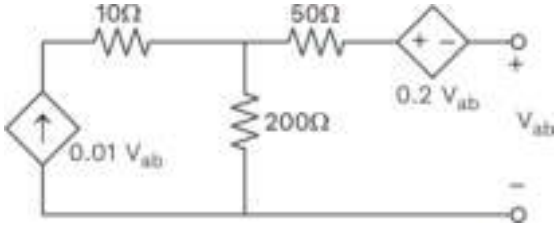
Q.37 Find the Norton equivalent current with respect to the terminal P and Q is ..... MSQ



(A) 4Ω (B) 6V

(C) 3Ω (D) 4V

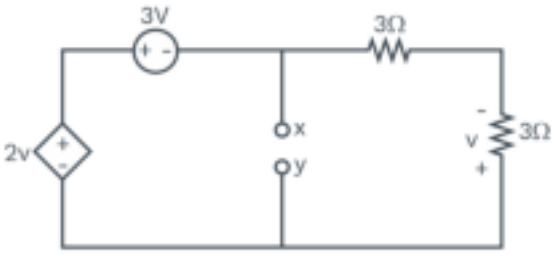
Q.38 Find the Thevenin's equivalent of the two terminal network shown in figure – MSQ



(A) 312.5 Ω (B) 2V

(C) 0V (D) 315 Ω

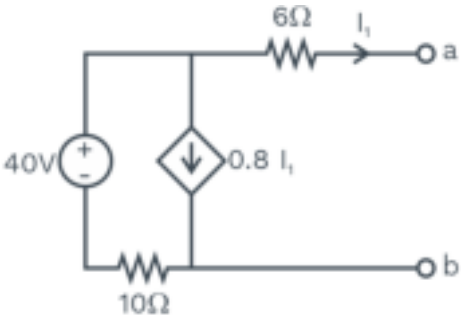
Q.39 For the given circuit find the R_{th} across the terminal xy.



(A) 2Ω (B) 0Ω

(C) 1Ω (D) 1.5Ω

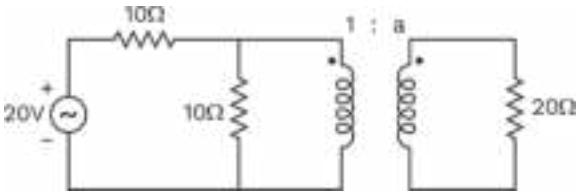
Q.40 Using Thevenin's theorem find the value of V_{th} & R_{th} across the terminal ab. MSQ



(A) 40V (B) 24Ω

(C) 18Ω (D) 32V

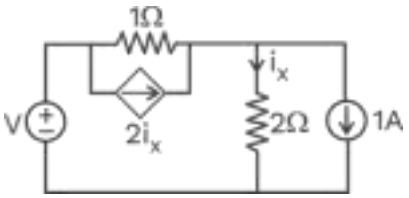
Q.41 For what value of 'a' maximum power transferred to 20 Ω and also determine maximum value of power in watts respectively.



(A) 5, 2 W (B) 10, 2 W

(C) 2, 5 W (D) 2, 10 W

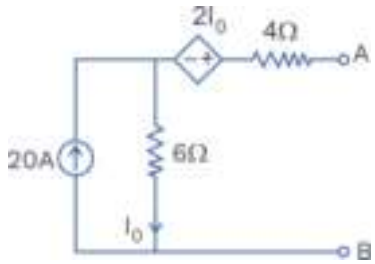
Q.42 The effective resistance seen by the voltage source is



(A) 1Ω (B) –1Ω

(C) 2Ω (D) –2Ω

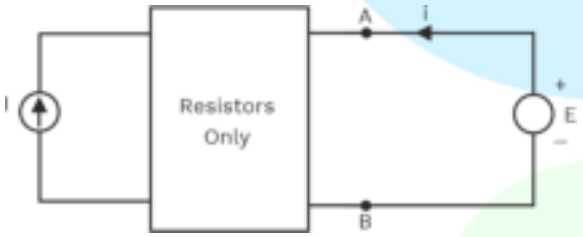
Q.43 The Thevenin's equivalent resistance seen across the terminal A and B is:



(A) 10 Ω (B) 14 Ω

(C) 12 Ω (D) 8 Ω

Q.44 Current i = 0 when E = 4V, I = 2A and i = 1A when E = 8V, I = 2A. The Thevenin voltage and the resistance looking into the terminals AB are



(A) 4V, 2Ω

(B) 4V, 4Ω

(C) 8V, 2Ω

(D) 8V, 4Ω

Q.45 For the circuit shown below. Find the Thevenin resistance.



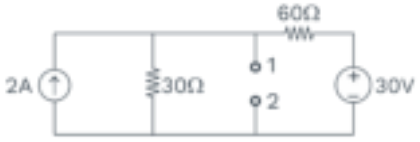
(A) 8 Ω

(B) 1.75 Ω

(C) 0.175 Ω

(D) 5.714 Ω

Q.46 Determine R_{Th} and V_{Th} at terminals 1-2 for the circuit of figure.



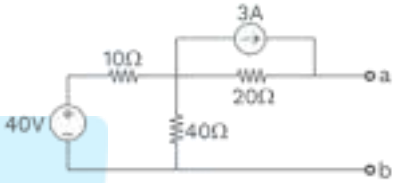
(A) V_{Th} = 5V, R_{Th} = 8Ω

(B) V_{Th} = 20V, R_{Th} = 50Ω

(C) V_{Th} = 50V, R_{Th} = 20Ω

(D) None of these

Q.47 Find the V_{Th} at terminals a – b of the circuit in figure.



(A) 100 V (B) 56 V

(C) 42 V (D) 92 V

Q.48 In any bilateral network, if a source of emf E in any branch produces a current I in any other branch, then same emf acting in the second branch would produce the same current I in the first branch.

The above statement is associated with-

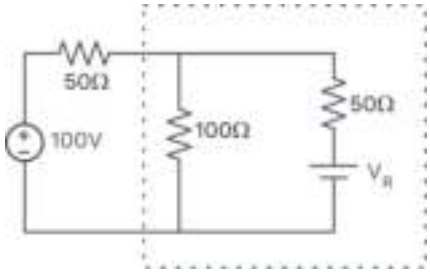
(A) Reciprocity theorem

(B) Superposition theorem

(C) Compensation theorem

(D) Thevenin theorem

Q.49 Calculate the value of V_R in case of maximum power transfer theorem.



Q.50 The current through a branch in a linear network is 2A, when the input source voltage is 10V. If the source voltage is reduced to 1V and the polarity is reversed then the current through the branch will be ..... Amp.

(A) 5 (B) –0.2

(C) –5 (D) 0.2

Q.51



When the network satisfies the reciprocity theorem, then find the value of I -

Q.52

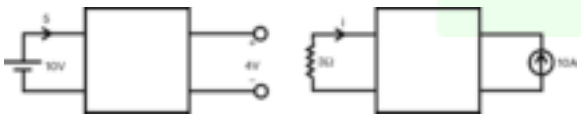


Calculate the value of I = ?

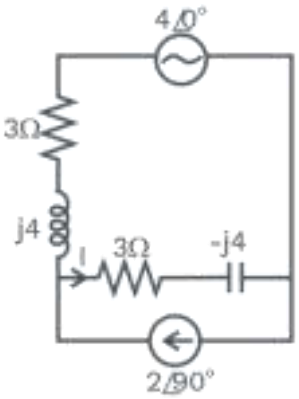
(A) 3 A (B) 6 A

(C) 2 A (D) 4 A

Q.53 When given network satisfies reciprocity theorem find the value of I?



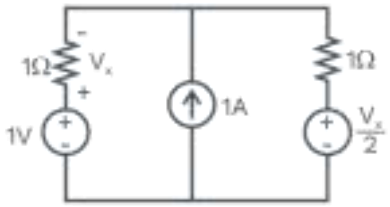
Q.54 For the given circuit find the value of I ?



(A) $j-\frac{4}{3}$ (B) $\frac{2}{3}$

(C) $j-2$ (D) $j-\frac{2}{3}$

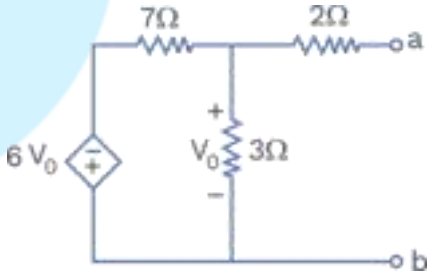
Q.55 Find the value of V_x using superposition theorem –



(A) $\frac{2}{5}$ (B) $\frac{-2}{5}$

(C) 0 (D) $\frac{4}{5}$

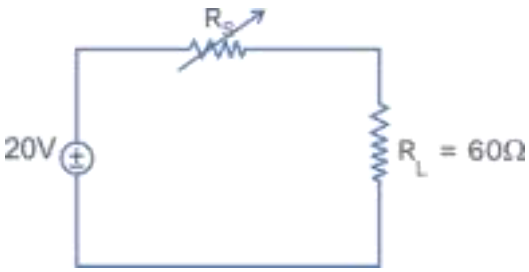
Q.56 The value of current source in Norton's equivalent circuit of below circuit is .....Amp.



(A) 4.5 A (B) 0 A

(C) 5 A (D) 5.5 A

Q.57 If R_S in the given figure is variable between 40Ω to 130Ω then the maximum Power transferred to load R_L will be:

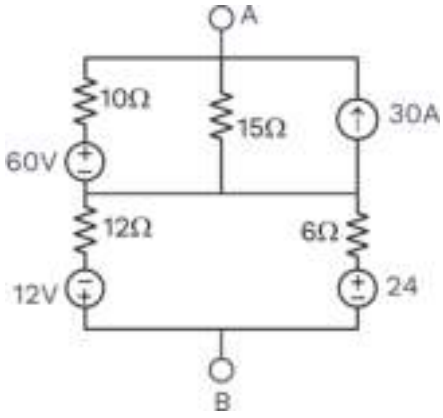


(A) 2.4 watt (B) 2 watt

(C) 5 watt (D) 1.6 watt

Q.58 Simplify the network into single voltage source in series with a

resistance between ab terminal of the circuit –



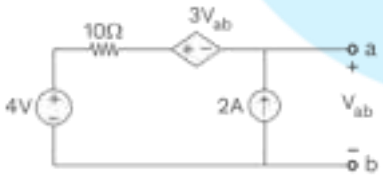
(A) 220 V in series with 5Ω

(B) 228 V in series with 10Ω

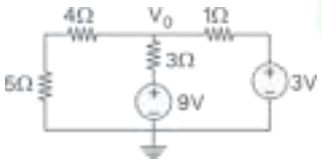
(C) 220 V in series with 10Ω

(D) 228 V in series with 5Ω

Q.59 For the circuit in figure, find the terminal voltage V_{ab} in volt using superposition.



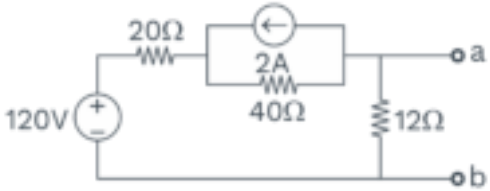
Q.60 Use superposition, find V_0 in the circuit of figure.



(A) 3 V (B) 4.14 V

(C) 11.3 V (D) 5 V

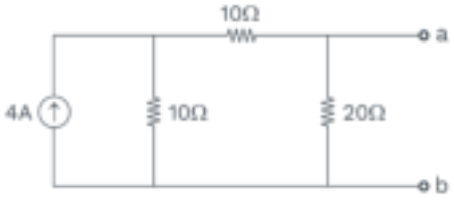
Q.61 Find the Norton current to terminals a – b in the circuit shown in figure



(A) 0.667 A (B) 6.67 A

(C) 1 A (D) 66.7 A

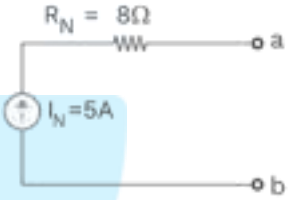
Q.62 Find the Norton equivalent at terminals a – b of the circuit in figure



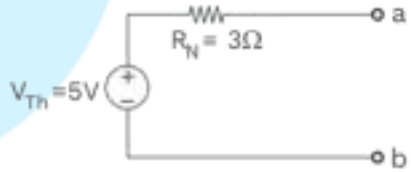
(A)



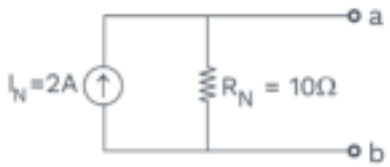
(B)



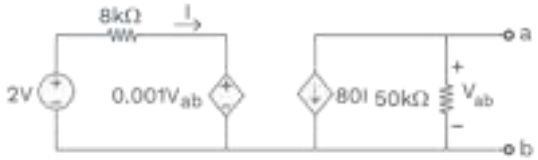
(C)



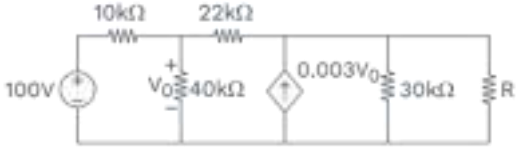
(D)



Q.63 For the circuit shown in figure. Norton equivalent resistance at terminal a – b is _____ kΩ.



Q.64 For the circuit shown in figure. Find the maximum power transferred to resistor R in the circuit of figure.



(A) 0 W (B) 5 W

(C) 3 W (D) ∞ W

Q.65 A 100 V source has a series internal resistance of 10Ω. The maximum power that can be delivered to load is:

(A) 3000 W (B) 2500 W

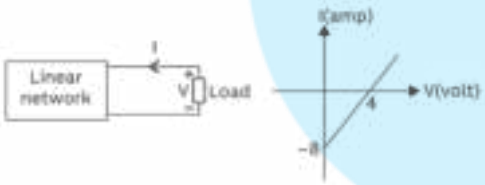
(C) 1500 W (D) 3500

Q.66 The concept on which superposition theorem is based-

(A) Non-linearity (B) Duality

(C) Linearity (D) Reciprocity

Q.67 For the linear network shown below, V-I characteristics is given in the figure. MSQ



Which of the following is/are correct:

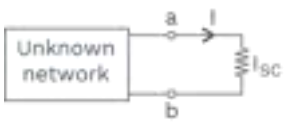
(A) The value of Norton equivalent resistance is 0.5Ω.

(B) The value of Norton equivalent current is –8A.

(C) The value of Norton equivalent resistance is 2Ω.

(D) The value of Norton equivalent current is 8A

Q.68 In the following, circuit some measurements were made at the terminals a, b and are noted in the table below:



R	I
11 Ω	1 A
3 Ω	2 A

Which of the following is/are correct: MSQ

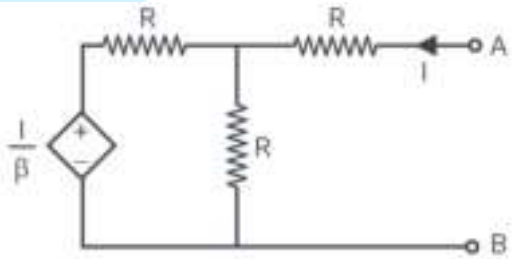
(A) The Thevenin equivalent voltage of the network across the terminal a–b will be 10V.

(B) The value of R that will cause the current I to be 2A is 3Ω.

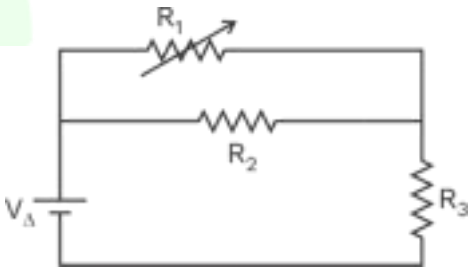
(C) The Thevenin resistance of the unknown network across the terminal a – b will be 5Ω.

(D) None of the above

Q.69 If R = 1 Ω and R_N (Norton equivalent resistance) seen form A-B terminal is 3 Ω. Then the value of β is _____.



Q.70 For what value of R_1 maximum power is dissipated in R_2.



(A) $R_1 = \infty$ (B) $R_1 = 0$

(C) $R_1 = \frac{R_2 \times R_3}{R_2 + R_3}$ (D) $R_1 = R_3$

Q.71 Consider the figure, the value of 'V' by using data provided in figure 1 is _____ Volts.

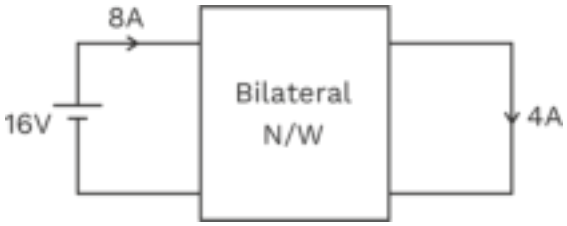


Figure-1

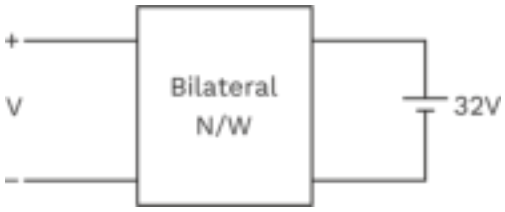
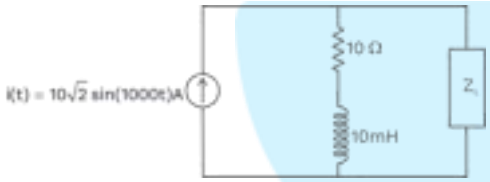


Figure-2

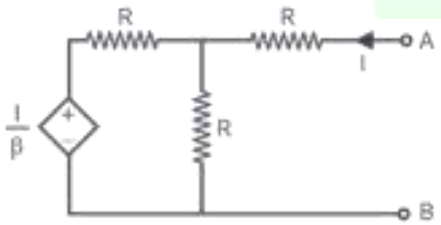
Q.72 In the circuit shown below, what will be the maximum power that can be transferred to the load Z_L.



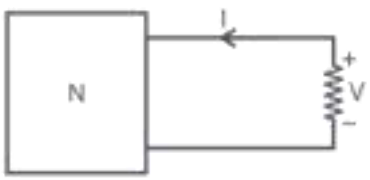
(A) 250 W (B) 500 W

(C) 200 W (D) 1 kW

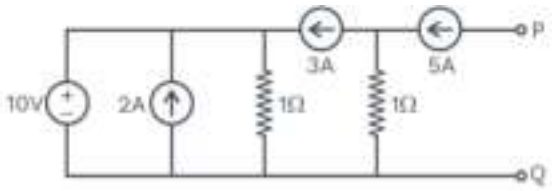
Q.73 If R = 1 Ω and R_N (Norton equivalent resistance) seen form A-B terminal is 3 Ω. Then the value of β is _____.



Q.74 In V-I relation for the network shown is, V = 4I – 20. What will be the value of load resistance in Ω so that maximum power transfer in it?



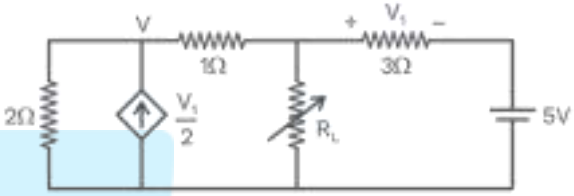
Q.75 Consider the given circuit and find the Thevenin's equivalent across P and Q.



(A) 2V, 0.5Ω (B) 5V, 1

(C) 2V, 1Ω (D) None

Q.76 Consider the circuit shown in figure.

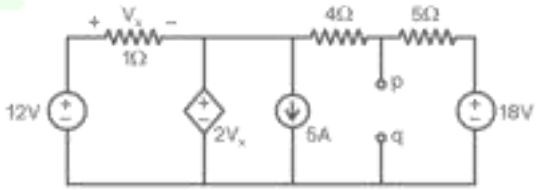


Find the maximum power dissipated in R_L resistor

(A) 0 W (B) 4 W

(C) 8 W (D) 16 W

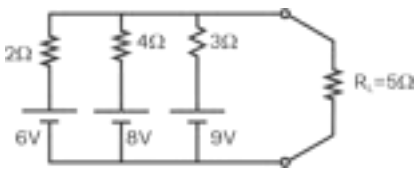
Q.77 The Norton's equivalent current and resistance between p and q terminal for the following circuit are:



(A) 5.6 A, 2.2 W (B) 6.5 A, 2.2 W

(C) 5.6 A, 3.2 W (D) 6.5 A, 3.2 W

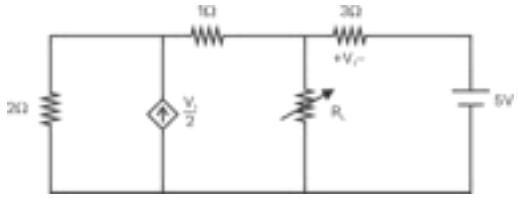
Q.78 Find the current through 'RL' in the circuit shown below:



(A) 2.25A (B) 1.75A

(C) 2.00A (D) 1.25A

Q.79 Consider the circuit shown in figure.



The maximum power dissipated in RL is

(A) 0W (B) 4W

(C) 8W (D) 16W

Q.80 In the given network the admittances of each branch is shown in figure, The Thevenin's Impedance across the terminal AB is.



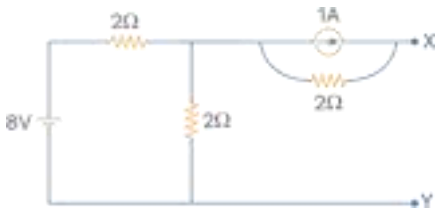
(A) $23.52\angle 10.86^0$

(B) $23.52\angle -10.86^0$

(C) $10.86\angle 23.52^0$

(D) $10.86\angle -23.52^0$

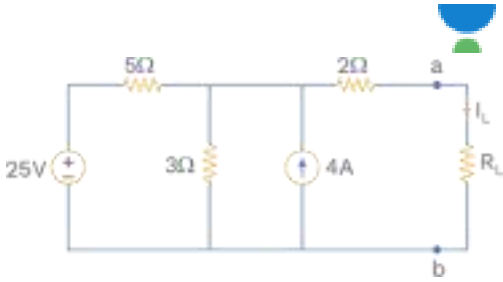
Q.81 For the circuit shown in the figure below, the value of the Norton equivalent source current and the resistance are respectively-



(A) 2A, 3Ω (B) 2A, 4Ω

(C) 1A, 3Ω (D) 0A, 1.5Ω

Q.82 In the circuit shown in figure, find the current I_L. MSQ



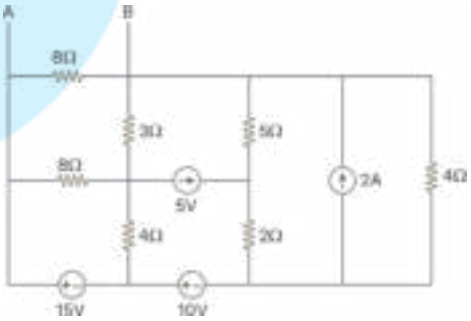
(A) For R_L = 5Ω, current I_L = 1.90 Amp

(B) For R_L = 8Ω, current I_L = 1.36 Amp

(C) For R_L = 3Ω, current I_L = 2.45 Amp

(D) For R_L = 13Ω, current I_L = 1 Amp

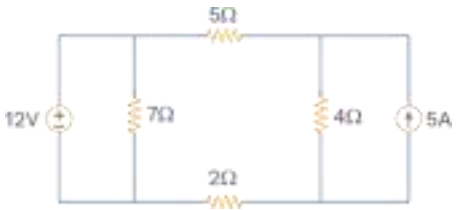
Q.83 What will be the Thevenin's resistance across the terminals AB of the circuit shown in the figure below?



(A) R_{Th} = 5.32Ω (B) R_{Th} = 33.24Ω

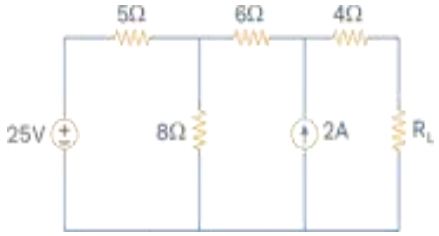
(C) R_{Th} = 1.44Ω (D) R_{Th} = 8Ω

Q.84 For the circuit shown below, the current through the 4Ω resistor (in Amp) is_____ (up to two decimal places).

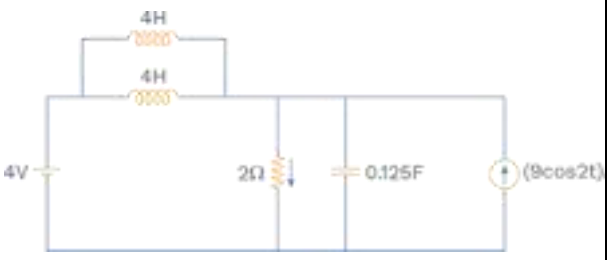


Q.85 The maximum power transferred through the load resistance R_L (in Watts) in the given circuit

is_____. (Up to two decimal places)



Q.86 In the circuit shown below, the current through the 2Ω resistor is-



(A) 2Amp

(B) 9 cos 2t – 6 Amp

(C) 4 – 9 cos 2tAmp

(D) 2 + 9 cos 2tAmp

Q.87 For the circuit given below, it is given that-

When $R_L = \infty$, V = 12V

When $R_L = 0$, I = 3A

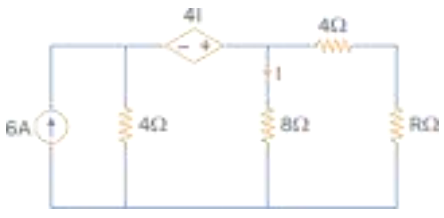
If $R_L = 6\Omega$, then V and I are respectively-



(A) 7.2V, 1.2A (B) 8.3V, 1.38A

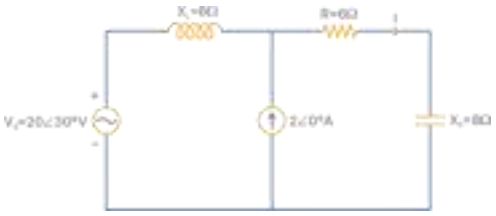
(C) 6.27V, 1.05A (D) 3.12V, 0.52A

Q.88 A circuit is shown in the figure below.



The maximum power dissipated in the external resistance 'R' is _____ W. (Rounded off to the nearest integer)

Q.89 In the circuit given below, the current through the 6Ω resistor is-



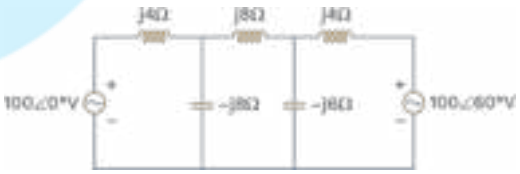
(A) I = 7.89 ∠82.3° Amp

(B) I = 4.42 ∠70.2° Amp

(C) I = 12.63 ∠36.8° Amp

(D) I = 2.56 ∠33.3° Amp

Q.90 In the given circuit, the steady-state current through the j8Ω inductor is-



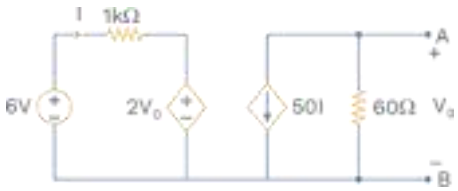
(A) I = 9.45∠–169.1° A

(B) I = 13.63∠–123.8° A

(C) I = 32.62∠–63.2°A

(D) I = 26.81∠–85.9°A

Q.91 For the circuit shown in the figure, if Thevenin's theorem is applied across the terminals A-B, then



MSQ

(A) $R_{Th} = 15\Omega$

(B) $V_{Th} = 3.6$ Volts

(C) $R_{Th} = 12\Omega$

(D) $I_{AB} = -0.5$ Amp

Q.92 In the given circuit, the approximate value of Z_L (in ohms) for which the maximum power will be delivered, is- MSQ



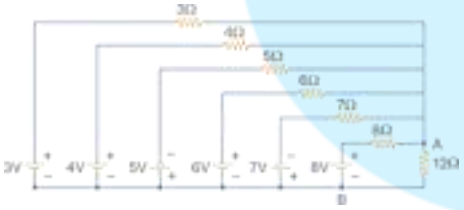
(A) $Z_L = 5.1176 + j0.9705$

(B) $Z_L = 9.63\angle 16.52°$

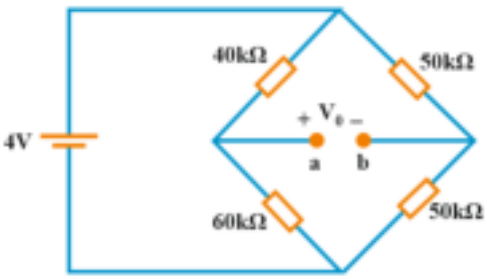
(C) $Z_L = 9.2315+j2.7392$

(D) $Z_L = 5.21\angle 10.74°$

Q.93 For the circuit shown below, the voltage across the 12Ω resistance is ___ Volts. (Up to two decimal places).

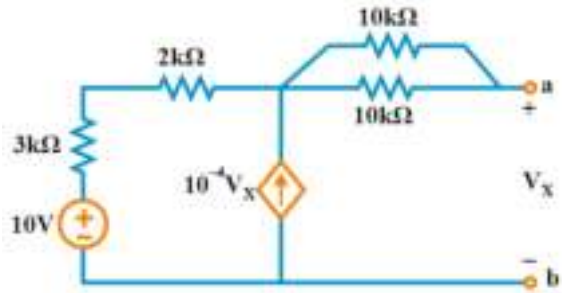


Q.94 A circuit is shown in the figure below.

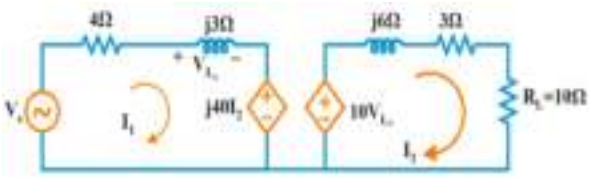


The output resistance between terminal 'a' and 'b' is: _____ kΩ. (Round off to the nearest integer)

Q.95 For the circuit shown in the figure below, the Thevenin equivalent resistance between terminals a and b is _____kΩ. (Rounded off to the nearest integer)



Q.96 A circuit is shown in the figure below. The source voltage, $V_s = 100\angle 36.86^\circ$ Volts.



The Thevenin equivalent voltage seen by the load resistance R_L is –

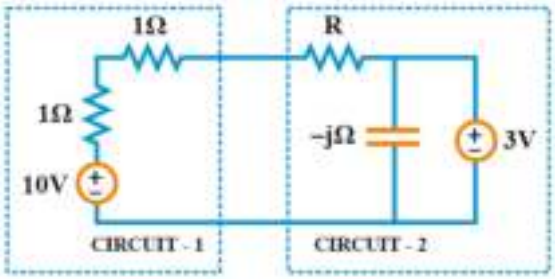
(A) $400\angle 16.86^\circ$

(B) $600\angle 90^\circ$

(C) $60\angle 36.86^\circ$

(D) $20\angle 106.28^\circ$

Q.97 A circuit is shown below.



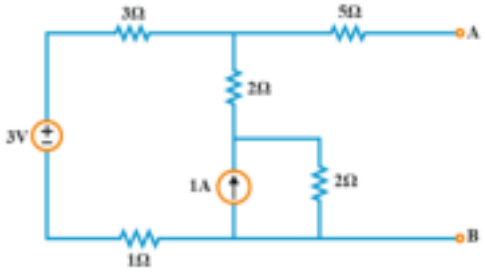
The voltage sources are in phase, the value of 'R' when maximum power is transferred from circuit-1 to circuit-2 is-

(A) 0.8Ω (B) 1.32Ω

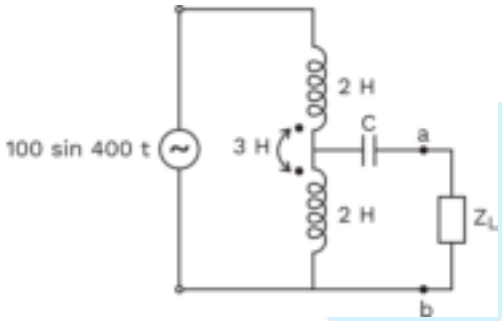
(C) 0.54Ω (D) 0.27Ω

Q.98 For the circuit shown below, the Norton equivalent resistance seen between terminals A and B is _____

Ω. (Rounded off to the nearest integer)



Q.99 If the voltage across Z_L is independent of the value of Z_L, then the value of C required in the circuit is –



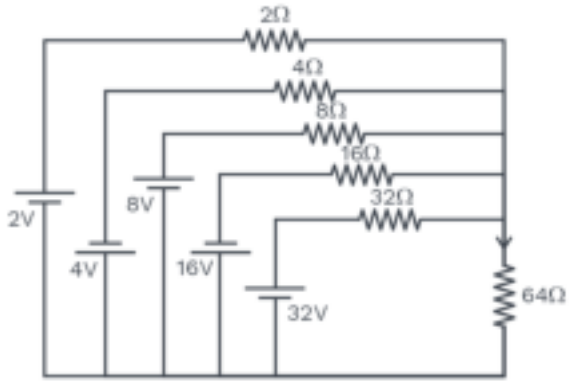
(a) 2.5 μF

(b) 1.25 μF

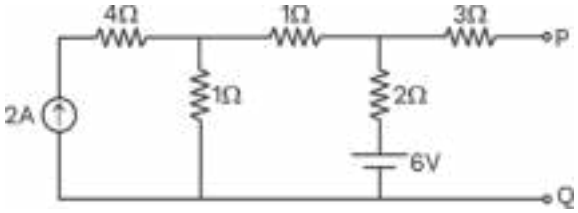
(c) 1.98 μF

(d) 2 μF

Q.100 Obtain the value of current I (in mA) in the circuit shown below:



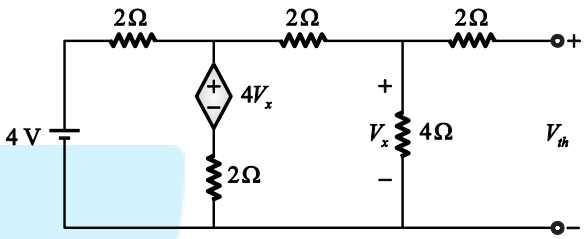
Q.101 Calculate the value of Thevenin's equivalent resistance across the terminal P and Q in the given circuit below ?



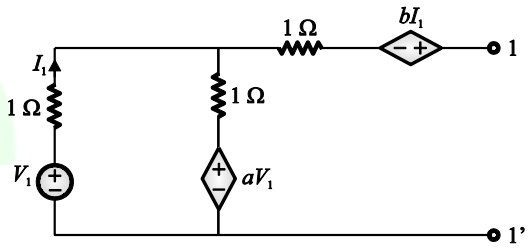
(A) 1Ω (B) 2Ω

(C) 3Ω (D) 4Ω

Q.102 Consider the circuit shown in the figure below. The value of Thevenin's equivalent voltage V_{th} is _____ V. (in integer)



Q.103 Consider the circuit shown in figure,



The short circuit current (I_{sc}) between the terminal 1-1' is

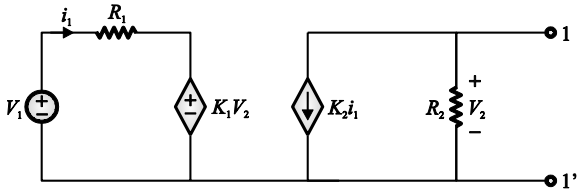
(A) $I_{sc} = \frac{V_1(a+b-ab)}{3+b}$

(B) $I_{sc} = \frac{V_1(1+a+b-ab)}{3-b}$

(C) $I_{sc} = \frac{V_1(1+a+b+ab)}{3-b}$

(D) $I_{sc} = \frac{V_1(1+a+b)}{3-b}$

Q.104 The Thevenin's impedance (Z_{th}) between the terminal 1-1' for the circuit shown below is



(A) $Z_{th} = \dfrac{R_1}{R_1 - K_1K_2R_1}$

(B) $Z_{th} = \dfrac{R_1R_2}{R_1 + K_1K_2R_2}$

(C) $Z_{th} = \dfrac{R_1R_2}{1 + K_1K_2R_2}$

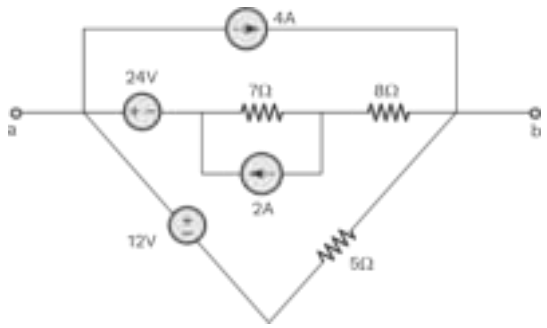
(D) $Z_{th} = \dfrac{R_1R_2}{R_1 - K_1K_2R_2}$

Q.105 The current through a branch in a linear network is 4A when the input voltage is 16V. If the source voltage is reduced to 2V, and polarity is reversed, then the current through the branch will be-

(A) 0.25A (B) 0.5A

(C) -0.25 (D) -0.5A

Q.106 A circuit is shown in the figure below.



Using Thevenin theorem, which of the following statements is/are true? MSQ

(A) Thevenin equivalent voltage is 3.5 V

(B) Thevenin equivalent resistance is 20Ω.

(C) Thevenin equivalent voltage is 0.285V.

(D) Thevenin equivalent resistance is 3.75 Ω.

Q.107 Superposition theorem is not applicable to network MSQ

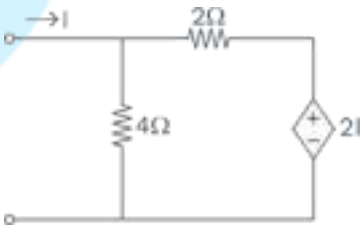
(A) Non-linear elements

(B) Linear elements

(C) Unilateral elements

(D) Bilateral elements

Q.108 The circuit shown in figure is equivalent to a load of-



(A) $\frac{4}{3}\Omega$ (B) 4 Ω

(C) 2 Ω (D) $\frac{8}{3}\Omega$

Q.109 The maximum power is delivered from a source to a load when the source resistance is-

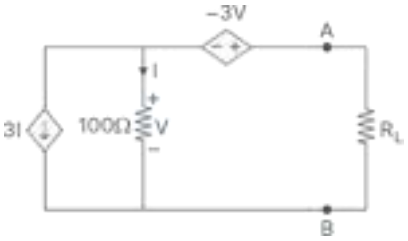
(A) Equal to load resistance

(B) Greater than the load resistance

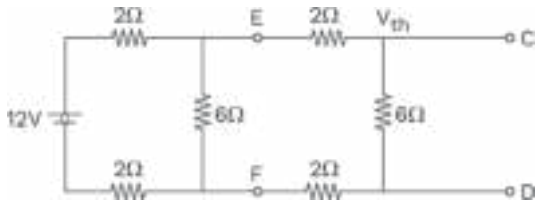
(C) Less than the load resistance

(D) Equal to zero

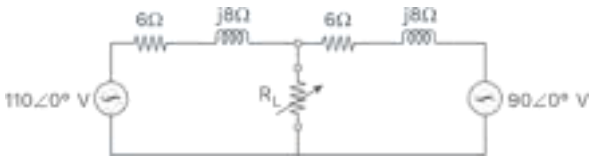
Q.110 The maximum power delivered to resistance R_L is _____ W.



Q.111 For the circuit shown below, determine the Thevenin's voltage is _____ V.



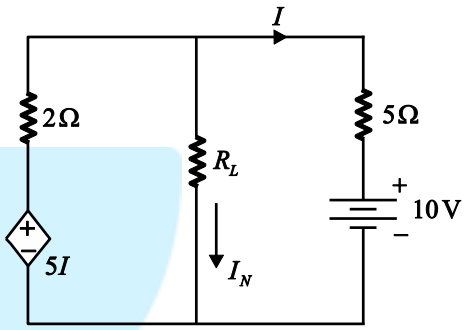
Q.112 Two ac sources feed a common variable resistive load as shown in figure. Under maximum power transfer condition, the power absorbed by the load resistance R_L is _____ W.



(A) 625 W (B) 1250W

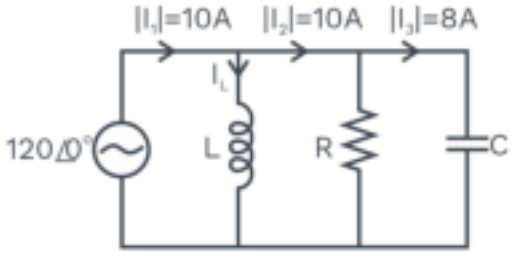
(C) 1000 W (D) 250 W

Q.113 Consider the circuit shown in figure. The Norton's equivalent current (in Amperes) through the load resistor R_L is _____. (in integer)



4 Phasor Analysis & 3 Phase Circuits

Q.1 For the given circuit –

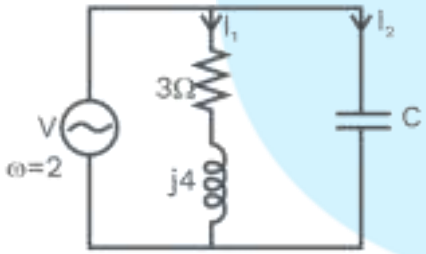


Find all possible value of I_L ?

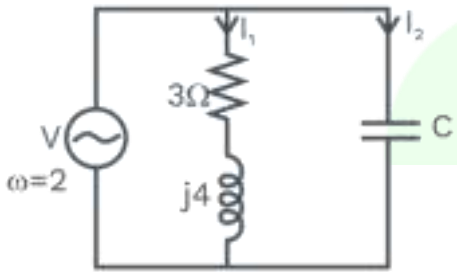
(A) 1 A (B) 0 A

(C) 16 A (D) 15 A

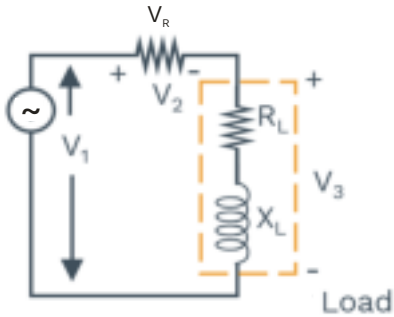
Q.2 I_1 leads I_2 by which angle?



Q.3 I_1 leads I_2 by which angle?



Q.4 Find the power factor of the load?

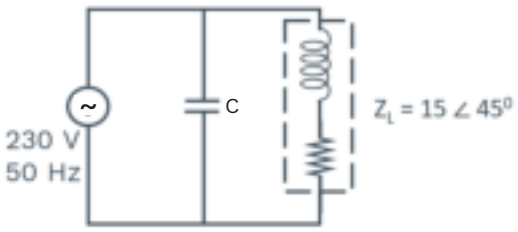


If $V_1 = 220\,V$, $V_R = 120V$, $V_3 = 136\,V$

(A) .5 (B) .475

(C) 1 (D) .8

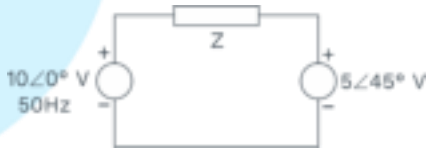
Q.5 The circuit shown in figure, what value of C will cause a unity power factor at the ac source-



(A) $100^{\mu f}$ (B) $150^{\mu f}$

(C) $145^{\mu f}$ (D) $149^{\mu f}$

Q.6 Find the magnitude of current through Z if $Z = (2 + j0.5)\,\Omega$.



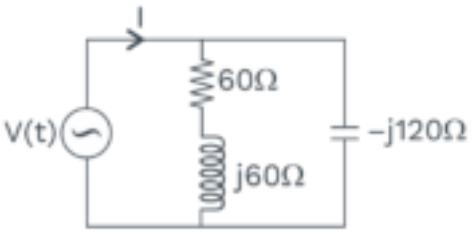
(A) $3.58\angle -42.7°A$

(B) $5.29\angle -37.9°A$

(C) $10.2\angle -29.5°A$

(D) $112\angle -57°A$

Q.7 For the AC circuit given below, the calculate the value of current if $V(t) = 120\sqrt{2}\sin\omega t$.

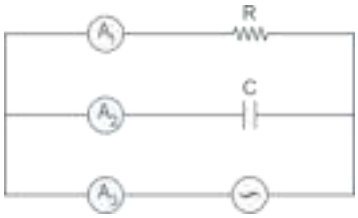


(A) 1 + j1 (B) 0 – j1

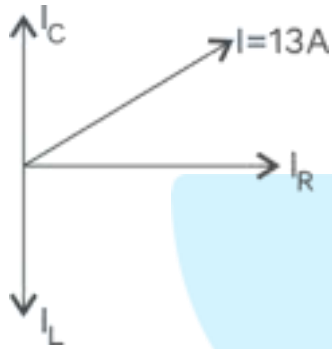
(C) 1 + j0 (D) 0

Q.8 In the figure below, A_1, A_2, A_3 are ideal ammeters. If A_1 and A_3 reads

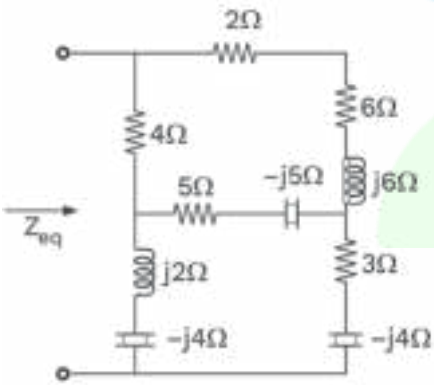
5A and 13A respectively, reading of A_2 will be _____ A.



Q.9 The phasor diagram of parallel RLC circuit is given below, If I_R and I_L are 5 A and 15 A respectively, the value of I_C is _____ A.



Q.10 The equivalent impedance Z_{eq} of the circuit given below is _____ .



(A) $\frac{14}{15} - j\frac{16}{5}\Omega$ (B) $\frac{16}{5} - j\frac{14}{15}\Omega$

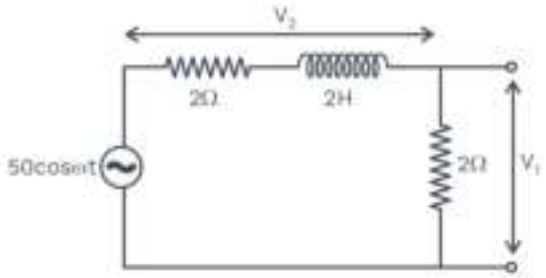
(C) $\frac{14}{15} + j\frac{16}{5}\Omega$ (D) $\frac{16}{5} + j\frac{14}{15}\Omega$

Q.11 A water boiler at home switched ON to AC mains supplying power at 230V/60Hz. The frequency of instantaneous power consumed by water boiler is-

(A) 0 Hz (B) 60 Hz

(C) 30 Hz (D) 120 Hz

Q.12 Determine the positive angular frequency ω (in rad/sec) for which the magnitude of the phase difference of the voltage V_1 and V_2 is equal to $\frac{\pi}{3}$ radian in the circuit given below.



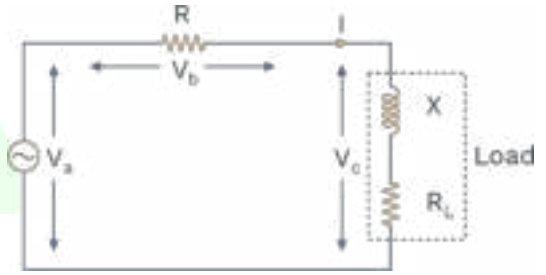
(a) 1.43 rad/sec

(b) 2.94 rad/sec

(c) 4.78 rad/sec

(d) 1.732 rad/sec

Q.13 A circuit is shown in the figure below. MSQ



The voltmeter reads V_a = 230V, V_b = 107V and V_C = 151V. Which of the following statement(s) is/are TRUE?

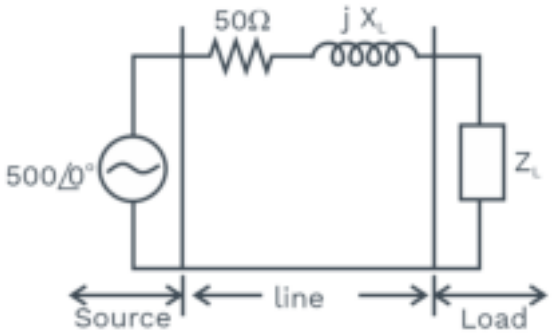
(A) The power factor of the load is 0.577

(B) The power factor of the load cannot be determined.

(C) If the load resistance is 7Ω, then power consumption in load is 1085W.

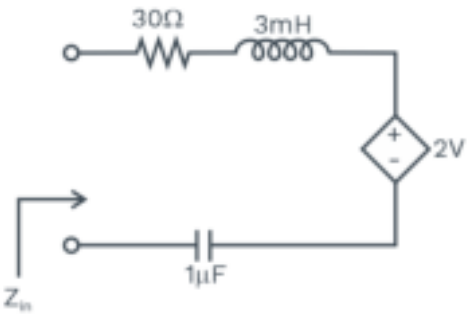
(D) If the load resistance is 7Ω, then power consumption in load is 650W.

Q.14 The load impedance in figure absorbs

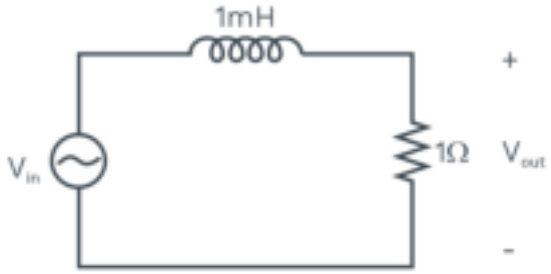


2500W & generates 5000 magnetising VAR. The sinusoidal voltage source develops 7500W find the value of inductive line reactance?

Q.15 For the circuit shown in figure the magnitude of wavefront input impedance Z_{in} at 10K rad/sec _ ___ Ω.



Q.16 Determine the value of ω for which the magnitude of the output voltage is 10% of its value at dc.



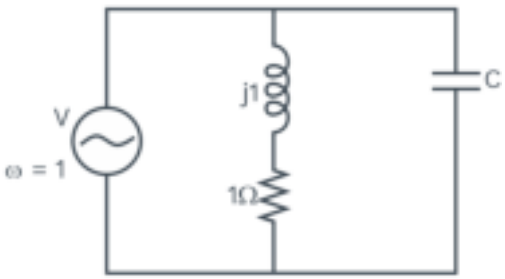
(A) 10 k rad/sec.

(B) 9 k rad/sec.

(C) 9.94 k rad/sec.

(D) 10.5 k rad/sec.

Q.17 Determine the value of C such that the power factor of the circuit is 0.8 lagging?

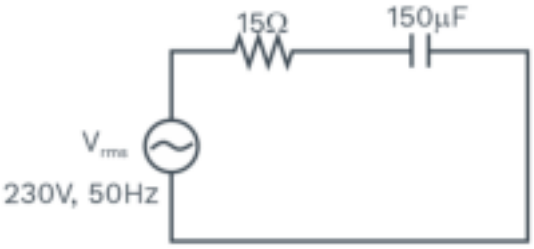


Q.18 230V supply is connected to two parallel connected loads. One load draws 10kW, 0.8pf leading & another load drawn 41KVA, 0.8pf lag. Find complex power determined by source.... in KVA

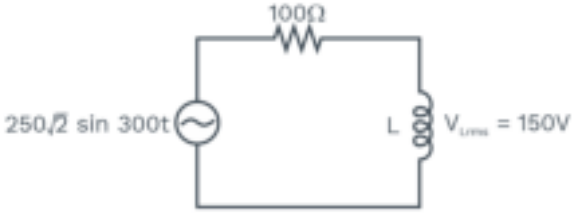
(A) 15-j1.5 (B) 15+j1.5

(C) 18+j1.5 (D) 18-j1.5

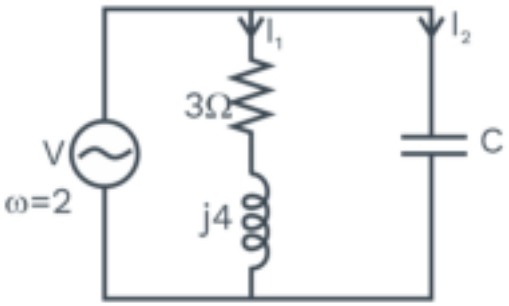
Q.19 If a resistor of 15Ω & a capacitor of 150μF are connected in series across 230V, 50Hz supply then calculate power factor –



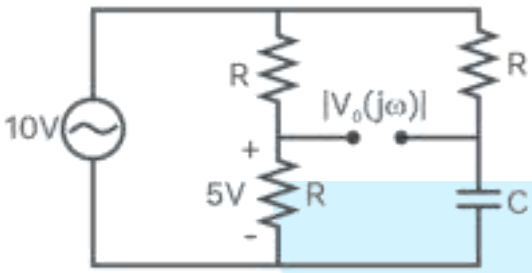
Q.20 Find the value of L ?



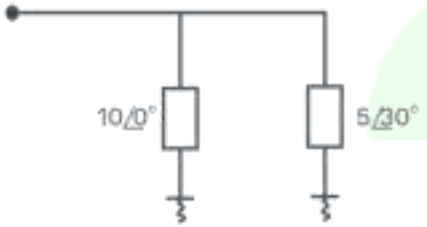
Q.21 I_1 leads I_2 by which angle?



Q.22 Find the value of $|V_0(j\omega)|$



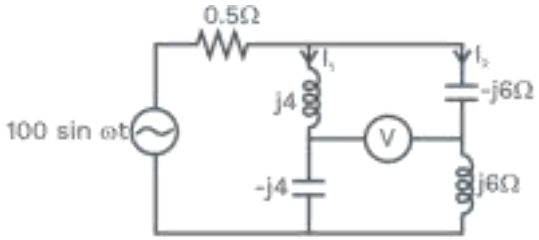
Q.23 A three phase 2 wire system with an voltage of 189V supplies two balanced load one in delta with $30\angle 0$ and the other in Y with Z_Y = $5\angle 30°$ total power is ___.



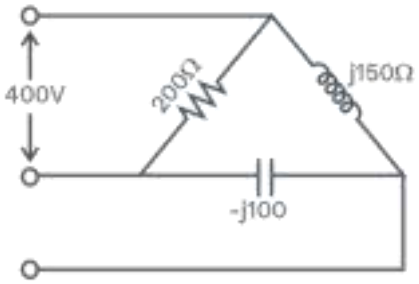
(A) 10W (B) 9.75W

(C) 9.23W (D) 10.5W

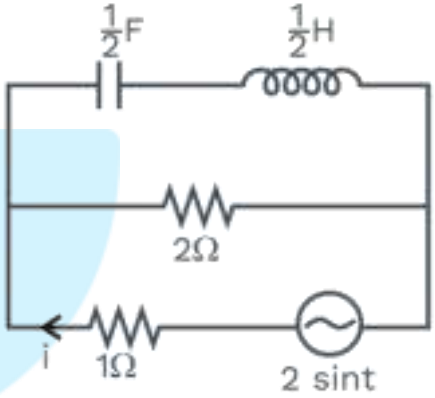
Q.24. Find the reading of ideal voltmeter?



Q.25 The set of 3 equal resistor, each at value RΩ connected in Y across R-Y-B of figure consume the same power as the unbalanced delta connected load shown. Find the value of R in ohm.



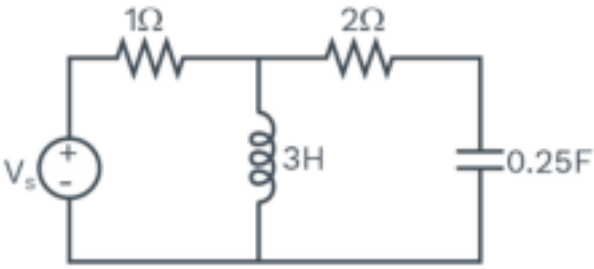
Q.26 The rms value of the current in the circuit shown below is –



(A) ½ (B) 1

(C) $1/\sqrt{2}$ (D) $\sqrt{2}$

Q.27 If V_s = 8 cos (2t – 40°)V in the given circuit, find the average power delivered to each of the passive elements. MSQ

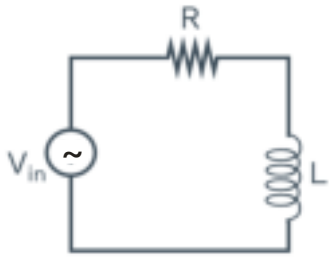


(A) 0 W (B) 1.82 W

(C) 1.41 W (D) 5 W

Q.28 The RL CKt of the figure is fed from a constant magnitude, variable frequency sinusoidal voltage source V_{in} at 100 Hz, the R & L

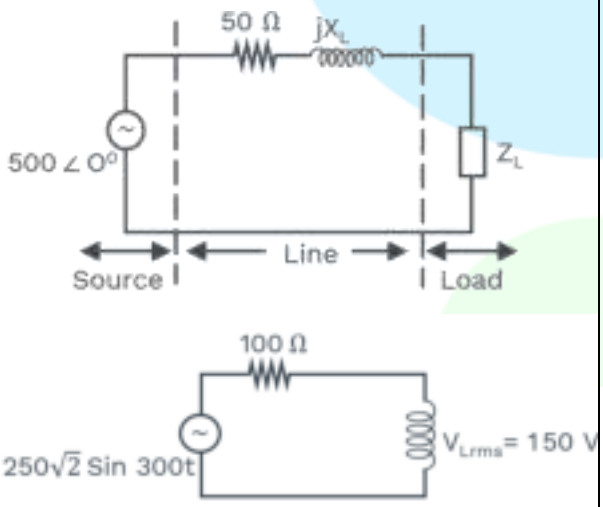
elements each have a voltage drop V_{rms}. If the frequency of the source is changed to 50 Hz. Then new voltage drop across resistance R is_____



(A) $\sqrt{\frac{5}{8}}U_{rms}$ (B) $\sqrt{\frac{4}{3}}U_{rms}$

(C) $\sqrt{\frac{8}{5}}U_{rms}$ (D) $\sqrt{\frac{3}{2}}U_{rms}$

Q.29 The load impedance in figure absorbs



2500 W and generates 5000 magnetizing VAR. The sinusoidal voltage source develops 7500 W find the value of inductive line reactance?

(A) 30 Ω (B) 35 Ω

(C) 45 Ω (D) 50 Ω

Q.30 The energy (in joules) consumed by the impedance (6+j8)Ω in 5 cycles for the circuit shown below is___

(A) 2.714 (B) 3.542

(C) 4.32 (D) 2.5

Q.31 Determine the value of ω for which the magnitude of the output voltage is 10% of its value at dc.

(A) 9.5 k rad /sec.

(B) 9.94 k rad /sec.

(C) 9 k rad /sec.

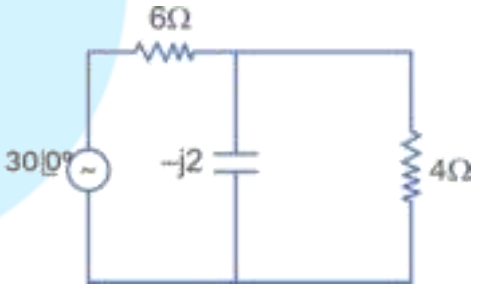
(D) 9.23 k rad /sec

Q.32 Find the value of L?

(A) .25 H (B) .5 H

(C) 1.5 H (D) 1 H

Q.33 Find the power factor of the circuit as seen by the source



Q.34 The applied voltage in a parallel RLC circuit is given by V = 50 sin $\left(5000t + \frac{\pi}{4}\right)$. If the value of R, L and C be given as 20Ω, 1.6 × 10^{-3} H and 20 μF, total current supplied by the source is-

(A) 5.12∠18.5°A

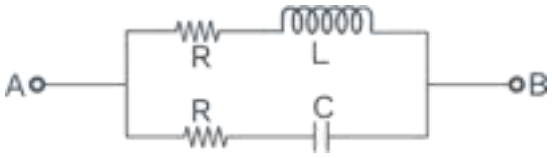
(B) 1.97∠18.5°A

(C) 5.12∠15.2°A

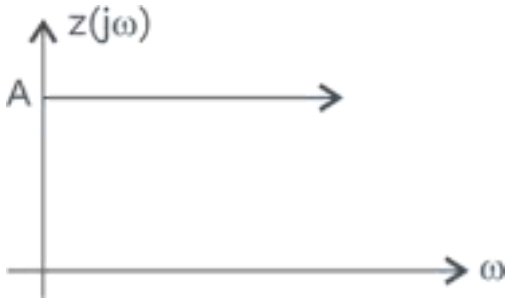
(D) 1.97∠15.2°A

Q.35 For the circuit shown below.

Assume $\frac{L}{C} = R^2$



If we plot impedance function V_s ω its look like.



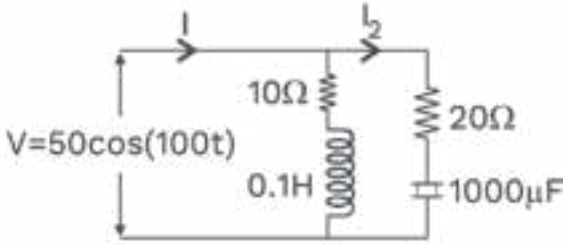
The value of A is-

(A) R (B) $\frac{RL}{C}$

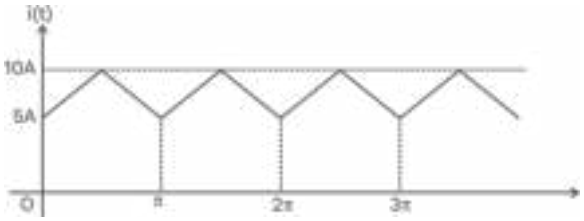
(C) $\frac{RC}{L}$ (D) $\frac{R}{2}$

Q.36 An RLC series circuit has a resistance R of 20 Ω. The current I through the circuit leads the applied voltage of 45°. If the voltage across the capacitor is thrice the voltage across the inductor. The value of capacitive reactance _____ Ω.

Q.37 In the circuit shown the phase angle difference between I, I_2 is _____ degrees.



Q.38 Find the average value of the given current waveform



(A) 10A (B) 2.5A

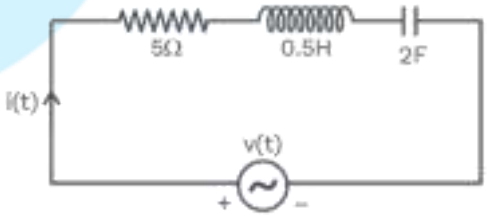
(C) 5A (D) 7.5A

Q.39 A series RC circuit excited by a dc voltage source of 20 V and with time constant τ. Find the expression for voltage across 'C' at time t = τ.

(A) $20(1-e^{-1})V$ (B) 20 (1–e) V

(C) $20 - e^{-1}$ V (D) (20 – e)V

Q.40 Determine the steady state response of the current in the circuit of figure shown below.



Here, v(t) = sin(t + 60°)

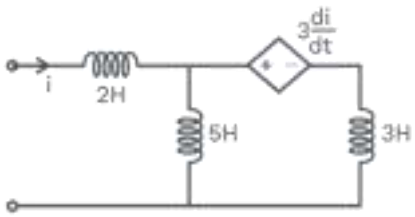
(A) 0.5 sin (t + 60°)A

(B) 0.2 sin (t + 60°)A

(C) 0.5 cos (t + 60°)A

(D) 0.5 sin (t + 30°)A

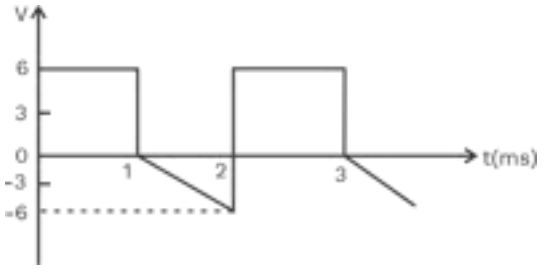
Q.41 Find the value of equivalent inductor as seen from open terminal for below diagram.



(A) 7.75H (B) 5.25H

(C) 5.75H (D) 6.25H

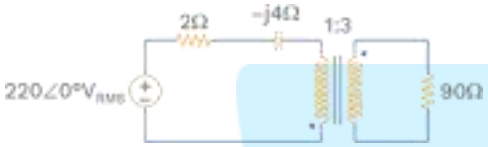
Q.42 Calculate the rms voltage for the waveform shown in figure:



(A) 2.74 (B) 3.74

(C) 1.74 (D) 4.74

Q.43 A circuit is shown in the figure below.



The complex power(VA) supplied by the source is-

(A) $17.39\angle -18.34°$

(B) $1500.82\angle 7.125°$

(C) $3630\angle 18.43°$

(D) $3826.24\angle -18.434°$

Q.44 The active power and reactive power of a circuit are 1000 W and 1500 VAR respectively, drawn from a 220V, 50 Hz AC supply. The current drawn from the source is _____ Amp. (Up to three decimal places)

Q.45 A three phase motor can be regarded as a balanced Y-load. A three phase motor draws 5.6kW when the line voltage is 220V and the line current is 18.2A. Determine the power factor of the motor.

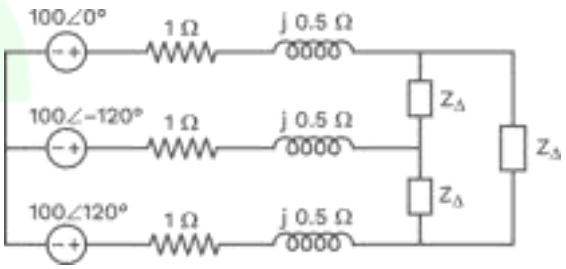
(A) 0.5 (B) 0.67

(C) 0.80 (D) 0.98

Q.46 A three phase balanced load star connected has per phase impedance of $Z_Y = 8 + j\ 6\Omega$. If the load is connected to 208V lines, determine the total average power absorbed by load ___ inkW. Power is measured by two wattmeter method.

Q.47 A three phase line has on impedance of $1 + j\ 3\ \Omega$ per phase. The line feeds a balanced delta-connected load, which absorbs a total complex power 12 + j 5 KVA. If the line voltage of the load ends has a magnitude of 240 V, calculate the magnitude of the line voltage at the source end _____ in volt.

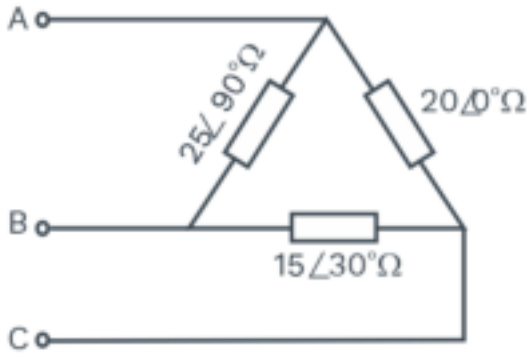
Q.48 For the three phase circuit shown below, find the average power absorbed by the delta-connected load with $Z_\Delta = 21 + j\ 24\ \Omega$.



(A) 2.34 kW (B) 1.98 kW

(C) 2 kW (D) 1.5 kW

Q.49 A three phase supply, with an effective line voltage 240V, has an unbalanced Δ-connected load as shown in figure below, then the total reactive power in VAR drawn from the supply is –



(A) 2304VAR (B) 4224VAR

(C) 6205VAR (D) 1920VAR

Q.50 Three identical resistance are connected in a star against a balanced three phase voltage supply. If one of the resistance be removed then the percentage reduction in power is _____.

(A) 33.33% (B) 50%

(C) 67.37% 17.32%

Q.51 A three phase system with line voltage of 300 V that supplies 1200 W to A delta connected loop of lagging p.f of 0.8 then find the phase impedance

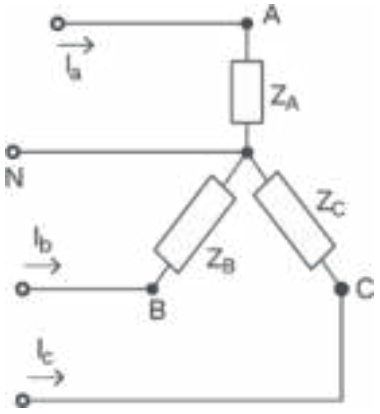
(A) 90 ∠53.14° (B) 150 ∠36.86°

(C) 180 ∠36.86° (D) 180 ∠53.14°

Q.52 An unbalanced star connected load with impedance of 100∠90°, 100∠–30° and 20∠–30°Ω is supplied from a 3-phase, 4-wire, 173 volts balanced supply. The voltage of 'R' phase to neutral is 100∠–150° V and having negative phase sequence.

The current in the neutral wire is _____ A.

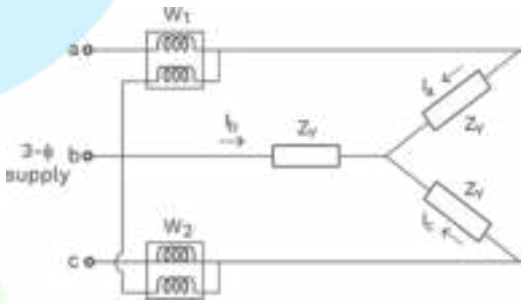
Q.53 The unbalanced star connected load has balanced voltages of 100V. If Z_A = 15Ω, Z_B = 10 + j5Ω and Z_C = 6 – j8Ω then the current in the neutral line is-



(A) 5 A (B) 7 A

(C) 10 A (D) 12 A

Q.54 The three-phase balanced load has impedance per phase of Z_Y = 8 + j6Ω. If the load is connected to 208 V lines, then the reactive power is _____ kVAR.



Q.55 The following loads are supplied by a 3-ϕ AC generator.

1. 20 kW at unity power factor
2. 30 kVA at 0.8 p.f (lagging)
3. 35 kVA at 0.9 p.f. (leading)

The total KW supplied is

(A) 75.5 (B) 27.5

(C) 96.4 (D) 21.4

Q.56 A balanced delta-connected load (8 + j12)Ω/phase is connected to a 3-phase 230V balanced supply. Determine the real power drawn in _____ KW.

Q.57 A 3-phase, 4-wire system supplies power to a balance star-connected load, the current in each phase is 10A. Determine the current in neutral wire.

(A) 81.35 A (B) 65.64 A

(C) 0 A (D) 45.94 A

Q.58 A Δ/Y transformer has phase to phase voltage transformation ratio of $\frac{b(\text{Delta phase})}{1(\text{Y-phase})}$.

The line-to-line voltage ratio of $\frac{Y}{\Delta}$ is given by

(A) $\frac{b}{\sqrt{3}}$ (B) $\frac{\sqrt{3}}{b}$

(C) $b\sqrt{3}$ (D) b

Q.59 The power input to a three phase Induction motor is read by two wattmeters. The readings are 1000W and 500W. Find the power factor of the motor.

(A) 0.7680 (B) 0.866

(C) 0.577 (D) 0.5

Q.60 A 400V, $3-\phi$ balanced supply is fed to an $3-\phi$ unbalanced Δ connected load having

$Z_{RY} = 20\angle -50^0, Z_{YB} = 10\angle 120^0$ &

$Z_{BR} = 25\angle -30^0$ the Line current I_Y will be_____.

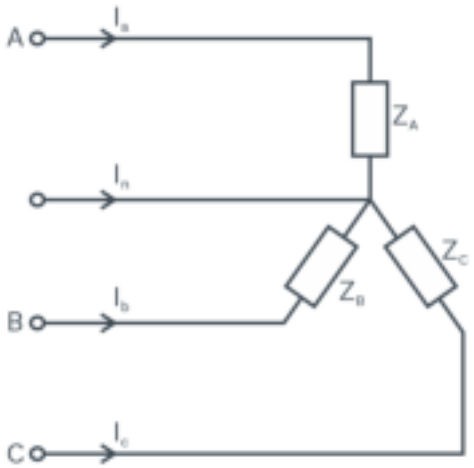
(A) $59.79\angle -136.67^0$

(B) $53.2\angle 130.21^0$

(C) $59.79\angle 136.67^0$

(D) $53.2\angle -130.21^0$

Q.61 The unbalanced Y-load as shown below has balanced voltage of 100V and acb sequence, calculate the neutral current if $Z_A = 15\Omega$, $Z_B = 10 + j5$, $Z_C = 6 - j8\Omega$.



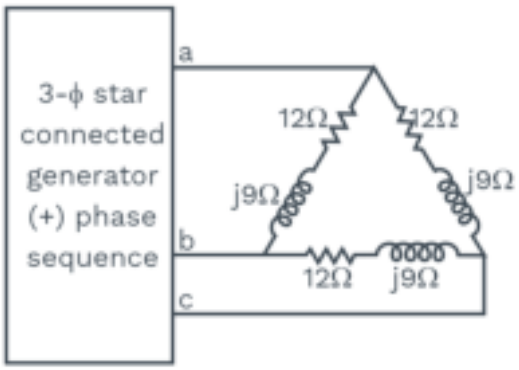
(A) $8.94\angle 93.44^\circ$

(B) $10.06\angle 178.4^\circ$

(C) $10\angle -66.87^\circ$

(D) $9.47\angle 0^\circ$

Q.62 If $V_{an} = 440\angle 60^\circ$ in the given network, find the phase current I_{CA}.



(A) $50.81\angle -66.87^\circ$ A

(B) $50.81\angle 0^\circ$ A

(C) $50.81\angle 53.13^\circ$ A

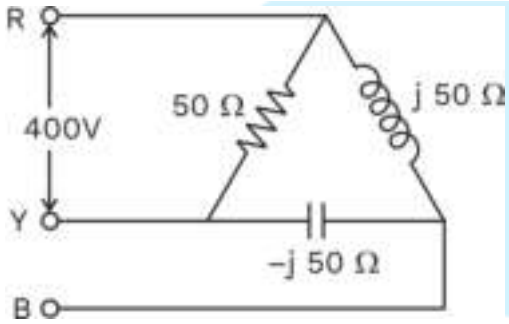
(D) $50.81\angle 173.13^\circ$ A

Q.63 Total power measured in a three-phase system feeding a balanced star connected load is 12 kW at a power factor of 0.6 leading. If the

line voltage is 440 V, calculate the load impedance Z_Y.

Q.64 A 3-wire, 3-phase supply feeds a delta connected load consisting of three equal resistors. What is the percentage reduction in load if one of the resistors is removed?

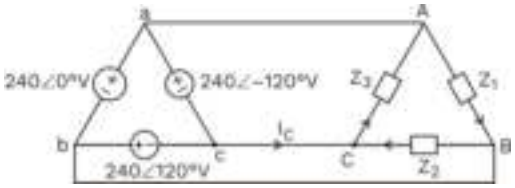
Q.65 The set of 3 equal resistors each of value R Ω. Connected in Y–across RYB consume the same power as the unbalanced Δ connected load shown in the figure. The value of R is _____ in ohm.



(A) 100 Ω (B) 50 Ω

(C) 120 Ω (D) 140 Ω

Q.66 Consider the delta-delta system as shown in figure. If Z_1 = 100 Ω, Z_2 = j 100 Ω and Z_3 = –j 100 Ω, then what is the value of line current I_C.



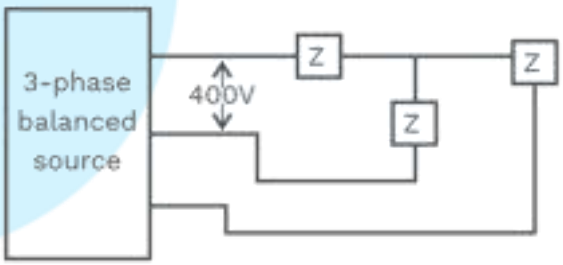
Q.67 Each phase of a balanced three-phase Δ -connected load consist of 0.1H inductor in series with the parallel combination of a 4μF capacitor and a 100Ω resistance. Assume a phase voltage of 230V at ω = 500 rad/sec. then the total power absorbed by the load is

(A) 1500W (B) 1497W

(C) 1491W (D) 1591W

Q.68 A balanced 3-Ø, star connected load of 150kW takes a leading current of 90A, when connected across a 3-phase 1100V, 50Hz supply. The capacitance of the load per phase is –

Q.69 If 3-phase balanced source in the figure delivers 1200W at a lagging power factor of 0.6, then the value of Z(in ohm) is approximately –



(A) 80 ∠-53.13° (B) 90 ∠ -32.44°

(C) 90 ∠ 32.44° (D) 80 ∠ 53.13°

Q.70 A balanced delta-connected load (20 + j16) ohm per phase is connected to a 3-phase 230 V balanced supply. The line current and the real power drawn respectively are-

(A) 19.9 A and 3.17 kW

(B) 15.55 A and 4.83 kW

(C) 19.9 A and 6.34 kW

(D) 11.5 A and 3.17 kW

Q.71 A 400V, $3-\phi$ balanced supply is fed to an $3-\phi$ unbalanced Δ connected load having

$Z_{RY} = 20\angle -50^0, Z_{YB} = 10\angle 120^0$ &

$Z_{BR} = 25\angle -30^0$ the Line current I_Y will be_____.

(A) $59.79\angle -136.67^0$

(B) $53.2\angle 130.21^0$

(C) $59.79\angle 136.67^0$

(D) $53.2\angle -130.21^0$

Q.72 Statement (I)

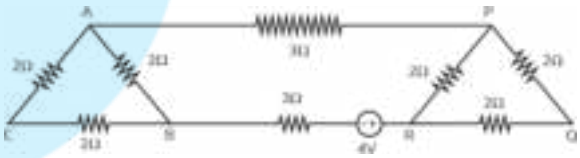
A 3-wire, 3 phase supply feeds a load which consists of three equal resistors connected in delta consuming X watts. If one of the resistors is removed then power consumption becomes X/2 watts.

Statement (II)

A 3-phase star connected balanced load consumes X watts of power from an ac source supply. If the same load is connected in delta across the same supply the power consumption is 3X watts.

1) Both statement (I) and statement (II) are individually true and statement (II) is correct explanation of statement (I).
2) Both statement (I) and statement (II) are individually true but statement (II) is not correct explanation of statement (I).
3) Statement (I) is true but statement (II) is false.
4) Statement (I) is false but statement (II) is true

Q.73 The potential difference between point A & P i.e. $V_{AP} = 12V$ then V_{BR} is

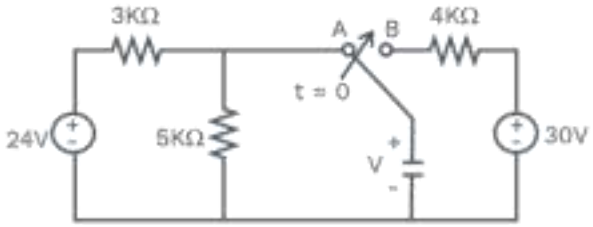


(A) 0V (B) -10V

(C) -16V (D) 10V

5 Transient Analysis

Q.1 In the given below switch is at position A for long time. At t = 0 the switch moves to B. Which of the statements is correct – MSQ



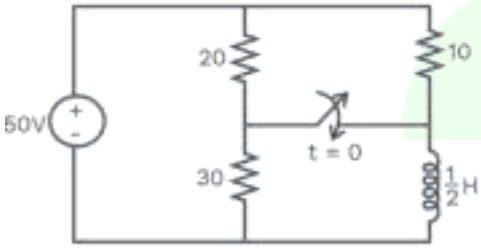
(A) $V_c\left(0^-\right) = 15V$

(B) $\left.\frac{dV_c}{dt}\right|_{t=0^+} = 7.5$

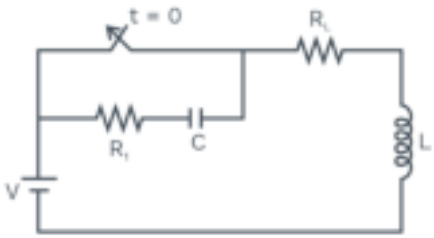
(C) $V_c(t) = 20 - 15e^{-0.5t}$

(D) V(1) = 20.9V

Q.2 Find the current passing through the inductor for t = 0



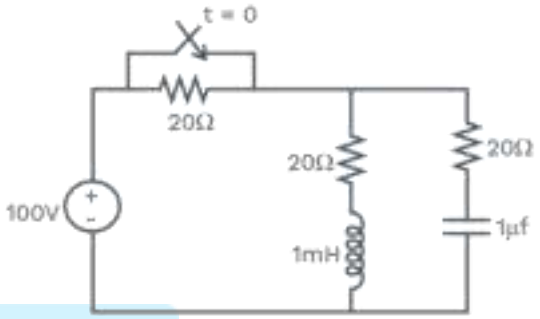
Q.3 Find the voltage across the switch V_s at $t = 0^+$



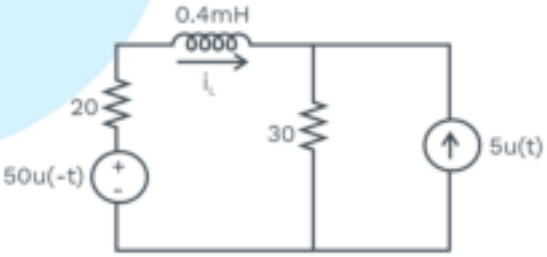
(A) $V\frac{R_L}{R_1}$ (B) $V(R_L + R_1)$

(C) $V.R_L R_1$ (D) $V\frac{R_1}{R_L}$

Q.4 Find $\frac{di}{dt}$ at $t = 0^+$ in mA__



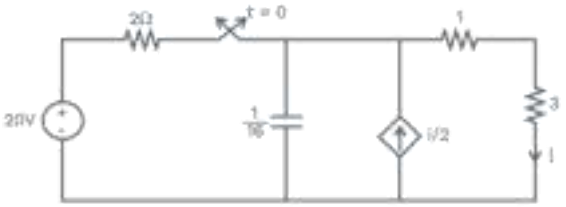
Q.5 Find the current i_L for all values of t.



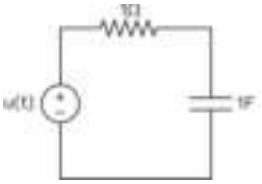
(A) $3 - 2e^{-0.125t}$ (B) $3 - 2e^{-8t}$

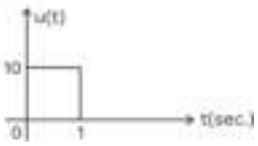
(C) $3 + 2e^{-0.125t}$ (D) $3 + 2e^{-8t}$

Q.6 Find the value of V_c for t > 0



Q.7 Find i(t) for t > 0

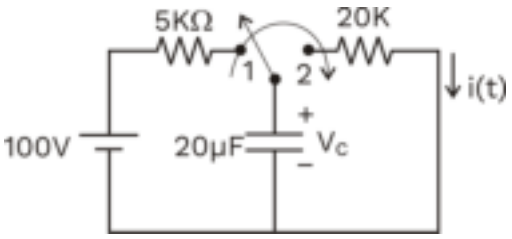




(A) $6.32\ e^{-t}A$ (B) $6.32\ e^{-(t-1)}$ A

(C) $-6.32\ e^{-(t-1)}$ A (D) $-6.32\ e^{(t-1)}$ A

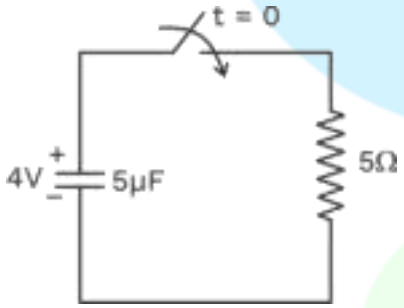
Q.8 The switch is at a position 1 for a long time and it is moved to position 2 at t = 0 find $i_C(t)$ for $t \geq 0$ in mA



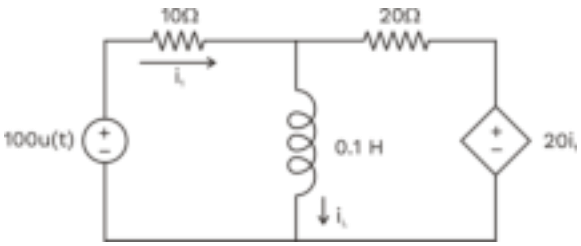
(A) $-100\ e^{-2.5\ t}$ (B) $-5\ e^{-2.5\ t}$

(C) $5\ e^{-2.5\ t}$ (D) $100\ e^{-2.5\ t}$

Q.9 The initial voltage across the capacitor is 4V.The switch is closed at t = 0, the charge loss by capacitor from t = 25 μ sec. to t = 100 μ sec. is _____ in Coulomb



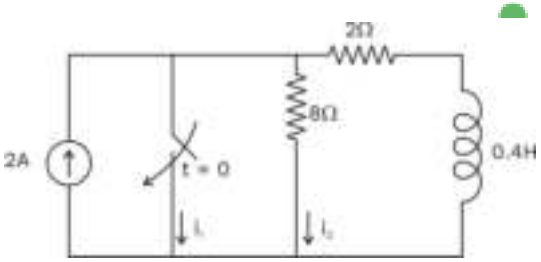
Q.10 For the given circuit find $i_1(t)$ for $t \geq 0$



(A) $20 - 20\ e^{-40t}$ (B) $10 - 8\ e^{-40t}$

(C) $20 + 20\ e^{-40t}$ (D) $10 + 8\ e^{-40t}$

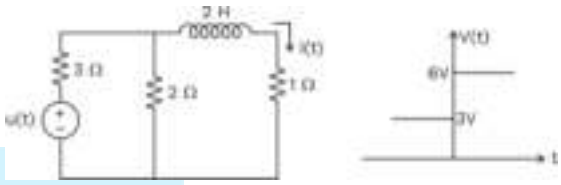
Q.11 The switch is opened for a long time and it is closed at t = 0 find $V_L(t)$ for $t \geq 0$



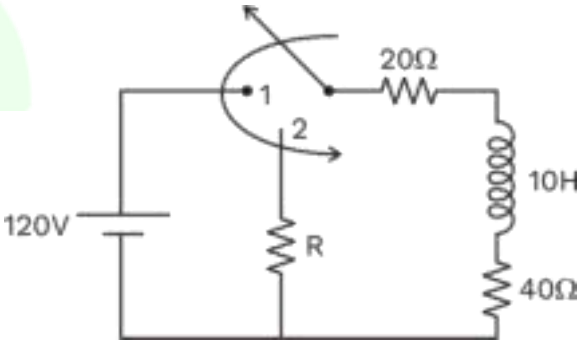
(A) $-3.2\ e^{-5t}$ (B) $1.6\ e^{-5t}$

(C) $2 - 1.6\ e^{-5t}$ (D) $2 + 1.6\ e^{-5t}$

Q.12 The voltage source has following characteristic is applied to the given circuit find the value of i(1) _____ in Amp.



Q.13 A coil of inductance 10 H and resistance 40 Ω is connected as shown in the figure. The switch is at a position 1 for a long time and it is moved to position 2 at t = 0. If at $t = 0^+$ the voltage across the coil is 120V, the value of resistance R is –



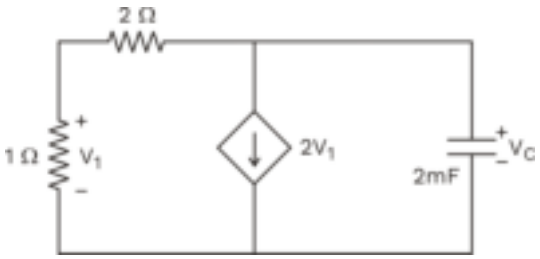
(A) 0 Ω

(B) 20 Ω

(C) 40 Ω

(D) 60 Ω

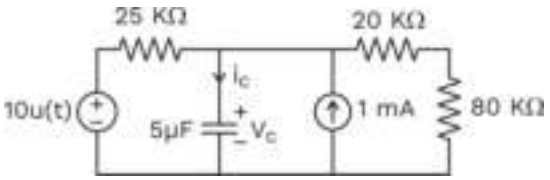
Q.14 For the given circuit if $V_C(0^-) = 12$ V then find $V_C(t)$ for $t \geq 0$



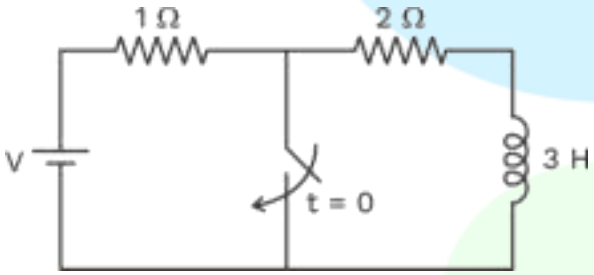
(A) $12\ e^{-0.5\ t}$ (B) $-12\ e^{-0.5\ t}$

(C) $15\ e^{-0.5\ t}$ (D) $-15\ e^{-0.5\ t}$

Q.15 Find the value of $V_C(0.08)$ _____ volt.



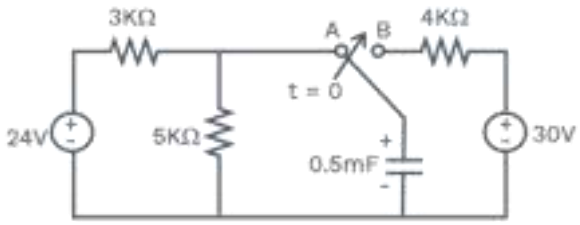
Q.16 Switch opened for long time and closed at t = 0 find the difference between discharging time constant and charging time constant _____ (sec.)



(A) 1 sec. (B) 1.5 sec.

(C) 0.5 sec. (D) 2 sec

Q.17 In the given below switch is at position A for long time. At t = 0 the switch moves to B. Which of the statements is correct – MSQ



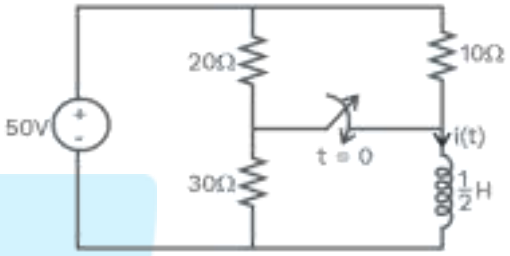
(A) $V_c\left(0^-\right) = 15V$

(B) $\left.\frac{dV_c}{dt}\right|_{t=0^+} = 7.5$

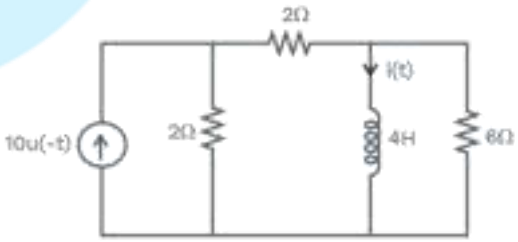
(C) $V_C(t) = 20 - 15e^{-0.5t}$

(D) $V\left(1\right) = 20.9V$

Q.18 Find the current passing through the inductor for t = 0



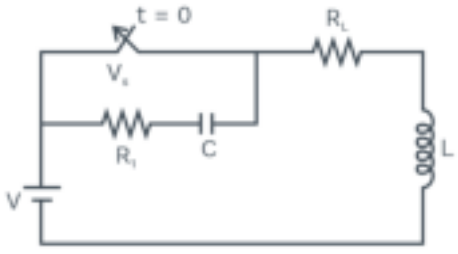
Q.19 Find the current response of inductor for t > 0.



(A) $5e^{-1.6t}$ (B) $-5e^{-0.48t}$

(C) $5e^{-0.48t}$ (D) $-5e^{-1.6t}$

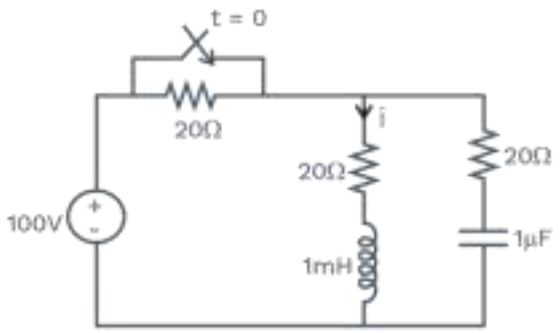
Q.20 Find the voltage across the switch V_s at $t = 0^+$



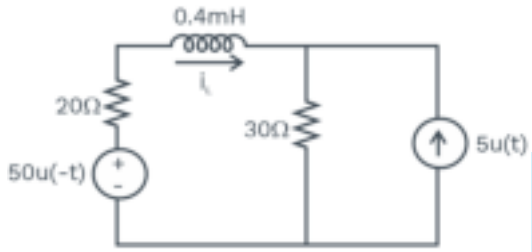
(A) $V\frac{R_L}{R_1}$ (B) $V(R_L + R_1)$

(C) $V.R_LR_1$ (D) $V\frac{R_1}{R_L}$

Q.21 Find $\frac{di}{dt}$ at $t = 0^+$ in mA__



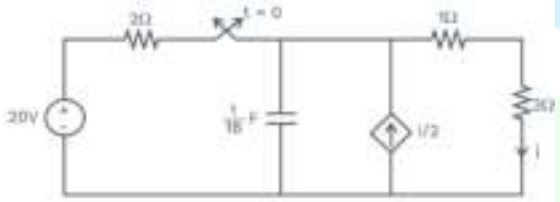
Q.22 Find the current i_L for all values of t.



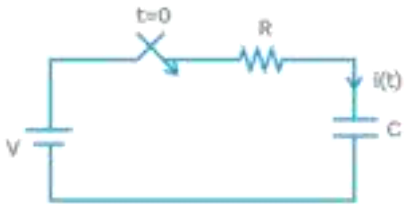
(A) $3 - 2e^{-0.125t}$ (B) $3 - 2e^{-8t}$

(C) $3 + 2e^{-0.125t}$ (D) $3 + 2e^{-8t}$

Q.23 Find the value of i for t < 0



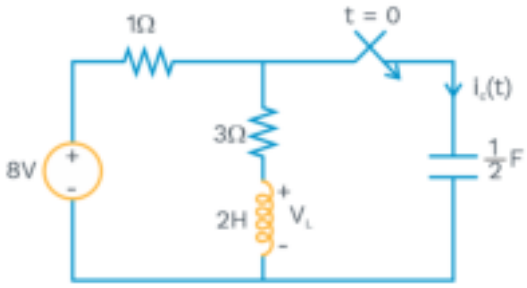
Q.24 The switch is opened for a long time then closed at t = 0. Determine the ratio of energy stored in a capacitor to the energy supplied by the source when $t \to \infty$.



(A) 1 (B) $\frac{1}{2}$

(C) 2 (D) ∞

Q.25 The switch is opened for a long time and it is closed at t = 0. Find which of the statement is correct–



(A) $i_L(0^-) = 2A$

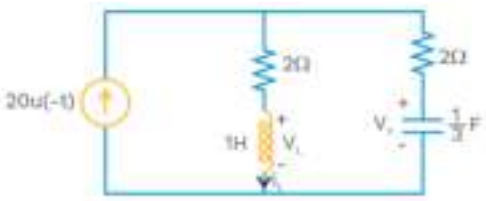
(B) $\frac{dV_c(0^+)}{dt} = 12V / sec.$

(C) $\frac{di_L(0^+)}{dt} = -3A / sec.$

(D) $\frac{di_L(0^+)}{dt} = 3A / sec.$

Q.26 _____

For the given circuit find the response of $i_L(t)$ –



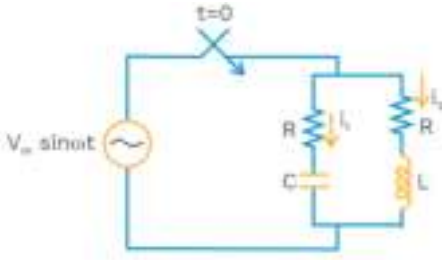
(A) $\left(10e^{-t} - 10e^{-3t}\right)u(t)$

(B) $\left(-10e^{-t} - 10e^{-3t}\right)v(t)$

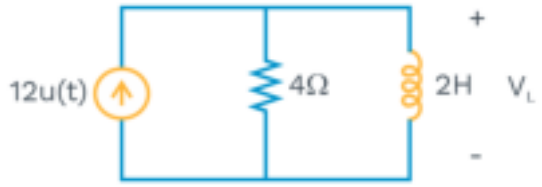
(C) $\left(10e^{-t} + 10e^{-3t}\right)v(t)$

(D) $\left(10e^{t} - 10e^{3t}\right)v(t)$

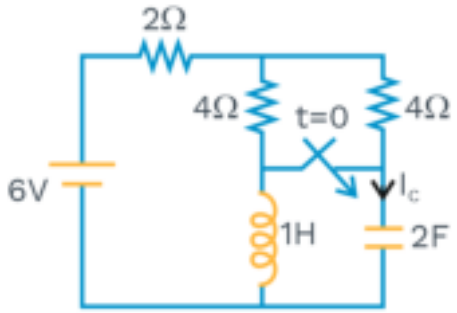
Q.27 For the given circuit find the value of $\frac{di_2(0^+)}{dt}$ _____ in A/sec.



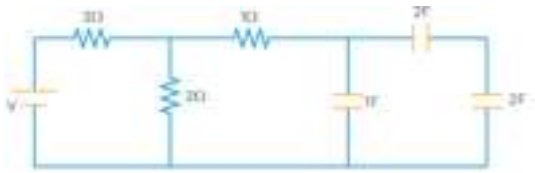
Q.28 For the given circuit find the value of $\frac{dV_L(0^+)}{dt}$ _____ in V/sec.



Q.29 The switch is opened for a long time and it is closed at t = 0. Find the value of $i_c(0^+)$____ in (Amp.)



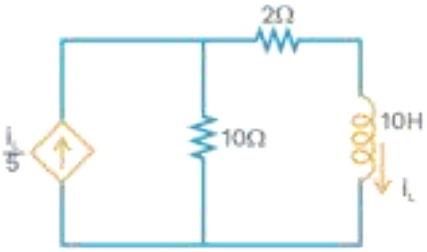
Q.30 Calculate the time constant for the given network _____ in (sec.)



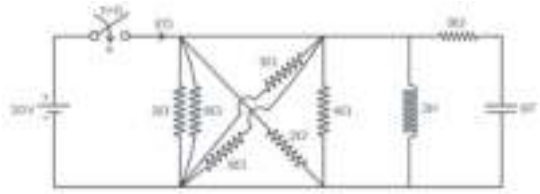
(A) 2 sec. (B) 4 sec.

(C) $\frac{1}{2}$ sec. (D) $\frac{1}{4}$ sec.

Q.31 Find the time constant for the given circuit is sec. _____.

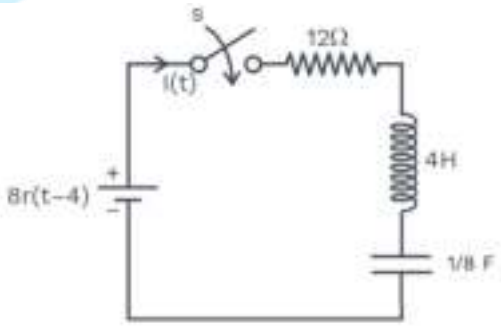


Q.32



For the above circuit shown above, if the inductor and capacitor are initially un-energized, then the value of I(t) at t = 0^+ is_____ A (up to two decimal places).

Q.33



For the circuit shown above the Switch is closed at t = 0, the value of current I(t) for t > 0 is

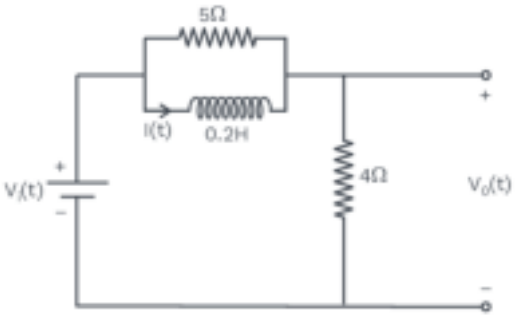
(A) $u(t-4) - 2e^{-(t-4)}u(t-4) + e^{-2(t-4)}u(t-4)$

(B) $u(t-2) - 2e^{-(t-4)}u(t-4) + e^{-2(t-4)}u(t-4)$

(C) $u(t-4) + 2e^{-(t+4)}u(t+4) + e^{-2t}u(t)$

(D) $u(t-4) - 2e^{(t-4)} + e^{-2(t-4)}u(t-4)$

Q.34

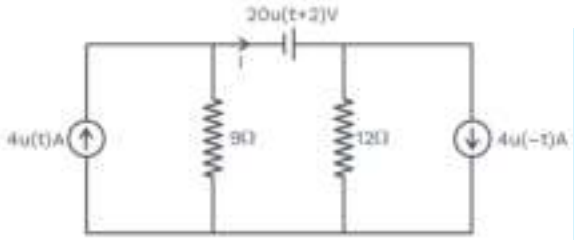


For the above circuit, if I(0) = 6A and $V_i(t)=0$, the voltage $V_0(t)$ for t > 0 will be

(A) $14.32e^{-t/0.06}$V (B) $13.32e^{-t/0.09}$V

(C) $13.32e^{-1/0.06}$ (d) $12.31e^{-t/0.9}$

Q.35



For the above circuit the value of current I at t = +3 sec.

(A) 3.85 A (B) 7.23 A

(C) 2.67 A (D) 4.47 A

Q.36 If a step signal is applied across a RC circuit, the capacitor voltage is given by, $v_c(t) = (1-e^{-t})V$, the RC circuit being initially at rest.

If there is an initial voltage of 1V on the capacitor, and the input signal is 5u(t), find the expression for $v_c(t)$.

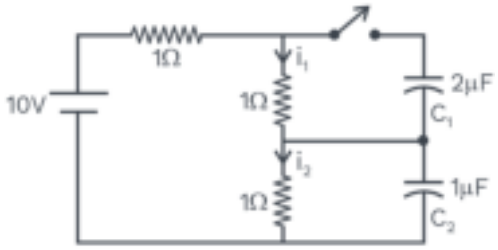
(A) $(5 - 5e^{-t})$ V

(B) $(5 - 4\ e^{-t})$ V

(C) $(4 - 5\ e^{-t})$ V

(D) $(4 - 4\ e^{-t})$ V

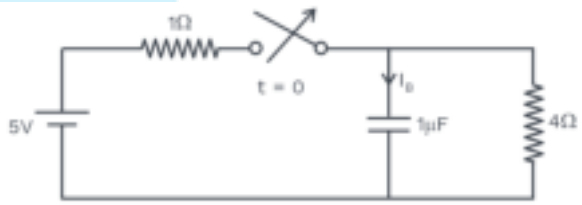
Q.37 In the below circuit, what will be the value of current i_1 and i_2 respectively at time ($t = 0^+$). (Switch is closed at $t = 0^-$)



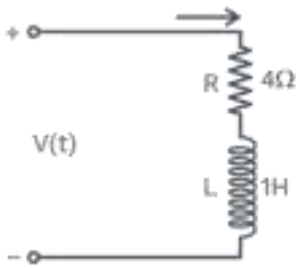
(A) 2.5A, 0A (B) 5A, 2.5A

(C) 2.5A, 2.5A (D) 0A, 2.5A

Q.38 The switch in the circuit has been closed for a long time. It is opened at t = 0. At $t = 0^+$, the current through the 1μF capacitor is..... (in A)



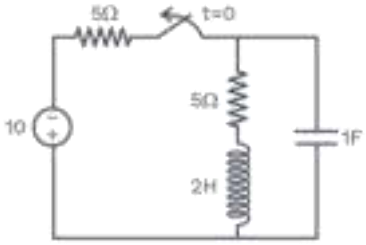
Q.39 Consider an L-R circuit in which a current $I = 5e^{-2t}$ A is flowing. The voltage across the R–L circuit is given by:



(A) $20e^{-2t}$ V (B) $-10e^{-2t}$ V

(C) $10e^{-2t}$ V (D) $5e^{-2t}$ V

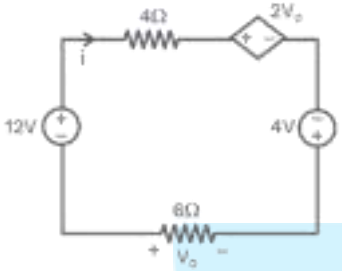
Q.40 In the given circuit, determine the value of $i_L(0^-)$ and $V_C(0^-)$. Switch is opened at t=0.



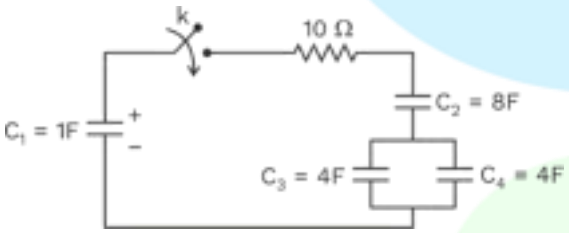
(A) 1A, 10V (B) 1A, 5V

(C) 0A, 5V (D) 0A, 10V

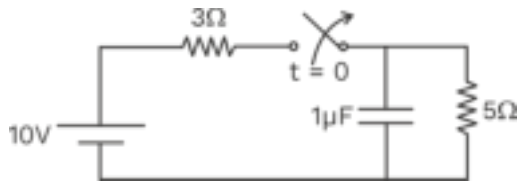
Q.41 Determine V_0 (in V) in the circuit shown in figure.



Q.42 The steady state energy in C_2 is _____ Joules (initial voltage across C_1 is 10V).



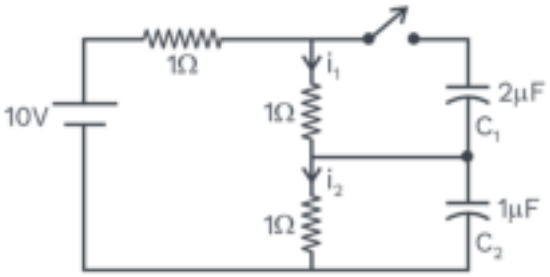
Q.43 The switch in the circuit remains closed for a long time. It is opened at t = 0. What will be the current flowing through 1 μF capacitor at time $t = 0^+$?



(A) 2.25 A (B) 2.5 A

(C) 1.25 A (D) 5 A

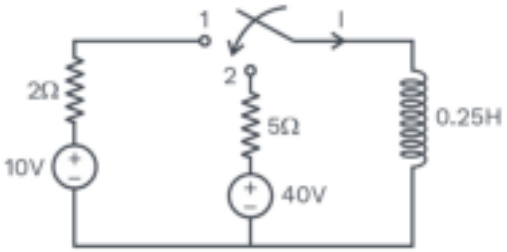
Q.44 In the below circuit, what will be the value of current i_1 and i_2 respectively at time ($t = 0^+$). (Switch is closed at $t = 0^-$)



(A) 2.5A, 0A (B) 5A, 2.5A

(C) 2.5A, 2.5A (D) 0A, 2.5A

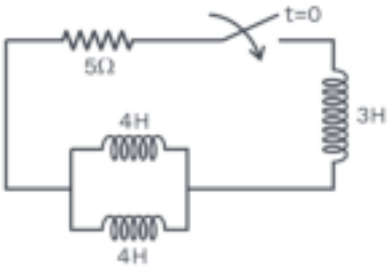
Q.45 In circuit switch is moved from 1 to 2 at t = 0. Find I.



(A) 8 – 3e–10t (B) 8 – 3e–20t A

(C) 8 + 3e–20t A (D) 8 + 3e–10t

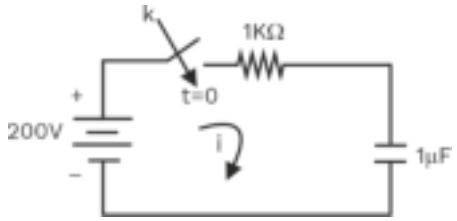
Q.46 In the circuit, the switch is closed at t = 0+. When the 3H inductor has initial current I0 = 5A. The voltage across the resistor is



(A) 25e–t V (B) 10e–t

(C) 20e–t (D) 30e–t

Q.47 In the network shown in the given figure., the switch k is closed at t = 0 with the capacitor uncharged. The value of $\frac{di(t)}{dt}$ at $t = 0^+$ will be



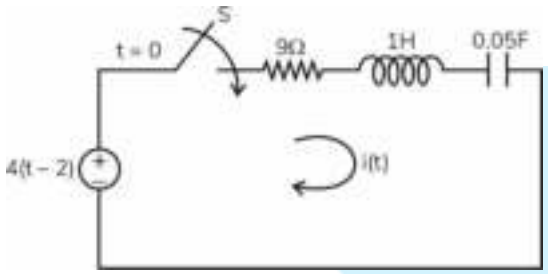
(A) 100 Amps/sec

(B) -200 Amps/sec

(C) 1000 Amps/sec

(D) -2000 Amps/sec

Q.48 In the given figure, the circuit is excited by the step delay voltage source. The switch S is closed at t = 0. The expression of current i(t) at t > 0 will be



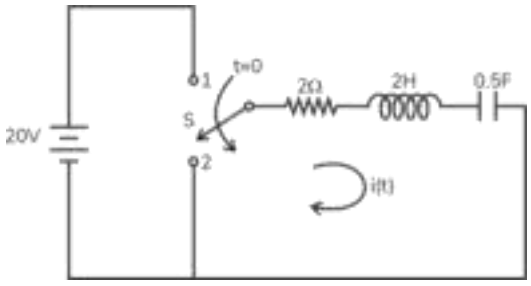
(a) $\left(e^{-4t-2}-e^{-5t-2}\right)u(t)$

(b) $\left(e^{-4t-2}-e^{-5t-2}\right)u(t-2)$

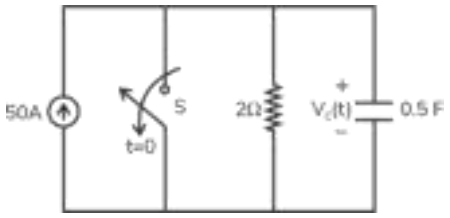
(c) $\left[e^{-4(t-2)}-e^{-5(t-2)}\right]U(t)$

(d) $\left[e^{-4(t-2)}-e^{-5(t-2)}\right]u(t-2)$

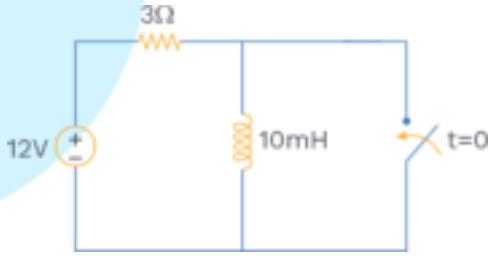
Q.49 In the given circuit the switch is at position 1 for a long time and moved to position 2 at t = 0. The value of $\frac{d^2i\left(0^+\right)}{dt^2}$ is____



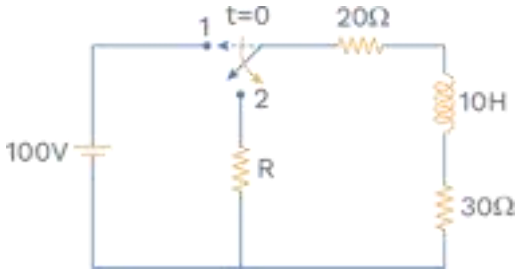
Q.50 In the given circuit the switch S is closed for a long time and is opened at t = 0. The value of $\frac{d^2v_c\left(0^+\right)}{dt^2}$ is _____ $(V/sec)^2$.



Q.51 The circuit shown in the figure below has achieved a steady-state. The current (in Amp) through the inductor at $t = 0^+$, if the switch is closed at t = 0 sec is_____(Rounded off to the nearest integer).

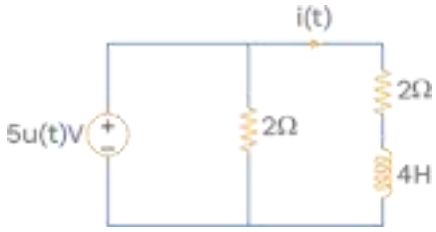


Q.52 For the circuit shown below, there exist a coil with inductance of 10H and resistance of 30Ω.The switch is connected at point 1 for a very long time and it is switched to point 2 at t = 0 sec.



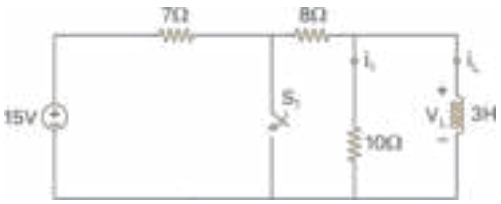
For t > 0, the voltage across the coil is 100V. The value of resistance R is _____ Ω. (Rounded off to the nearest integer)

Q.53 Consider the circuit shown below.



The current i(t) at 2 sec is ____(in Amp) (upto 2 decimal places)

Q.54 The circuit shown in the figure, if switch S_1 was open for long time and it is closed at t = 0 sec, then



MSQ

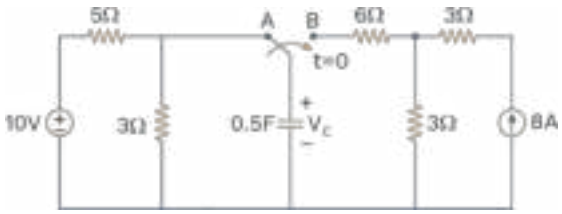
(A) $i_L(t = 0.5\ s) = 0.47$ Amp

(B) $V_L(t = 0.5\ s) = -2.12$ Volts

(C) $V_L(t = 1\ s) = 2.42$ Volts

(D) $i_L(t = 1\ s) = 0.32$ Amp

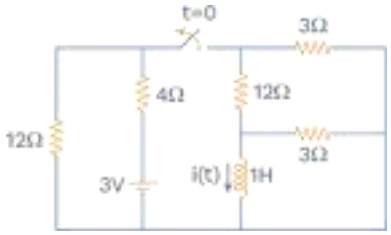
Q.55 The switch is connected to position A for a long time and changes its position to B at t=0. The capacitor voltage V_C at t = 1 sec is-



(A) 8.52 Volts (B) 7.78 Volts

(C) 4.36 Volts (D) 6.92 Volts

Q.56 The current i(t) in the circuit for t > 0 is $\alpha e^{-\Gamma t}$. Which of the following statement(s) is/are **TRUE**?



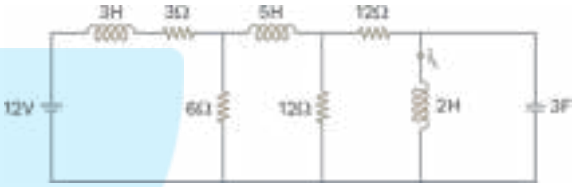
(A) The current i(t) for t > 0 is $4e^{-0.4t}$A.

(B) The value of α is $\frac{4}{3}$.

(C) The current through the inductor is 0A.

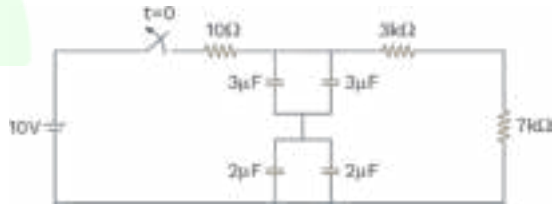
(D) The value of Γ is –2.5.

Q.57 A circuit is shown in the figure below.



The circuit has been operating for a long time. The current i_L flowing through the circuit is _____ A. (Upto one decimal places)

Q.58 Consider the circuit shown in the figure below.

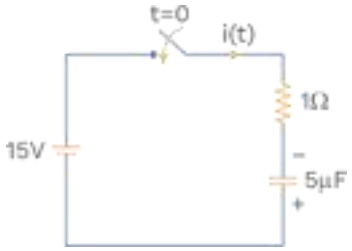


The switch in the circuit was closed for a long time and it is opened at t = 0. The time constant of the circuit is-

(A) 17 msec (B) 24 msec

(C) 25 msec (D) 48 msec

Q.59 A circuit is shown in the figure below.



If the initial charge on the capacitor is 250□C, then the current i(t) flowing through the circuit at t = 0 will be-

(A) $65\,e^{-(2\times10^5\,t)}$A (B) $50\,e^{-(2\times10^5\,t)}$A

(C) $15\,e^{-(5\times10^5\,t)}$A (D) $65\,e^{-(5\times10^5\,t)}$A

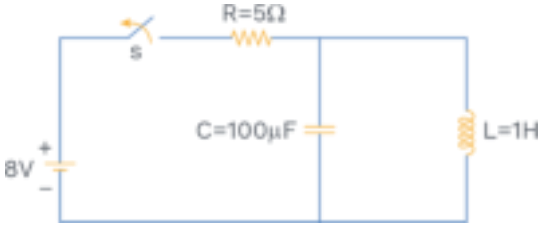
Q.60 A circuit is shown in the figure below.



The current i for t < 0 is _____ A. (Upto two decimal places)

Q.61 For a series RL circuit, the charging time needed for the inductor current to reach 40% of its steady-state value is given as 4 sec. If the value of the resistor is 8Ω, then the value of the inductor is ____ H. (Upto three decimal places)

Q.62 In the circuit shown in the figure, the switch was closed for a long time. At t = 0, the switch is opened. The current through the inductor for t > 0 is



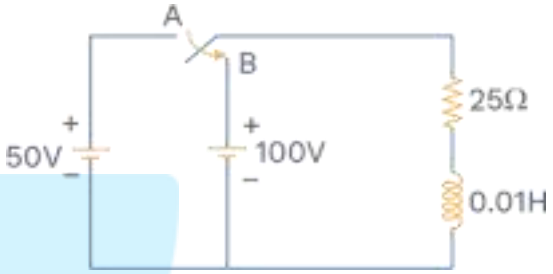
(A) 7.2 $\sin10^4$t Amp

(B) 1.6 sin100t Amp

(C) 1.6 cos100t Amp

(D) 7.2 $\cos10^4$t Amp

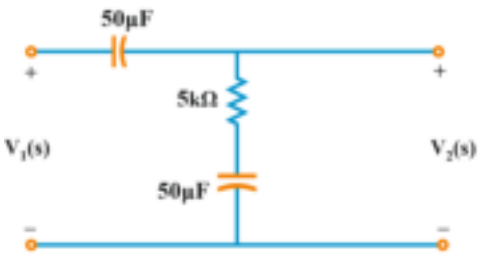
Q.63 In the R-L circuit shown in the figure, the switch is at position-A for a long time and switched to position-B at t = 0. The current through the inductor at t=0.5msec is _____ Amp. (Up to two decimal places)



Q.64 In the capacitive network shown below, the voltage across the capacitor is V_C, and the step input is V_i. The time (in □sec) when V_C becomes zero is_____ (Up to two decimal places)



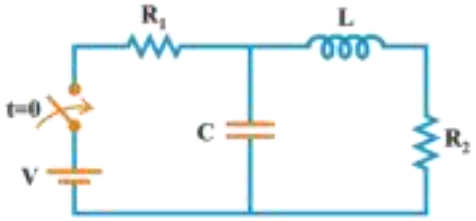
Q.65 For the circuit shown in the figure below, the transfer function $\frac{V_2(s)}{V_1(s)}$ is–



(A) $\frac{s+1}{s+4}$ (B) $1+\frac{s}{s+8}$

(C) $\frac{s+4}{s+8}$ (D) $\frac{s+8}{s+2}$

Q.66 The switch is closed at t=0 for the circuit shown in the figure below.



Which of the following statement(s) is/are true? MSQ

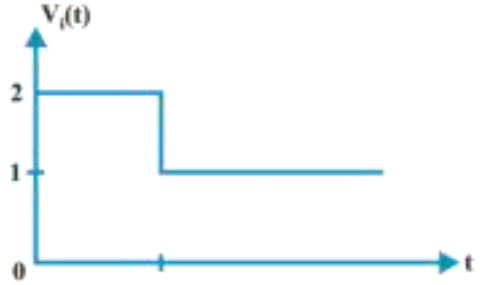
(A) The supply current at $t = 0^+$ is $\frac{V}{R_1}$

(B) The supply current at $t = 0^+$ is $\frac{V}{R_1 + R_2}$

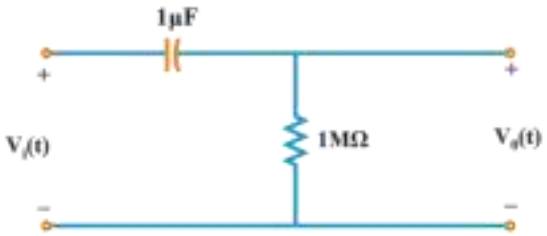
(C) The voltage of capacitor at $t \to \infty$ is $\frac{VR_2}{R_1 + R_2}$

D) The voltage of capacitor at $t \to \infty$ is $\frac{VR_2}{R_1}$

Q.67 The input voltage $V_i(t)$ is applied to the circuit is shown below.



The input voltage given above is constant at 2V for t > 0 sec, and at certain time t, it switches to 1V. The output voltage for the circuit given below, 5 sec after the change in output voltage is –



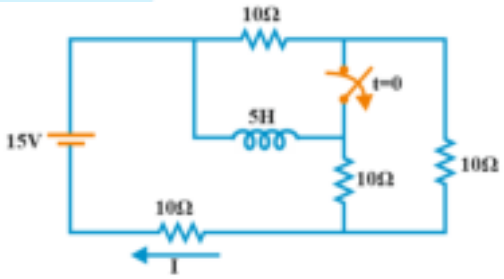
(A) $e^{2.5}$ V

(B) $-e^{-5}$V

(C) $2\left(1-e^{5}\right)$V

(D) None of the above

Q.68 For the circuit shown below, the switch is closed for a long time and is opened at t=0.



Which of the following statements is/are true? MSQ

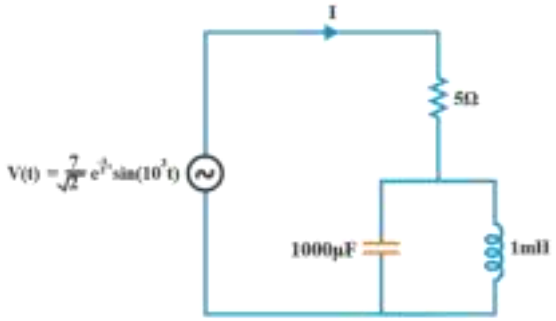
(A) The current I before the switch is opened is 1/2A.

(B) The current I before the switch is opened is 1A.

(C) The current I after the switch is opened is 0A.

(D) The current I after the switch is opened is 7/6A.

Q.69 The steady state current for the circuit shown below is-



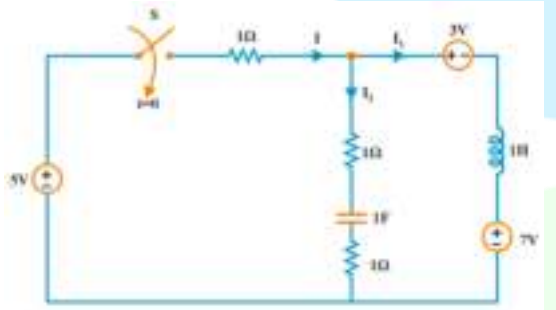
(A) 0A

(B) $\frac{7}{\sqrt{2}}e^{-\frac{8}{5}t}\sin\left(1000t-\frac{\pi}{4}\right)$A

(C) $\frac{7}{\sqrt{2}}$A

(D) sin(1000t)A

Q.70 The switch S is opened for a long time such that the circuit shown below achieves steady state. The switch closes at time t = 0. MSQ



Which of the following statements is/are true?

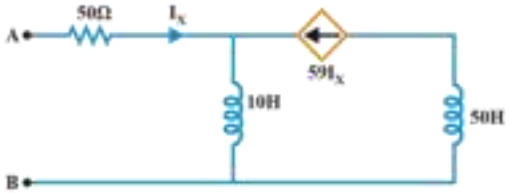
(A) The current I at $t=0^+$ is $-\frac{5}{3}$A

(B) At time $t=0^+$, $\frac{dI_1}{dt}$ is $-\frac{10}{3}$A / sec

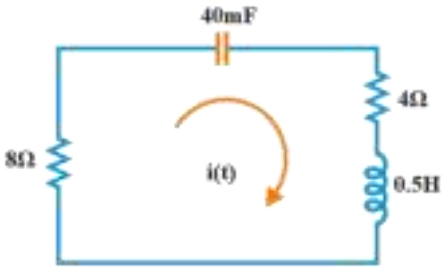
(C) The current I at $t=0^+$ is $-\frac{10}{3}$A

(D) At time $t=0^+$, $\frac{dI_1}{dt}$ is $-\frac{5}{3}$A/sec

Q.71 A circuit is shown in the figure below. The equivalent inductance between terminals A-B is _____ H. (Rounded off to the nearest integer)



Q.72 An RLC circuit is shown below.



The initial current across the inductor is 1.2A and the initial voltage across the capacitor is 50V The current i(t) is –

(A) undamped

(B) underdamped

(C) critically damped

(D) overdamped

Q.73 The transfer function of a filter is given as $H(s)=\frac{5(s^2+49)}{s^2+4.9s+245}$. The given transfer function represents-

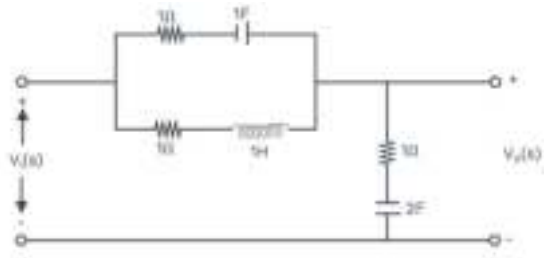
(A) A high Q bandpass filter.

(B) A low Q bandpass filter.

(C) A high pass notch filter.

(D) An all-pass filter.

Q.74 Consider the circuit shown in figure. The time constant of the circuit is___



(A) 2 sec. (B) 4 sec.

(C) 3 sec. (D) 1 sec.

Q.75



For passive network shown in figure input impedance is given by-

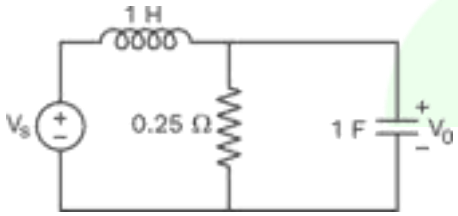
$$z(s) = \frac{4s(s+3)}{(s+5)}$$

If terminal is open circuited and V_{ab} (0) = 12 V, find a $V_{ab}(t)$?

(A) $12e^{5t}u(t)$ (B) $-12e^{-5t}u(t)$

(C) $12e^{-5t}u(t)$ (D) $e^{-5t}u(t)$

Q.76 The circuit shown in a figure is a –



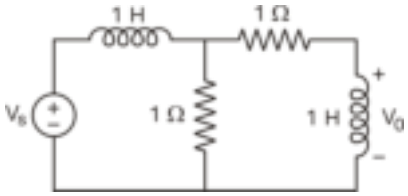
(A) law pass filter

(B) high pass filter

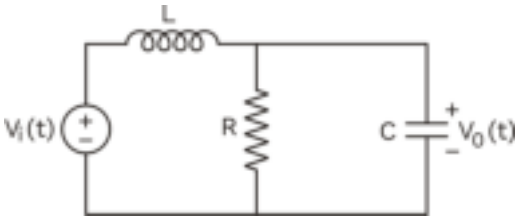
(C) band stop filter

(D) band reject filter

Q.77 Determine the bandwidth the band pass filter as shown below –



Q.78 What type of filter is shown below in figure–



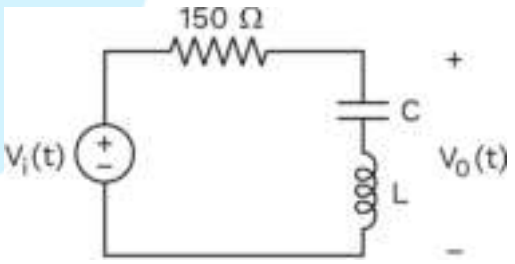
(A) high pass first order filter

(B) low pass first order filter

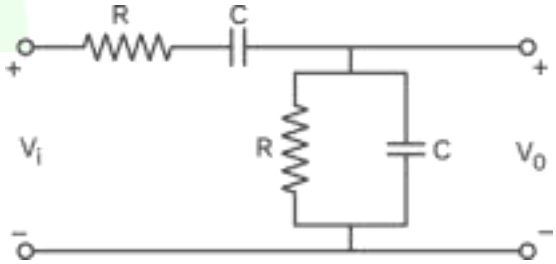
(C) low pass second order filter

(D) high pass second order filter

Q.79 If the band stop filter reject a 200 Hz sinusoidal while passing other frequencies, calculate the value of C(in Farad), if the R = 150 Ω and band width is 100 Hz.



Q.80 The RC circuit shown in the figure is:



(A) band pass filter

(B) low pass filter

(C) high pass filter

(D) band reject filter

Q.81 Consider a parallel RLC circuit with R = 5Ω, L = 9H, C = 4F. The type of damping exhibited by the circuit is –

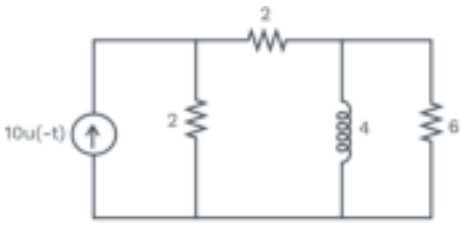
(A) Under damping

(B) Critical damping

(C) Over Damping

(D) Undamping

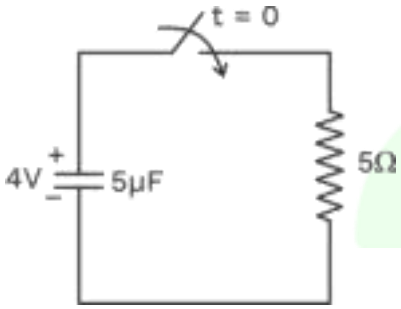
Q.82 Find the current response of inductor for t > 0.



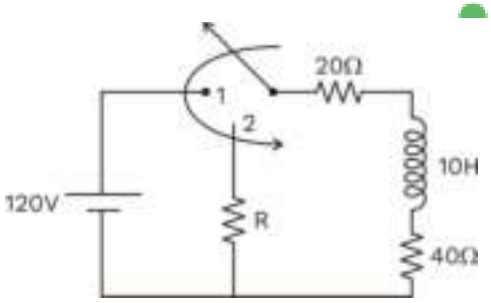
(A) $5e^{-1.6t}$ (B) $-5e^{-0.48t}$

(C) $5e^{-0.48t}$ (D) $-5e^{-1.6t}$

Q.83 The initial voltage across the capacitor is 4V.The switch is closed at t = 0, the charge loss by capacitor from t = 25 μ sec. to t = 100 μ sec. is _____ in Coulomb



Q.84 A coil of inductance 10 H and resistance 40 Ω is connected as shown in the figure. The switch is at a position 1 for a long time and it is moved to position 2 at t = 0. If at $t = 0^+$ the voltage across the coil is 120V, the value of resistance R is –



(A) 0 Ω (B) 20 Ω

(C) 40 Ω (D) 60 Ω

Q.85 An ideal voltage source will charge an ideal capacitor.

(A) Exponentially

(B) Linearly

(C) In infinite time

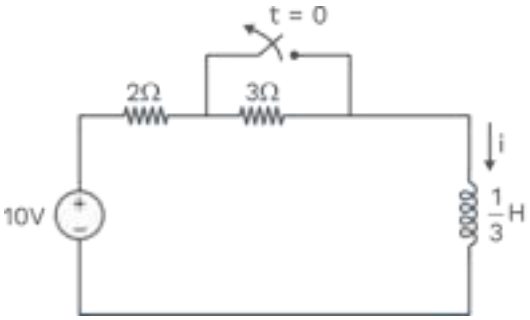
(D) Instantaneously

Q.86 The coil of a certain relay is operated by a 12-V battery. If the coil has resistance of 150Ω and an inductance of 30 mH and current needed to pull is 50 mA, calculate the relay delay time.

(A) 1 ms (B) 0.5 ms

(C) 0.196 ms (D) 3 ms

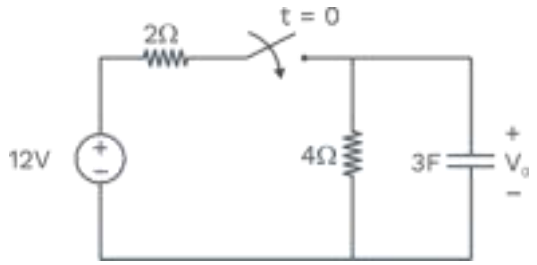
Q.87 Find the time constant for the circuit shown in figure.



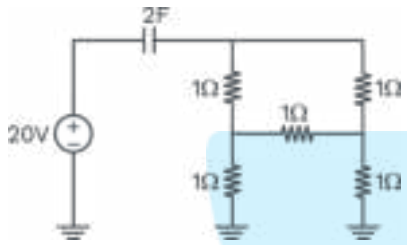
(A) 60 ms (B) 15 ms

(C) 30 ms (D) 66.67 ms

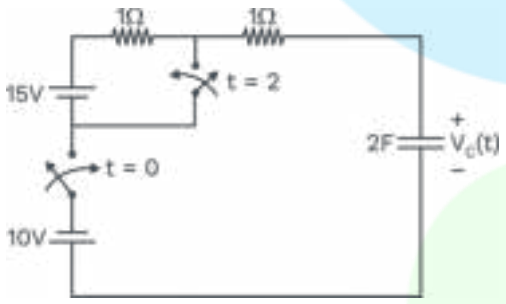
Q.88 If the switch in figure has been open for a long time and is closed at t = 0, the value of $V_0(t)$ at t = 4s is _____ Volt.



Q.89 The initial charge in the 2F capacitor present in the circuit shown is zero. The energy in Joules transferred from the DC source until steady state condition is reached equals _____.



Q.90 Given figure which of the following statements is/are true? MSQ



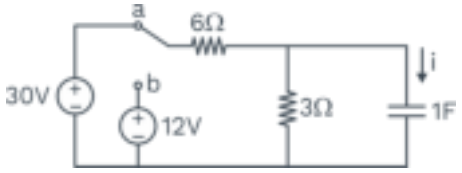
(A) $V_C(2^-) = 1.967$ V

(B) $V_C(0^-) = 15$ V

(C) $V_C(t) = -10 + 11.967\, e^{-0.5\,(t-2)}$, $t > 2$

(D) $V_C(t) = 5e^{-\frac{t}{4}}$, $0 < t < 2$

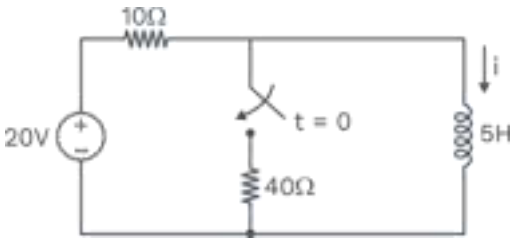
Q.91 The switch in figure has been at position a for a long time. At t = 0 it moves to position b. Calculate i(t) for all t > 0.



(A) $5 + 3e^{-\frac{t}{2}}$ A (B) $-3e^{-\frac{t}{2}}$ A

(C) $3 + e^{-\frac{t}{2}}$ A (D) $5e^{-\frac{t}{2}}$ A

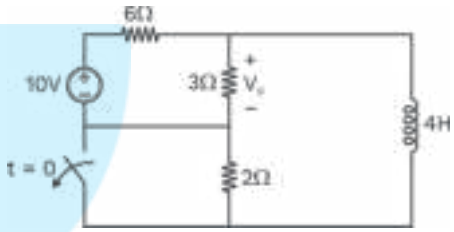
Q.92 For the circuit in figure find i(t) for t > 0.



(A) 2A (B) $2e^{-t}$

(C) $2e^{-\frac{8}{5}t}$ (D) $3e^{-5t}$

Q.93 Find time constant for t = 0 in the circuit of figure.



(A) 4 seconds (B) 5 seconds

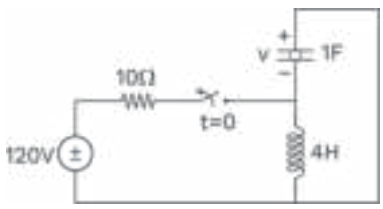
(C) 3 seconds (D) 1 seconds

Q.94 If R = 50Ω, L = 1.5 H, what value of C will make a RLC series circuits overdamped-

(A) C = 2.4 mF (B) C < 2.4 mF

(C) C > 2.4 mF (D) C = 0

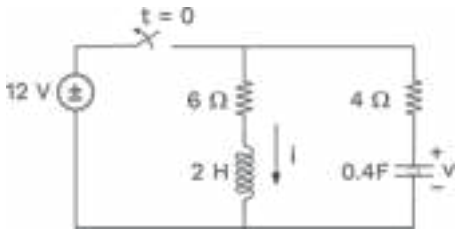
Q.95 Obtain v(t) for t > 0 in the circuit of figure.



(A) 24 cos 0.5 t V (B) 12 sin 0.5 t V

(C) 3 sin 3 t V (D) 24 sin 0.5 t V

Q.96 For the circuit in figure,



Which of the following statements is/are true? MSQ

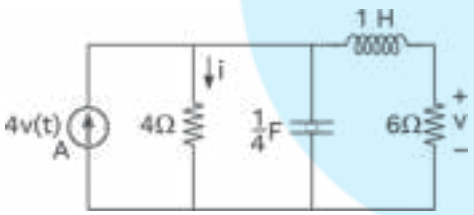
(A) $\frac{di(0^+)}{dt} = -4$ A/sec

(B) $i(0^+) = 2A$

(C) $i(\infty) = 2A$

(D) $\frac{dv(0^+)}{dt} = -5$ V/sec

Q.97 For the circuit shown in figure. The value of i(□) is _____ A.



Q.98 In series RLC circuit, setting R = 0, will produce.

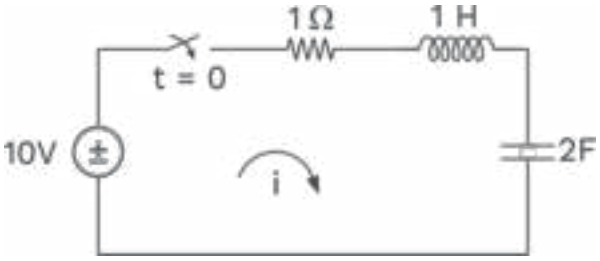
(A) An overdamped response

(B) An undamped response

(C) A critically response

(D) An underdamped response

Q.99 In below given electrical network find the value of current i for t > 0



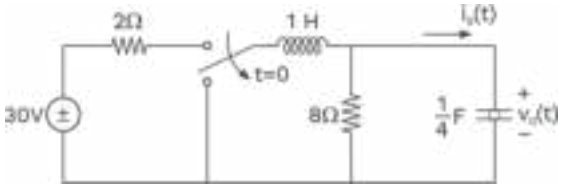
(A) 10 e^{-t} sin t

(B) 50 e^{-t} sin t

(C) 10 $e^{-0.5t}$ sin 0.5 t

(D) None of these

Q.100 In the circuit of figure calculate $v_0(t)$ for t > 0.



(A) 3[1 – 5e^{-t} cos t] V

(B) e^{-t} sin 30t V

(C) (24 cos 1.9043 t + 3.023 sin · 1.9843t)e^{-t} V

(D) (10 cos 3t + 5 sin 2t]e^{-t} V

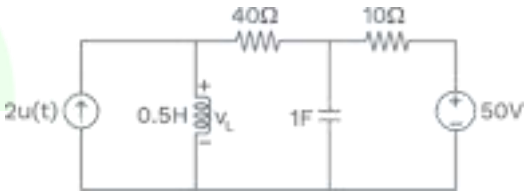
Q.101 For the circuit in figure calculate the value of R needed to have a critically damped response.



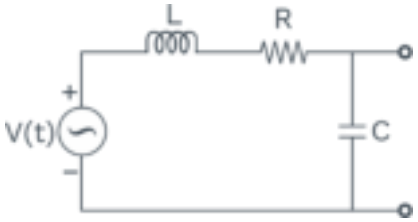
(A) 100 Ω (B) 10 Ω

(C) 120 Ω (D) 30 Ω

Q.102 For the circuit shown in figure. The value of $V_L(0^+)$ _____ Volt.



Q.103 The condition on R, L, C such that the step response y(t) in the figure has no oscillations is-

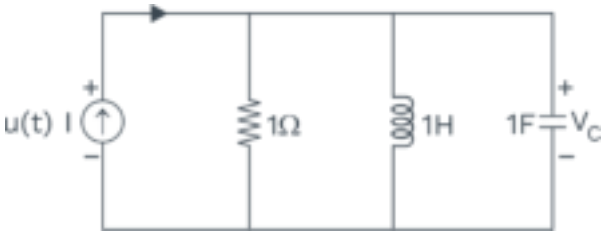


(A) $R \geq \frac{1}{2}\sqrt{\frac{L}{C}}$ (B) $R \geq \sqrt{\frac{L}{C}}$

(C) $R \geq 2\sqrt{\frac{L}{C}}$ (D) $R = 2\sqrt{\frac{1}{LC}}$

Q.104 A parallel R-L-C circuit is excited by a current source with unit step

signal. The voltage across capacitor is equal to-



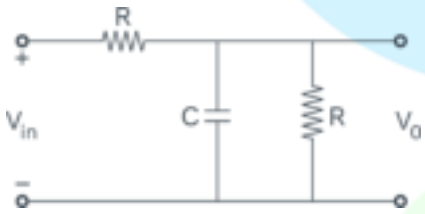
(A) $\frac{2}{\sqrt{3}} e^{-\frac{t}{2}} \sin\left(\frac{\sqrt{3}}{2} t\right)$

(B) $\frac{2}{\sqrt{3}} e^{-\frac{t}{2}} \cos\left(\frac{\sqrt{3}}{2} t\right)$

(C) $\frac{2}{\sqrt{3}} e^{-\frac{t}{2}}$

(D) $\frac{2}{\sqrt{3}}\left(e^{-\frac{t}{2}} - e^{\frac{\sqrt{3}}{2}t}\right)$

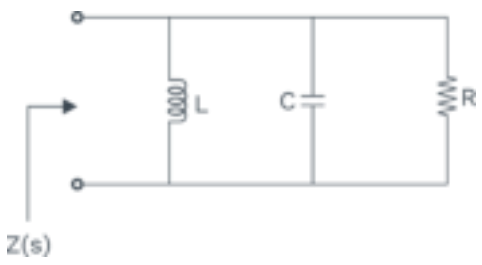
Q.105 Find the transfer function of the network $\frac{V_0(s)}{V_{in}(s)}$ is-



(A) $\frac{1}{2+sCR}$ (B) $\frac{1}{1+sCR}$

(C) $\frac{2}{1+2sCR}$ (D) None of these

Q.106 The driving point impedance of the below network is given below Z(s) = $\frac{0.2s}{s^2+0.1s+2}$. The component values are-



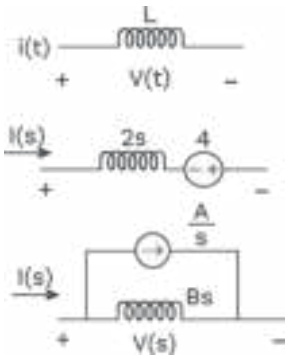
(A) L = 5H, R = 0.5Ω, C = 0.1F

(B) L = 0.1H, R = 0.5Ω, C = 5F

(C) L = 5H, R = 2Ω, C = 0.1F

(D) L = 0.1H, R = 2Ω, C = 5F

Q.107 Given two circuits are equivalent of the circuit

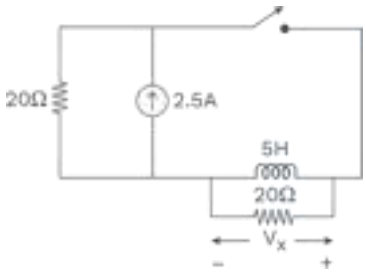


The value of A and B is-

(A) 2, 2 (B) 2, 4

(C) 4, 4 (D) 4, 2

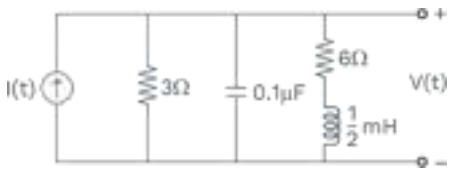
Q.108 In the figure the switch was closed for long time before opening at t = 0. The voltage at t = 0 is-



(A) 25 V (B) 50 V

(C) −50 V (D) 0 V

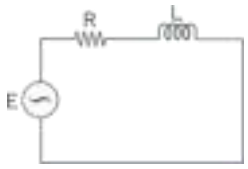
Q.109 In the circuit shown, I(t) is a unit step current. The steady state value of V(t) is-



(A) 2 V (B) 3 V

(C) 6 V (D) 9 V

Q.110 Find the value of the current in a RL circuit following step response using final value theorem?



(A) $i(\infty) = \frac{E}{R}$ (B) $i(\infty) = \frac{2E}{R}$

(C) $i(\infty) = \frac{E}{\sqrt{2}R}$ (D) $i(\infty) = \sqrt{\frac{E}{R}}$

Q.111 The current through a circuit element is $\frac{4s^2}{s+7}$. The current in time domain at 0 and ∞ are-

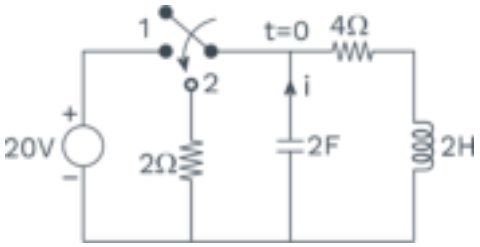
(A) 1, 1 (B) 0, 0

(C) ∞, 0 (D) 0, ∞

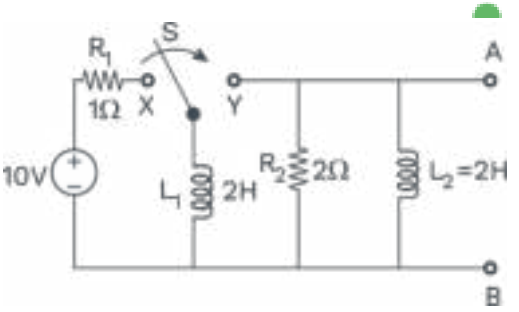
Q.112 A 10 V step voltage is applied across a RC series circuit at t = 0. The value of $\left.\frac{di}{dt}\right|_{t=0^+}$ will be _____ (A/sec).

[Assume R = 100Ω, C = 100 μF

Q.113 For the circuit shown below initially switch is at 1, at t = 0 switch is moved to 2. Then the value of $i(0^+)$ is _____ A.



Q.114 Initially, Switch S is at x. At t = 0, the switch is shifted to y, the voltage at t > 0 across AB is [Assume zero initial condition for L_2]



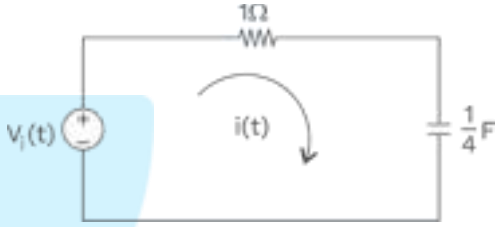
(A) $V_{A-B}(t) = 20e^{-2t}$ V

(B) $V_{A-B}(t) = 10e^{-4t}$ V

(C) $V_{A-B}(t) = 30e^{-2t}$ V

(D) $V_{A-B}(t) = 40e^{-2t}$ V

Q.115 Assuming $V_i(t) = tu(t)$, i(t) of the RC circuit given below will be-



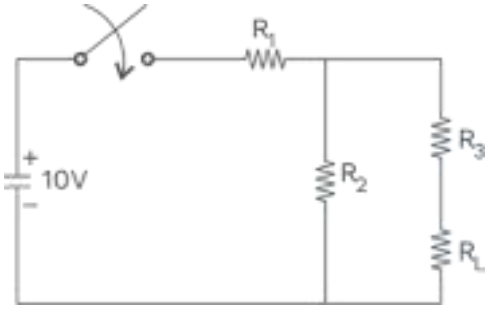
(A) $i(t) = u(t) - \frac{1}{4}e^{-4t}u(t)$

(B) $i(t) = \frac{1}{4}u(t) - \frac{1}{4}e^{-4t}u(t)$

(C) $i(t) = u(t) - e^{-4t}u(t)$

(D) $i(t) = u(t) - \frac{1}{2}e^{-4t}u(t)$

Q.116 Find the expression of discharging voltage of the capacitor (C = 1F) at $t = 0^+$ following switching at t = 0. $R_3 = R_L = 3\Omega$, $R_2 = 6\Omega$, $R_1 = 1\Omega$.

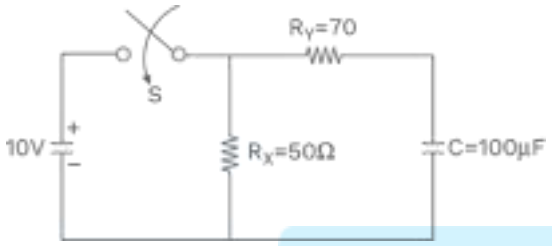


(A) $12e^{\frac{-t}{6}}$ V (B) $10e^{\frac{-t}{4}}$ V

(C) $12e^{\frac{-t}{4}}$ V (D) $10e^{\frac{-t}{6}}$ V

Q.117 Time taken by a capacitor of 1 μF in series with a 1 MΩ resistance to be charged up to 80% of the final value is _____ (sec.)

Q.118 In the figure given below, the switch K is closed at t = 0. The time, when the current from the battery reaches to 300 mA is _____ (ms).

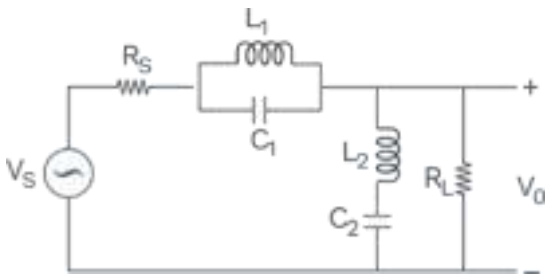


Q.119 In a series RC circuit, the values of R = 10Ω and C = 25 nF. A sinusoidal voltage of 50 MHz is applied and maximum voltage across the capacitor is 2.5 V. Find the maximum voltage across the series combination?

(A) 125 V (B) 196.2 V

(C) 188.5 V (D) 119.21 V

Q.120 The circuit shown below represents a-



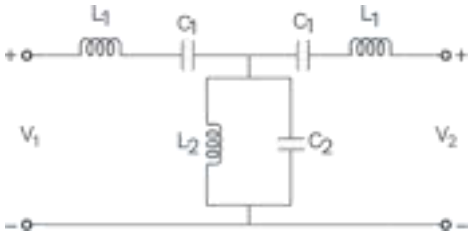
(A) Low pass filter (LPF)

(B) High pass filter (HPF)

(C) Band pass filter (BPF)

(D) Band stop filter (BSF)

Q.121 The circuit given below will act as-



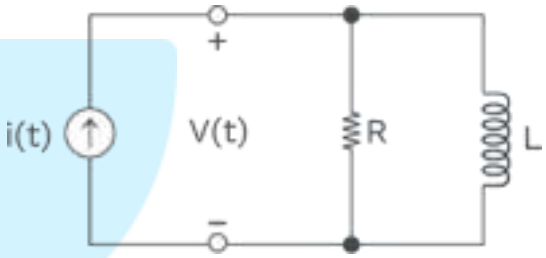
(A) Low pass filter

(B) High pass filter

(C) Band pass filter

(D) Band stop filter

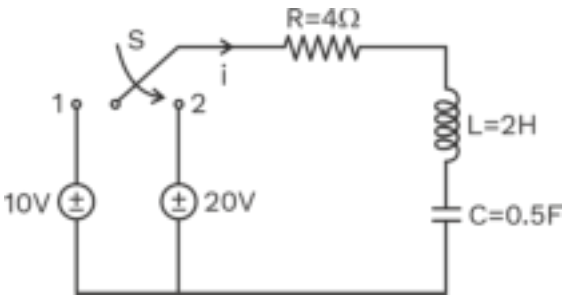
Q.122 What is the average power delivered to the L in the following circuit, if i(t) = 1.25 cos (5t – 15)A, R = 20Ω and L = 3H?



(A) 9.375 W (B) 25.63 W

(C) 0 W (D) 9.375 W

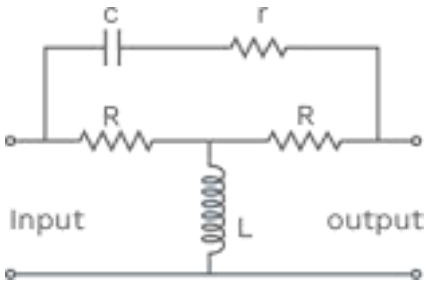
Q.123 If the switch is at position '1' for a long time and at (t = 0) it is moved to position '2', then the current i(t) for t > 0 will be



(A) $(5e^{-t} + 25e^{t})$A (B) $(5 + 25te^{-t})$A

(C) $(5te^{-t})$A (D) $(25te^{-t} + 5)$A

Q.124 Which type of filter circuit is shown in the figure given below.



(A) Low pass filter

(B) High pass filter

(C) Band pass filter

(D) Band stop filter

Q.125 Consider a circuit having zero at s = 1 and a pair of poles at s = 1 ± j1. If excited by a step voltage then what will be the current in the circuit ?

(A) Oscillatory with amplitude increasing with time

(B) Oscillatory with amplitude decreasing with time

(C) Non oscillatory and increasing with time

(D) Non oscillatory and decreasing with time

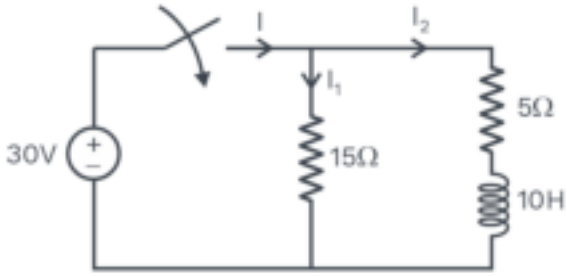
Q.126 A square pulse of 3V amplitude is applied to RC circuit as shown in figure. The capacitor is initially uncharged. Find the output voltage V_0 at time t = 2 sec.



(A) 3V (B) –3V

(C) 4V

Q.127 Find I2 in circuit shown below at t = 0.4 sec.



(A) 1.25 A (B) 1.17 A

(C) 1.90 A (D) 1.09 A

Q.128 A coil of resistance 10Ω and inductance 15H is suddenly connected to a dc supply of 45V. Find the initial rate of change of current.

(A) 3 A/sec (B) 1.67 A/sec

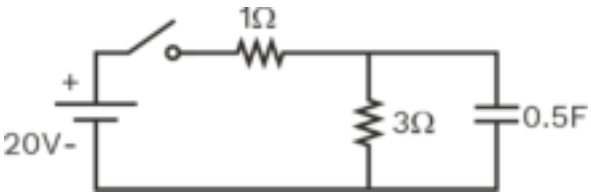
(C) 2.67 A/sec (D) 3.67 A/sec

Q.129 A tungsten filament bulb rated at 500 W, 110 V is to be connected in series with a capacitance across 220 V, 50 Hz supply. Find the value of capacitor such that the voltage across the bulb and the power consumed by the bulb are according to the rating of the bulb.

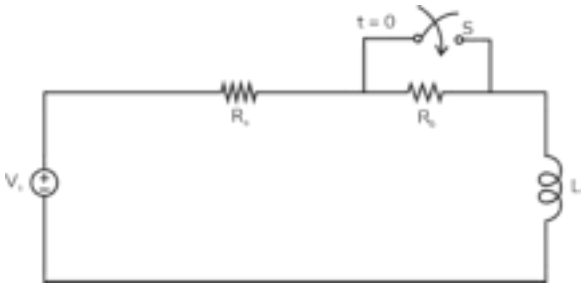
(A) 85.85 μF (B) 81 μF

(C) 79.85 μF (D) 75.85 μF

Q.130 The time constant of the network shown in figure is_____



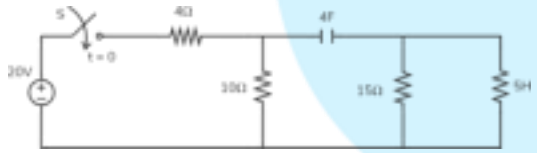
Q.131 Determine the rate of change of current $\frac{di}{dt}(0^+)$ in the network shown in figure. Consider switch is closed at t = 0.



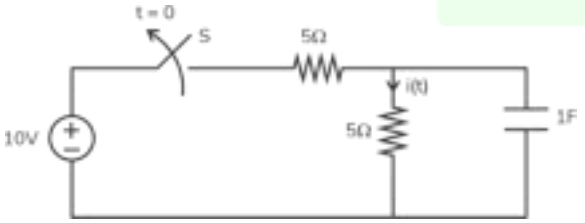
(A) $\frac{V_s}{R_a}$ (B) ∞

(C) $\frac{V_s}{L}\left(\frac{R_b}{R_a+R_b}\right)$ (D) $\frac{V_s}{L}\left(\frac{R_a}{R_a+R_b}\right)$

Q.132 Consider the circuit shown in figure, the with S is closed at time t = 0. The voltage across the inductor just after the switch is closed. $(\text{i.e. at } t = 0^+)$will be_____



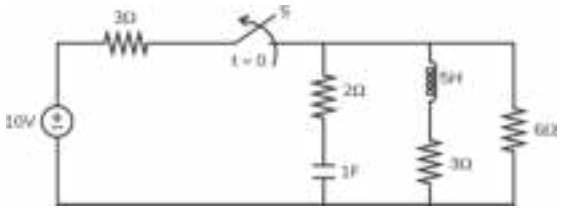
Q.133 The network shown in figure, the switch S has been closed for a long time. The current across the 5Ω resistor i(t) at time t = t when the switch is opened will be.



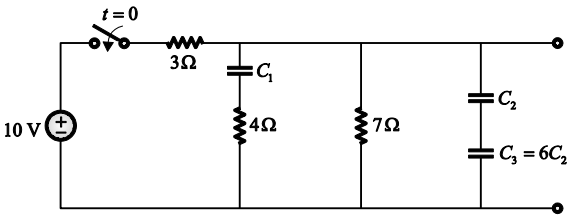
(A) u(t) (B) $5e^{-5t}u(t)$

(C) 3u(t) (D) $e^{-5t}u(t)$

Q.134 In the given figure, the switch S has been closed for a very long time and is opened at t = 0. The voltage across the capacitor at $t = 0^+$ will be_____.

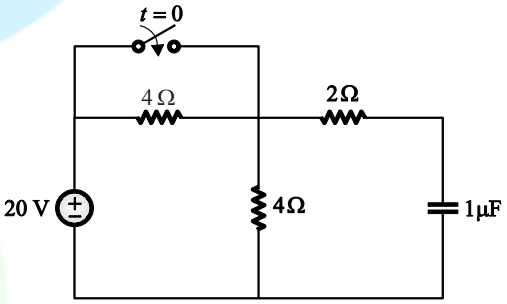


Q.135 In the circuit shown in the figure below, the switch is closed at $t = 0$.



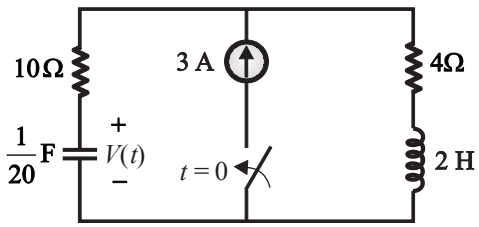
The ratio of energies stored in capacitor C_3 to that of C_2 in steady state is _____. (rounded upto three decimal places)

Q.136 Consider the circuit shown in the figure below.



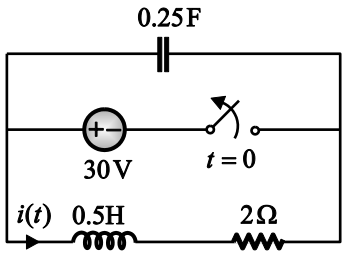
If the switch is closed at $t = 0$, then the value of $\frac{dV_c(0^+)}{dt}$ is_____ $\times 10^6$ V/sec. (in integer

Q.137 Consider the network shown in the figure below.



The value of voltage $V(t)$ (in V) at $t = 0.2\,\text{sec}$ is _____. (correct up to three decimal places)

Q.138 Consider the circuit shown in figure below. The switch was initially closed and opened at $t = 0$. The value of current i(t) for $t \geq 0$ is



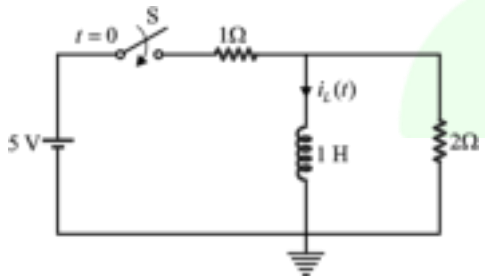
(A) $15e^{-2t}\cos 2t$

(B) $30e^{-2t}\cos 4t$

(C) $15(e^{-2t}\cos 4t + e^{-2t}\sin 4t)$

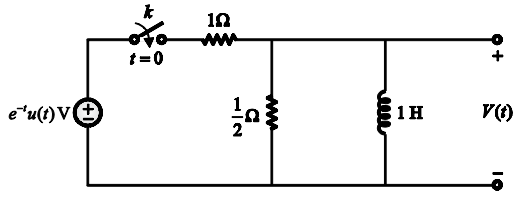
(D) $15(e^{-2t}\cos 2t + e^{-2t}\sin 2t)$

Q.139 In the circuit shown in figure switch *S* has been open for a long time. At $t = 0$, switch is closed.

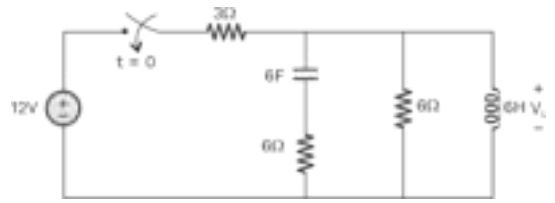


The value of $i_L''(0^+)$ is _____ A/sec^2. (rounded upto two decimal places)

Q.140 For the circuit shown in below figure, switch *k* is closed at $t = 0$ .The value of voltage $V(t)$ at t = 3 sec is _____ V. (correct upto two decimal place)



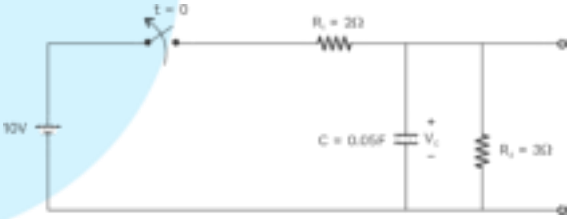
Q.141 For the circuit shown in the figure below, switch is closed at t = 0. The voltage V_L across the inductor at $t = 0^+$ is-



(A) -3V (B) 0V

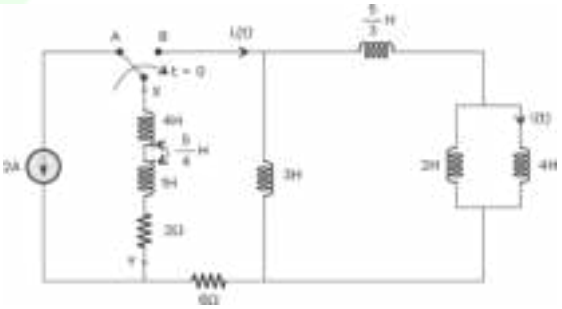
(C) 3V (D) 6V

Q.142 Circuit is shown in the figure below.



The switch is closed for t < 0. The value of $\left.\frac{d^2V_c}{dt^2}\right|_{t=0^+}$ is_____ V/s^2.

Q.143 A network is shown below.



Which of the following statements is/are true MSQ

(A) The current i(t) at t = 1 sec is 3.81 mA

(B) The current i(t) at t = 1 sec = 28.19 mA

(C) The equivalent inductance in branch X – Y is $\frac{5}{2}$H

(D) The current $i_1(t) = \frac{5}{4}e^{-2t}u(t)$

Q.144 A circuit is shown in the circuit below. The switch 1 and 2 are open t=0, switch 1 is closed and switch 2 is open t = 4, switch 1 and 2 are closed



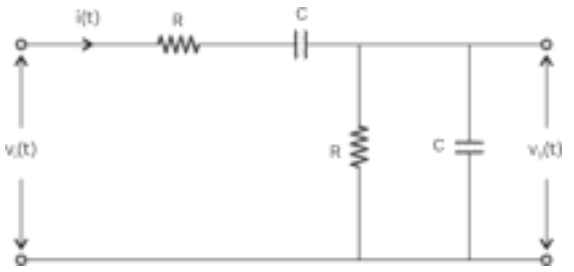
The current i(t) is

(A) 1.5 – 1.5 $e^{-2.5t}$ (B) 3.75 – 1.5 e^{-t}

(C) 3.75 – 2.25 $e^{-(t-4)}$ (D) 1.5 – 2.25 $e^{(t-4)}$

Q.145 The step voltage of 50V is applied to a coil having R = 5Ω in series with L = 10H. The time taken by the circuit current to reach half of its final value is_____sec. (upto 3 decimal places)

Q.146 Find the transfer function of the given network for R = 1Ω and C = 1H.



(A) $\frac{1}{s^2+2s+1}$ (B) $\frac{s}{s^2+2s+1}$

(C) $\frac{1}{s^2+3s+1}$ (D) $\frac{s}{s^2+3s+1}$

Q.147 There are no transients in pure resistance circuits because they-

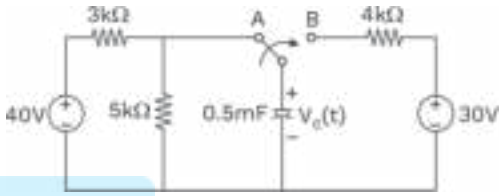
(A) are linear circuit

(B) have no stored energy

(C) obey Ohm's law

(D) offer high resistance

Q.148 For the circuit shown in figure below, the switch has been position A for a long time. At t = 0, the switch is moved to B. Then, the capacitor voltage $V_c(t)$ for t>0 is _____.



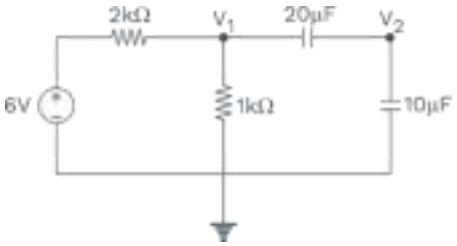
(A) $V_c(t) = (24 - 4e^{-2t})$ V

(B) $V_c(t) = (30 - 15\ e^{-0.5t})$V

(C) $V_c(t) = (6 - 6\ e^{-2t})$V

(D) $V_c(t) = (30 - 5\ e^{-0.5t})$V

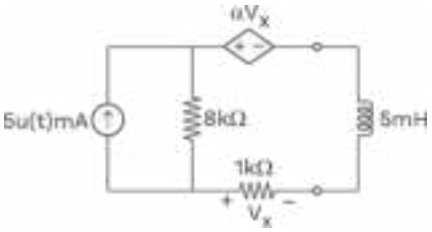
Q.149 In the DC circuit shown in the adjoined figure, the node voltage V_2 at steady state is-



(A) 0 V (B) $\frac{4}{3}$V

(C) $\frac{2}{3}$V (D) 1 V

Q.150 Let u(t) denotes a unit step function. Given circuit will be stable for-

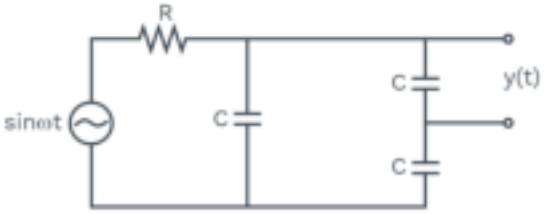


(A) $\alpha > 9$ (B) $\alpha = 9$

(C) $\alpha < 9$ (D) $\alpha > 11$

6 Steady State Analysis

Q.1 The steady state value of y(t) = A sin(ωt + Ø). If the amplitude (A) = 0.5, then the frequency of circuit in rad/sec. is _____.

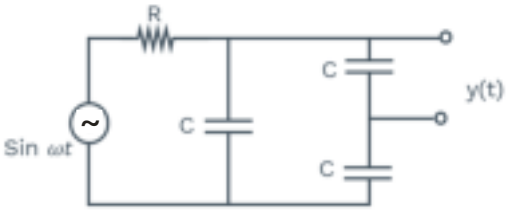


If R = 1Ω, C = 2F

(A) $\frac{1}{\sqrt{3}}$ rad / sec. (B) $\sqrt{3}$ rad / sec.

(C) $\sqrt{2}$ rad / sec. (D) $\frac{1}{\sqrt{2}}$ rad / sec.

Q.2 The steady state value of $y(t) = A\sin(\omega t + \phi)$. If the amplitude (A) = 0.5, then the frequency of circuit in rad./sec. is____



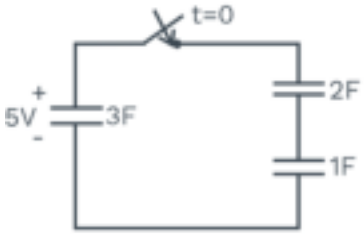
If R = Ω, C = 2F

(A) $\sqrt{3}$ (B) $\sqrt{2}$

(C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{2}}$

Q.3 In the circuit shown the steady state voltage across the 1F capacitor is –

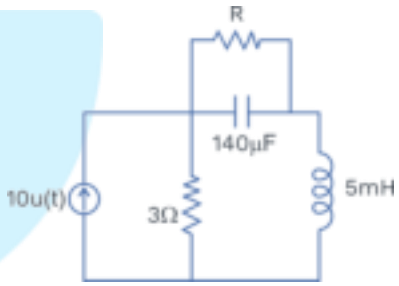
If initial voltage across 3F capacitor is 5 volt.



(A) 3V (B) $\frac{30}{11}$ V

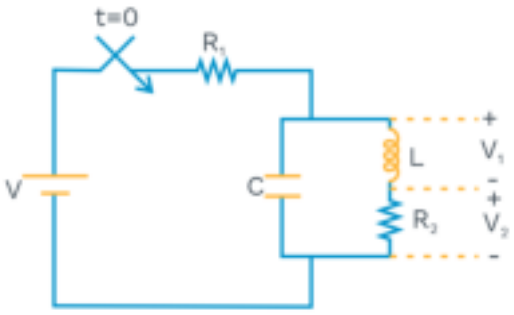
(C) $\frac{36}{11}$ V (D) $\frac{18}{11}$ V

Q.4 Consider the circuit given below:



If energy stored in capacitor and inductor is same in steady state condition, then value of R is

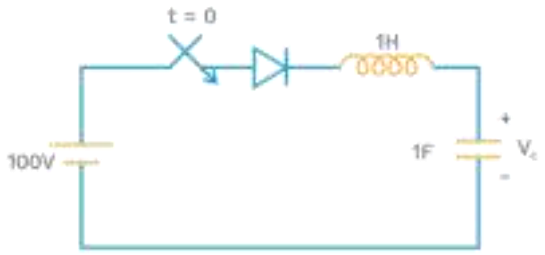
Q.5 The switch is opened for long time and it is closed at t = 0. Find value of $V_1(\infty) - V_2(\infty)$ -



(A) 0 (B) $-V\frac{R_2}{R_1+R_2}$

(C) $V\cdot\frac{R_2}{R_1+R_2}$ (D) $V\cdot\frac{R_1}{R_1+R_2}$

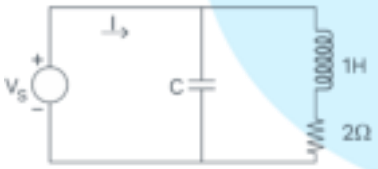
Q.6 The switch is opened for long time & it is closed at t = 0. Find the steady state voltage across the capacitor.



(A) 100V (B) 200V

(C) 0V (D) -100V

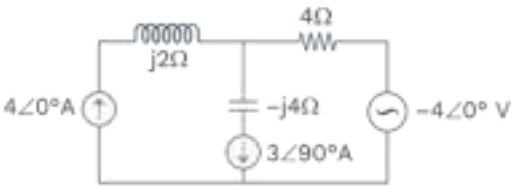
Q.7 In the below circuit, V_S = sin2t. C is chosen so that I and V_s are in phase. (I is the current drawn from V_s). The value of C is given by-



(A) $\frac{1}{2}$F (B) $\frac{1}{4}$F

(C) $\frac{1}{8}$F (D) $\frac{\sqrt{2}}{8}$F

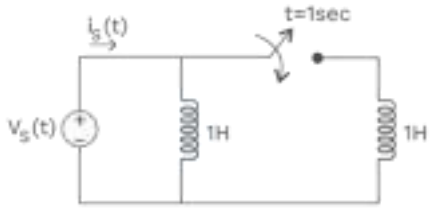
Q.8 Determine the power delivered by $3\angle 90°$ source in figure shown below-



(A) 18 W (B) 9 W

(C) 27 W (D) 36 W

Q.9 In the circuit shown below-



The source voltage $V_s(t)$ = cos (t) V for t ≥ 0 and 0 otherwise. The input current i_s (t) for t ≥ 1 is-

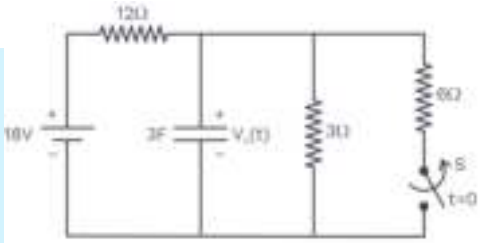
(A) 1 + sin (2t) A

(B) 1 + 2 sin (t – 1)A

(C) 2 sin (1) – sin (t) A

(D) 2 sin (t) – sin (1) A

Q.10

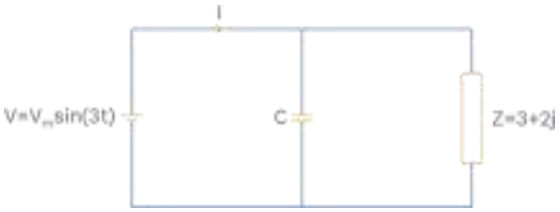


For the above circuit the switch is closed for a long time.

The value of $V_c(t)$ at t → 0^-, switch is open at t = 0_____V

(Up to two decimal places)

Q.11 A circuit is shown in the figure below.



If the current (I) is in phase with the voltage (V), then the value of capacitor (C) is _____ F. (upto three decimal places)

Q.12 For the circuit given below, the magnitude of the currents are I = 18A, I_R = 14A and I_L = 10A.



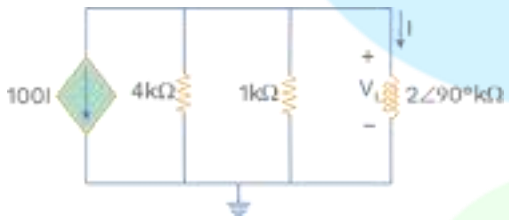
The power factor of the load is _____. (upto 2 decimal places)

Q.13 A circuit is shown in the figure below.



The value of capacitance for which the power factor is unity for the AC source is _____ μF. (upto 2 decimal places)

Q.14 For the given circuit, the voltage across the inductor for I = 1 mA is-



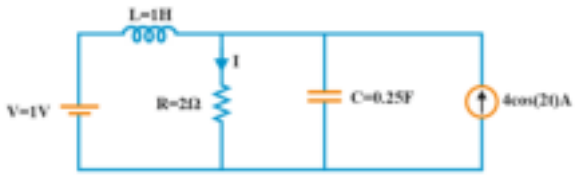
(A) V_L = −74.28∠21.80° volts

(B) V_L = 93.37∠84.26° volts

(C) V_L = −67.38∠29.72° volts

(D) V_L = 34.82∠44.40° volts

Q.15 For the circuit shown in the figure below, the current through 2Ω resistor is –



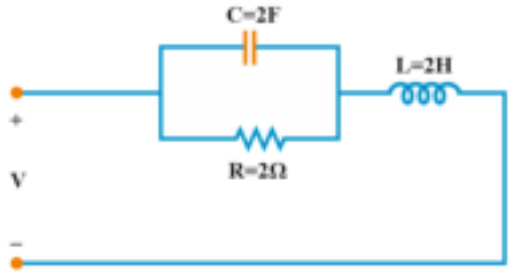
(A) [1 + 4 cos(2t)]A

(B) 0A

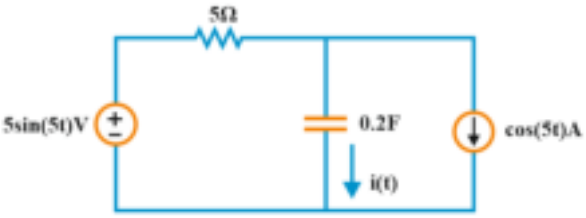
(C) 4.03A

(D) [0.5 + 4 cos(2t)] A

Q.16 The input frequency of the given circuit when the power factor is unity is _____ rad/sec. (Up to two decimal places).



Q.17 For the circuit shown below, which of the following statement(s) is/are true? MSQ



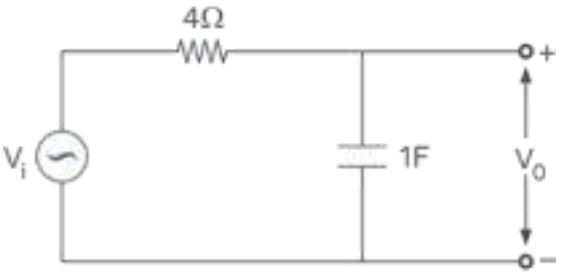
(A) The average power delivered by the two sources is 2.4W

(B) The average power delivered by the two sources is 4.8W

(C) The current i(t) through the capacitor is $1.38\sin\left(5t - 33.7^{\circ}\right)$ A

(D) The current i(t) through the capacitor is $\sin\left(5t + 33.7^{\circ}\right)$ A

Q.18 Determine the steady state output voltage (V_0) in the given circuit, if the input voltage is 4 + 3 cos (0.25t).



(A) 4 + 2.12 cos (0.25t – 45°)

(B) 1 + 2.12 cos (0.25t – 45°)

(C) 4 + 2.12 cos (0.25t – 14.5°)

(D) 4 + 2.12 cos (0.25t – 76°)

Q.19 If a sinusoidal wave has frequency of 50 Hz with 30 A r.m.s. current which of the following equation represents this wave?

(A) 60 sin 25t (B) 42.42 sin 314t

(C) 30 sin 50t (D) 84.84 sin 25t

Q.20 Calculate the time constant of RC circuit shown below-



(A) 25 msec (B) 20 msec

(C) 12 msec (D) 50 msec

Q.21 A voltage $v(t) = 6e^{-2t}$ is applied at t = 0 to a series RL circuit with L = 1H. If $i(t) = 6(e^{-2t} - e^{-3t})u(t)$ then R will have a value of _____.

(A) 0.663 Ω (B) 1 Ω

(C) 3 Ω (D) 0.333 Ω

Q.22 A current impulse 10δ (t) is forced through a capacitor C. The voltage across the capacitor will be -

(A) 10t (B) 10u(t) – C

(C) $\frac{5}{C}t$ (D) $\frac{10u(t)}{C}$

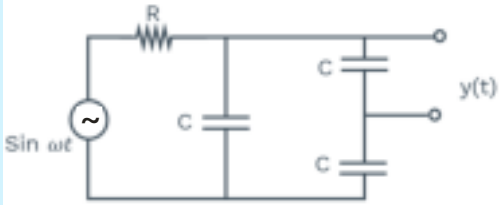
Q.23 In the network below if Z_L = 4 + j7.86Ω then determine the value of the capacitance that will make the power factor to be unity when placed in series with Z_L



(A) 337.4 μF (B) 377.5 μF

(C) 342.6 μF (D) 300.8 μF

Q.24 The steady state value of $y(t) = A\sin(\omega t + \phi)$. If the amplitude (A) = 0.5, then the frequency of circuit in rad./sec. is____

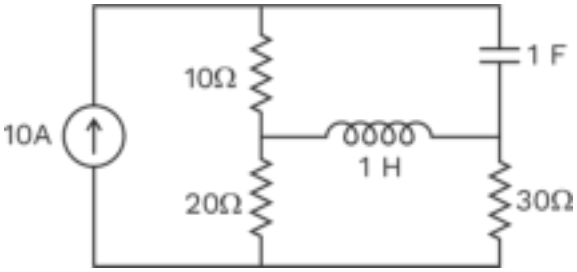


If R = 1 Ω, C = 2F

(A) $\sqrt{3}$ (B) $\sqrt{2}$

(C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{2}}$

Q.25 Total energy stored in the network given below at steady state is-



(A) 5 kJ (B) 7 kJ

(C) 8 kJ (D) 10 kJ

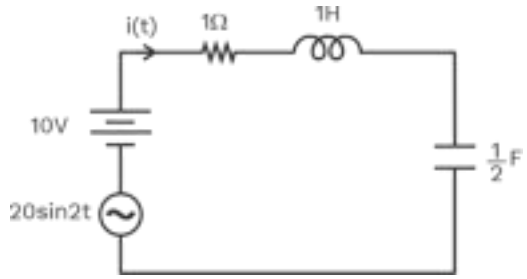
Q.26 An RLC series circuit has a resistance R of 20 Ω and a current which lags behind the applied voltage by 45°. If the voltage across

the inductor is twice the voltage across the capacitor, the value of inductive reactance is:

(A) 10 Ω (B) 20 Ω

(C) 40 Ω (D) 60 Ω

Q.27 In the following circuit, find the value of i(t) under steady state is _____



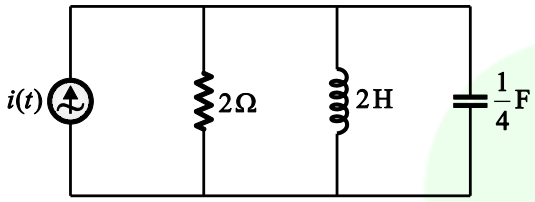
(A) 14.14 sin (2t-450)

(B) Zero

(C) 10

(D) 14.14 sin2t

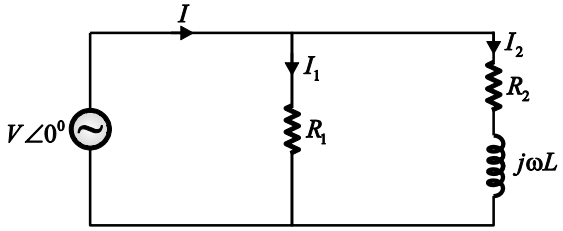
Q.28 The network is shown in below figure.



Where, $i(t) = 12\sin 100\pi t$ A

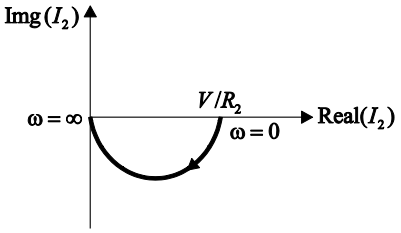
The difference of higher 3 dB frequency and lower 3 dB frequency (in rad /sec) is _____. (in integer)

Q.29 Consider the circuit shown in the figure below. The circuit is supplied with a variable frequency source.

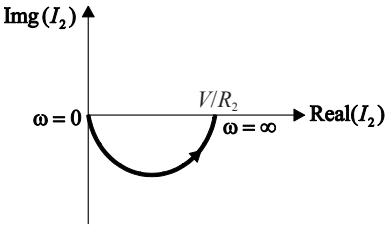


The locus diagram of the current I_2 is-

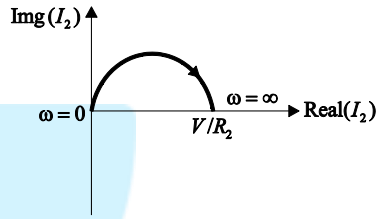
(A)



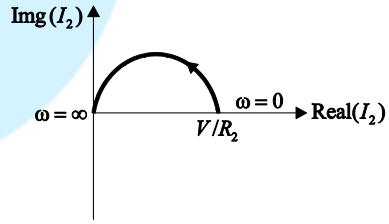
(B)



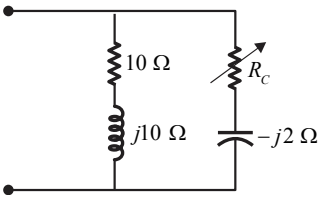
(C)



(D)



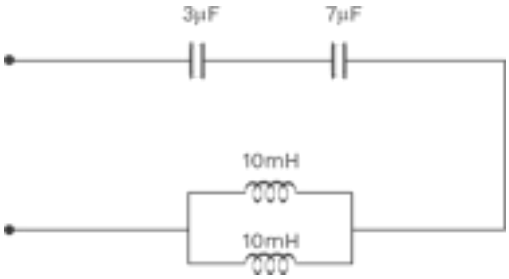
Q.30 The circuit shown in below figure is under resonance. The value of resistance R_C (in Ω) is _____. (in integer)



Q.31 In a series RL circuit, the inductance is 35mH and the magnitude of total impedance is 19.62 Ω. If the angle of lag of the input current from the applied voltage is 67.8°, then the value of angular frequency will be equal to

_____ rad/sec (up to two decimal places).

Q.32 Consider the circuit shown below.



The resonant frequency of the circuit is-

(A) 987 Hz (B) 1.553 kHz

(C) 5.683 kHz (D) 9.759 kHz

Q.33 Four wires 1, 2, 3 and 4 are connected at a point 'O'. The current in the first three wires is as:

$$i_1 = 6\sin\left(\omega t + \frac{\pi}{3}\right)A$$

$$i_2 = 5\sin\left(\omega t + \frac{\pi}{6}\right)A$$

$$i_3 = 3\sin\left(\omega t + \frac{\pi}{4}\right)A$$

If the fourth wire carries the resultant current, then the resultant current in polar form is-

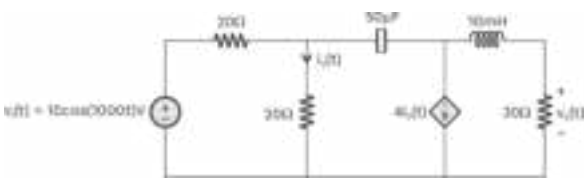
(A) $i_4 = (19.26\angle 43.9°)A$

(B) $i_4 = (19.26\angle 46.1°)A$

(C) $i_4 = (4.38\angle 46.1°)A$

(D) $i_4 = (4.38\angle 43.9°)A$

Q.34 For the given network



Find the value of output voltage $v_0(t)$.

(A) 6.154 cos(1000t + 72.26)V

(B) 6.154 cos(1000t - 72.26)V

(C) 0.362 cos(1000t + 72.26)V

(D) 0.362 cos(1000t - 72.26)V

Q.35 A coil of resistance 24 Ω has a reactance 32Ω when connected across a supply voltage V_s = 566 sin(126t). The value of supply current is-

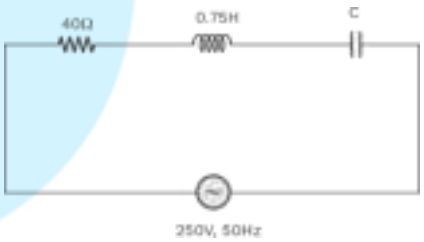
(A) 7.07sin(126t + 0.927°)

(B) 14.14sin(126t + 0.927°)

(C) 14.14sin(126t - 0.927°)

(D) 7.07sin(126t - 0.927°)

Q.36 A series RLC circuit is shown below



If the resonant frequency is 55Hz, then the voltage across the coil is _____V

Q.37 A relay coil is connected to a supply of 250V, 50Hz. If it has R = 30Ω and L = 0.5H, then which of the following statement(s) is/are correct? MSQ

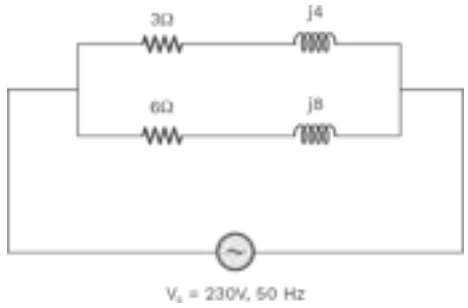
(A) Apparent power (S) = 275.6 VA

(B) Apparent Power (S) = 1.31 VA

(C) Power factor = 0.1876 (lagging)

(D) Power factor = 0.1876 (leading)

Q.38 For the circuit shown below

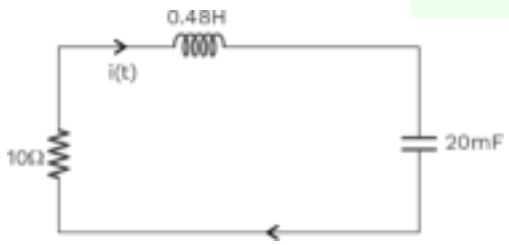


The power, drawn from the source is _____kW.

Q.39 A series RLC circuit consists of R = 16Ω, L = 5mH and C = 2μF. If it is desired to increase the bandwidth by 30% keeping the resonant frequency as fixed, then the value of quality factor is _____.

Q.40 The current i(t) in the circuit has its Laplace transform gives as, $I(s) = \frac{5(s+9)}{s(s+13)}$. The steady state value of the current is_____A. (up to two decimal places).

Q.41 A source for RLC circuit is shown below.



If the initial voltage is 60V, the initial current through the inductor is 1.8A. The current, i(t) is

(A) Overdamped

(B) Critically damped

(C) Under damped

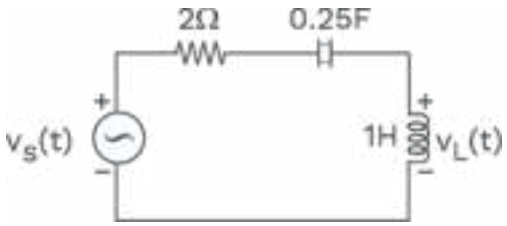
(D) Undamped

Q.42 What is the average power delivered to a network having the following voltage and current-

$v(t) = 100 \sin(\omega t + 30°)$

$i(t) = 40 \cos(\omega t - 30°)$

Q.43 The sinusoidal voltage applied to the circuit below is $v_s(t) = 10 \cos(2t)V$. The voltage $v_L(t)$ across the inductor is _____.



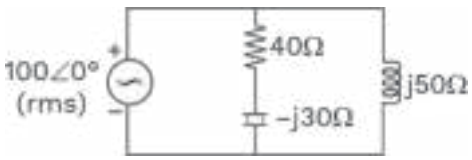
(A) –10 sin 2t V

(B) 10 cos 2t V

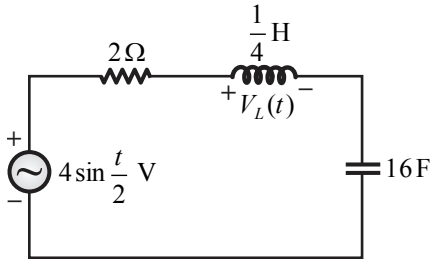
(C) 10 sin (2t – 45°)V

(D) 10 cos (2t – 45°)V

Q.44 For the network shown in the figure below, the apparent power supplied by the source is _____ VA.



Q.45 The inductor voltage $V_L(t)$ for the circuit shown below will be

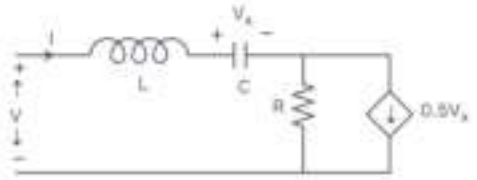


(A) $4 \cos \frac{t}{2}$ V (B) $\frac{1}{4} \cos \frac{t}{2}$ V

(C) $\frac{1}{4} \sin \frac{t}{2}$ V (D) $4 \sin \frac{t}{2}$ V

7 Resonance

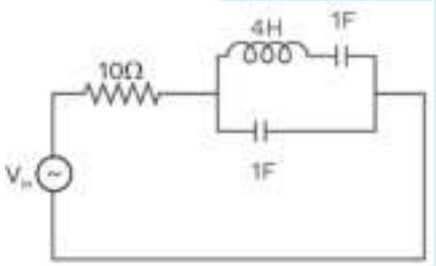
Q.1 Find the quality factor Q for the circuit given below – if R = 1Ω, L = 2H, C = 4F



(A) 1 (B) 0.5

(C) 1.414 (D) 0.707

Q.2



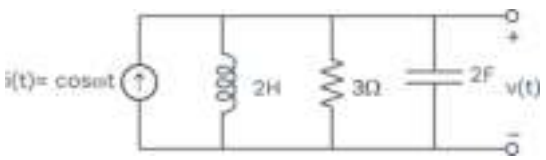
Determine the frequency in rad/sec. at which the current flowing through the circuit is minimum?

Q.4 A series RLC circuit operating different from resonance frequency. The operating frequency such that current leads the voltage and it is half the value of resonance current then, operating frequency of the circuit in rad/sec. is ....

(A) .707 (B) .5

(C) .456 (D) .58

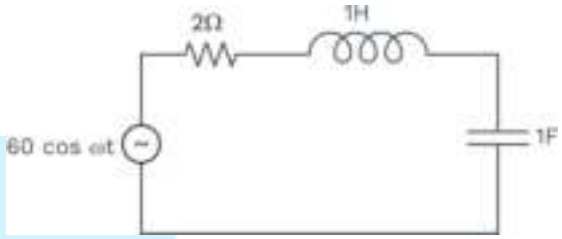
Q.4 For a parallel RLC network shown in below, the steady state output voltage v(t) for ω = 1 rad/Sec. is -



(A) sin t (B) 3 cos t

(C) cos t (D) $\sqrt{2}$ cos t

Q.5 What is the voltage across capacitor at resonance?

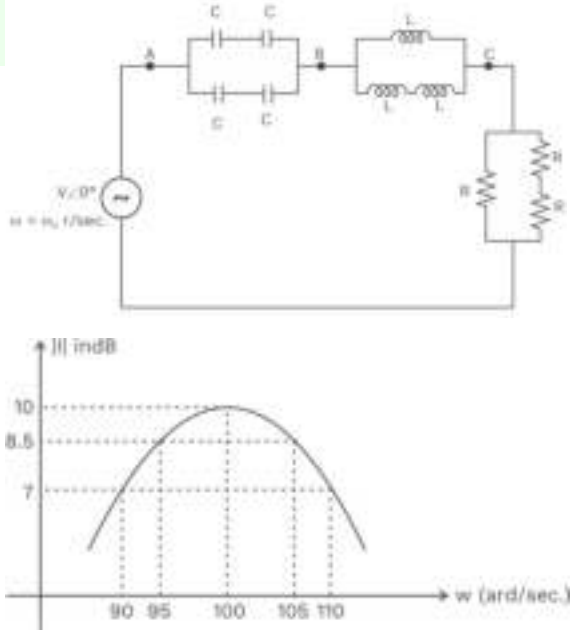


(A) 30∠90° (B) 3∠90°

(C) 30∠ –90° (D) 3∠ –90°

Q.6 Given that –

$|V_B - V_A| = |V_C - V_B| = 100$ volt and magnitude plot of current 'I' is given for the circuit is –



Determine the value of (a) Quality factor MSQ

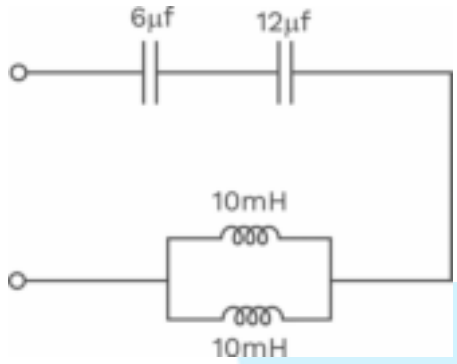
(b) The value of 'V' in volt

(A) 20 V (B) 10 V

(C) 5 (D) 15

Q.7 A series RLC circuit is excited with a 15V ac voltage source. Voltage across capacitor at ω = 3.65 K rad/Sec. is 500 volts. L = 15mH and C = 5 μf. find the value of R.....?

Q.8 Consider the circuit shown below



The resonant frequency of circuit will be in KHz .....?

(A) 2 (B) 1.89

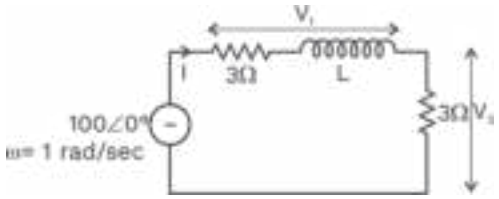
(C) 1.53 (D) 1.126

Q.9 The total steady state energy stores by the inductor and the capacitor together in a parallel RLC circuit at any instant at resonant frequency is given buy -

(A) $\frac{1}{2}\frac{L}{C}V_m^2$ (B) $\frac{1}{2}\frac{C}{L}V_m^2$

(C) $\frac{1}{2}CV_m^2$ (D) CV_m^2

Q.10 Consider the figure given as shown below:

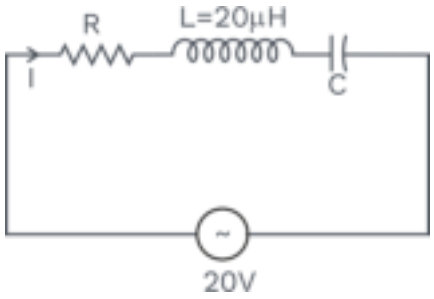


The value of L so that V_1 phasor leads V_2 phasor by 45° is:

(A) 4H (B) 7H

(C) 6H (D) 3H

Q.11 Consider the electric circuit given below:



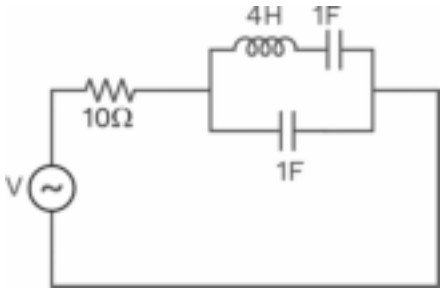
The quality factor is 20 and resonant frequency 20 kHz.

The value of current I at f = f_r where f_r = resonant frequency.

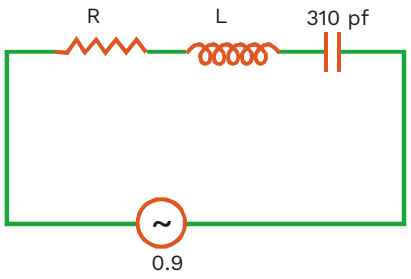
(A) πA (B) $\frac{500}{\pi}$ A

(C) 2πA (D) 4πA

Q.12 In the below circuit, determine the frequency in radian per second at which current is maximum or impedance is minimum.....?



Q.13 For the circuit shown in the figure if quality factor 100 and resonant frequency (f_o) 185 kHz is given then the value of circuit current at f = f_0 is ____ in (mA).



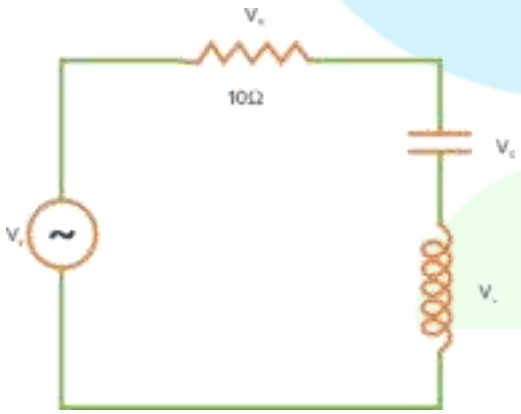
Q.14 A parallel RLC circuit has R = 2 KΩ, L = 220 mH, C = 50 μf the quality factor of the circuit at resonance is -

Q.15 In a series RLC circuit which is excited by a 30V sinusoidal voltage source of variable frequency the circuit exhibits resonance at 150 Hz and has 3 dB bandwidth of 15 Hz the magnitude of voltage across capacitor at resonance is -

(A) 30 V (B) 200 V

(C) 300 V (D) 100 V

Q.16 In the given circuit the magnitude of V_L, V_C are twice that of V_R. Given that f = 50 Hz the inductance of coil is –

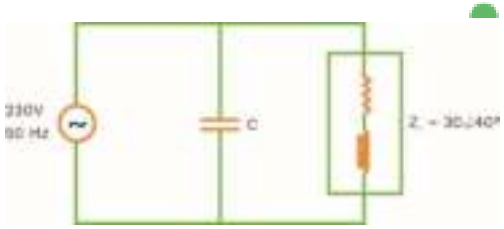


Q.17 A first order low pass filter is given with R = 60 Ω and C = 6 μf. What is frequency at which the gain of voltage transfer function of the filter is 0.30.

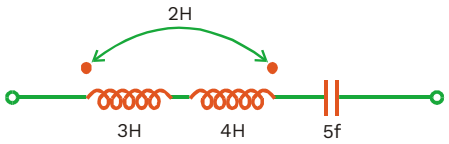
(A) 4.92 kHz (B) 24.6 kHz

(C) 2.46 kHz (D) 1.40 kHz

Q.18 In the circuit shown in fig. what value of C will cause a unity power factor at the 230 V source..... in (μf).



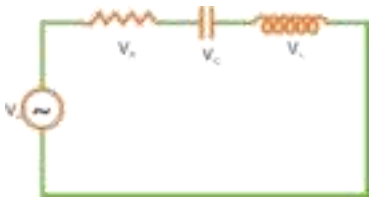
Q.19 The resonant frequency of the given series circuit is –



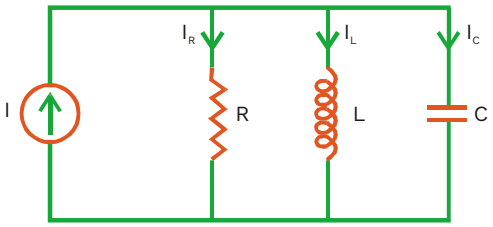
(A) $\frac{1}{2\pi\sqrt{3}}$ (B) $\frac{1}{2\pi\sqrt{5}}$

(C) $\frac{1}{4\pi\sqrt{15}}$ (D) $\frac{1}{2\pi\sqrt{15}}$

Q.20 In a AC series RLC circuit, the voltage across R and L is 25V. Voltage across L & C is 12V and voltage across source is 15V. What is voltage across capacitor C?



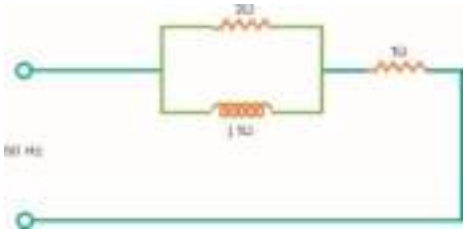
Q.21 The circuit is at resonance which of the following relation is true MSQ



(A) $|I_R| < I$ (B) $|I_L + I_C| > I$

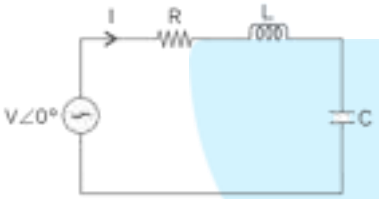
(C) $|I_R + I_L| < I$ (D) $|I_R + I_C| > I$

Q.22 Consider the circuit shown in the figure below



In order that the input current is in phase with the voltage a capacitor should be inserted in the series in the circuit. The value of this capacitor will be____mF.

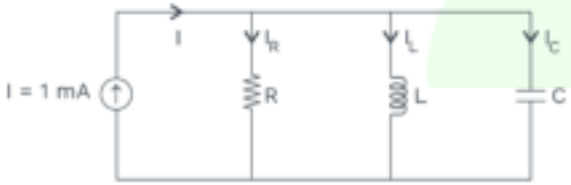
Q.23 The resistance of the series RLC circuit is doubled and inductance is halved, then bandwidth becomes-



(A) Half (B) Double

(C) No change (D) Four times

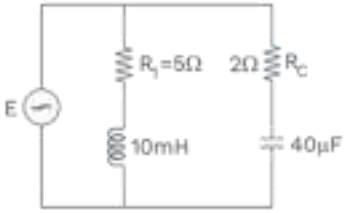
Q.24 The circuit shown in figure is at resonance, which of the following reactions is true?



(A) $|I_R| < 1$ mA (B) $|I_R + I_C| > 1$ mA

(C) $|I_L + I_C| > 1$ mA (D) $|I_R + I_L| < 1$ mA

Q.25 Find the resonant frequency-



(A) 50 Hz (B) 100 Hz

(C) 240 Hz (D) 120 Hz

Q.26 In a series RLC circuit with frequency of resonance as f_0, voltage across resistor, capacitor and inductor are V_R, V_C and V_L respectively. Then at any frequency, 'f' which of the following is true- MSQ

(A) $|V_C| > |V_L|$ if $f > f_0$

(B) $|V_C| > |V_L|$ if $f < f_0$

(C) V_R leads source voltage $f < f_0$

(D) $|V_L| = |V_C| = |V_R|$ if $f = f_0$

Q.27 Consider a series R-L-C circuit at lower half power frequency.

(A) Current leads the applied voltage.

(B) Current lags the applied voltage.

(C) Current is in phase with applied voltage

(D) Current is out of phase by 180° with applied voltage.

Q.28 A series RLC circuit is supplied by 220V, 50 Hz. At resonance, the voltage across the capacitor = 550 V, I = 1A, determine R, L and C-

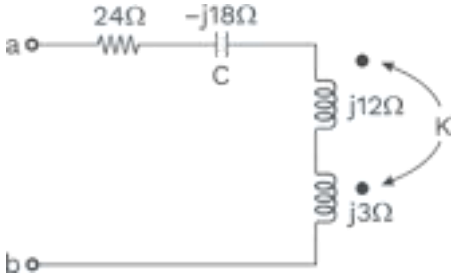
(A) R = 110 Ω, C = 3.63 μF, L = 1.75 H

(B) R = 220 Ω, C = 3.63 μF, L = 0.75 H

(C) R = 220 Ω, C = 5.75 μF, L = 1.75 H

(D) R = 110 Ω, C = 5.75 μF, L = 0.75 H

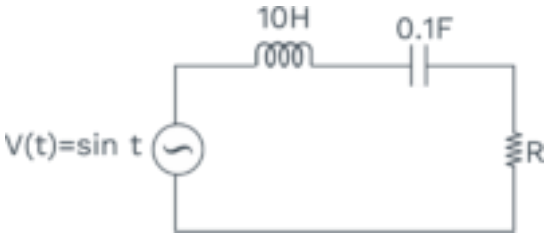
Q.29 In series circuit at resonance, find the value of coupling coefficient-



(A) zero (B) 1.0

(C) 0.5 (D) 0.25

Q.30 In the figure shown below, the peak value of voltage across R will be-



(A) 0.707 (B) 1

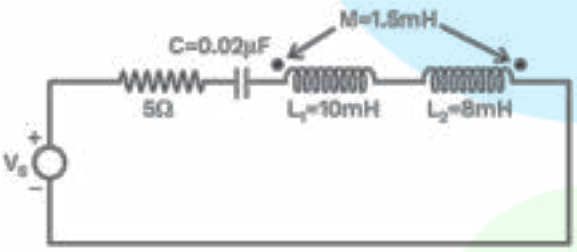
(C) 1.414 (D) Depends on R

Q.31 The response of a parallel circuit resonance is to magnify-

(A) Power (B) Current

(C) Voltage (D) Frequency

Q.32 Determine the resonant frequency of the network given:

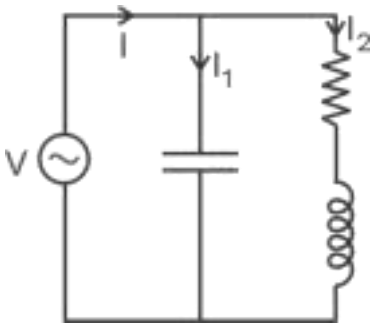


(A) 10.3 kHz (B) 5.67 kHz

(C) 9.18 kHz (D) 8.34 kHz

Q.33 A series RLC circuit has a resonance frequency $\Omega_0 = 10^4$ rad\sec, quality factor $\theta = 45$ and value of resistance R = 420Ω. Determine the value of C____(in PF).

Q.34 For the circuit shown, parallel resonance would occur when



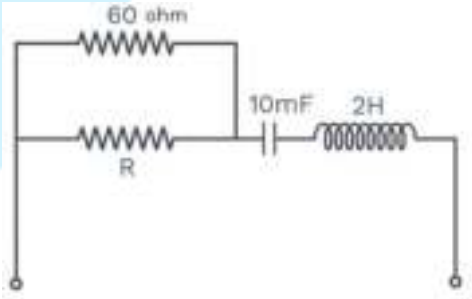
(A) Current I_1 is equal to I_2

(B) When current I is maximum

(C) When capacitor current I_1 is maximum

(d) When supply current I is in phase with V

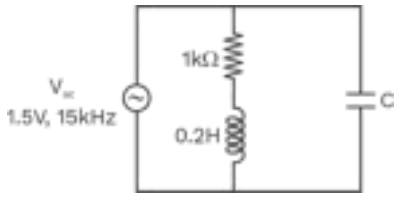
Q.35 Determine the value of R for critically damped network given below -



(A) R = 60Ω (B) R = 53.49Ω

(C) R = 47.28Ω (D) R = 32.34Ω

Q.36 For the circuit shown below, calculate C when the supply current is minimum.



(A) 888.9 pF (B) 672.4 pF

(C) 468.23 pF (D) 561.884 pF

Q.37 In a series R, L, C circuit with applied voltage of 10V, the response $i(t) = 3\,e^{-2t} - 3\,e^{-4t}$, then R, L, C value are respectively.

(A) 6 Ω, 8 H, 10 F

(B) 1.667 Ω, 10 H, 0.075 F

(C) 10 Ω, 1.667 H, 0.075 F

(D) 0.075 Ω, 10 H, 1.667 F

Q.38 In case of series RLC resonance circuit, what will be the total reactance at the lower half power frequency?

(A) R (B) $-R$

(C) $\sqrt{2}R\angle 45°$ (D) $\sqrt{2}R\angle -45°$

Q.39 Consider the following statements:

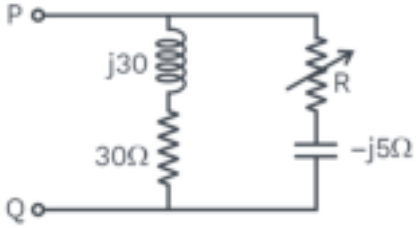
1. In case of critically damped circuit, the quality factor is $Q = \frac{1}{2}$.
2. The damping ratio decreases, if we increase the quality factor.
3. In a parallel resonance circuit, quality factor becomes half times when resistance is doubled.
4. In series resonance, the circuit behaves as voltage magnifier circuit.

Which of the above statements are correct?

(A) 1 and 3 only (B) 2, 3 and 4

(C) 1, 2, and 3 (D) 1, 2, and 4

Q.40 Find the value of resistance 'R' for which circuit is under resonance (in W).



(A) 17.85 W (B) 16.95 W

(C) 16.58 W (D) 17.2 W

Q.41 Consider the following statements:

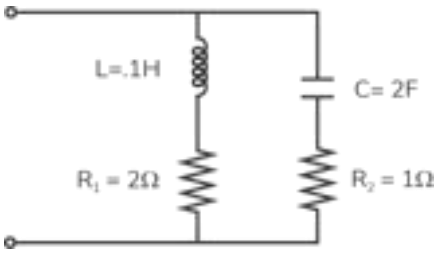
1. For a series RLC circuit, Impedance is minimum and current is maximum for resonance condition.
2. For a series RLC circuit, Impedance is maximum and current is minimum for resonance condition.
3. For a parallel RLC circuit, Impedance is maximum and current is minimum for resonance condition.
4. For a parallel RLC circuit, Impedance is minimum and current is maximum for resonance condition.

Which of these statements are correct?

(A) 1 and 3 only (B) 2 and 4 only

(C) 2 and 3 only (D) 1 and 4 only

Q.42 The frequency of resonance in the given figure will be.



(A) 5.46 rad/sec.

(B) 4.56 rad/s.

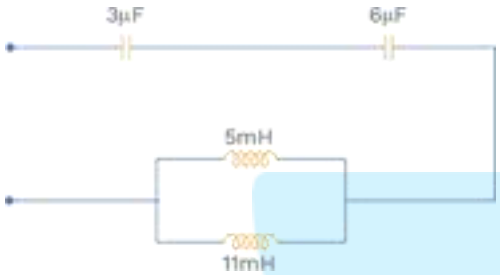
(C) 5 rad/sec.

(D) 4.92 rad/s

Q.43 A series RLC circuit consists of a resistance (R) = 2Ω in series with an

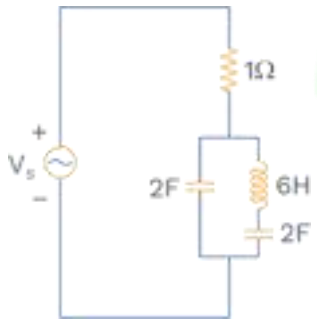
inductor (L) and capacitor (C) of values 25μH and 100□F respectively. The value of the inductor is changed to L_1 such that the resonant frequency is one-fourth of the original resonant frequency. The value of the inductor L_1 is _____ μH. (Rounded off to the nearest integer)

Q.44 Consider the circuit shown below.



The resonant frequency of the circuit is _____ kHz. (Upto 3 decimal places)

Q.45 The resonant frequency of the given circuit is _____rad/sec. (Up to two decimal places).

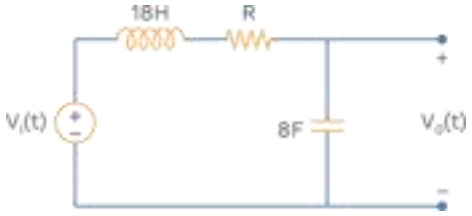


Q.46 An RLC series circuit has a resonant frequency of 3.2 MHz and abandwidth of 16 kHz. If the capacitor is 160 pF, then the effective resistance of the circuit is-

(A) 10.62Ω (B) 3.92Ω

(C) 6.25Ω (D) 1.55Ω

Q.47 A circuit is shown in the figure below.



If the input $V_i(t)$ to the circuit is unit step, then the minimum value of R(in ohms) for which there are no oscillations in the step response, is _____(Rounded off to the nearest integer)

Q.48 A series RLC resonant circuit has a resistance of 5Ω, an inductance of 50mH, and its quality factor is 50. Which of the following statement **NOT TRUE**?

(A) The bandwidth of the series RLC circuit is 100 rad/sec.

(B) The resonant frequency is 5000 rad/sec.

(C) The value of the capacitor is 0.5 μF.

(D) The ratio of half power frequencies is 0.980 or 1.02

Q.49 A coil has inductance (L) of 10mH along with the resistance of 'R' Ω. The coil is connected in series with a capacitor of 100μF. The circuit draws a power of 40 W when it is connected to a voltage source of $12\sqrt{2}\cos(1000t)$V. Which of the following statements is/are true?

MSQ

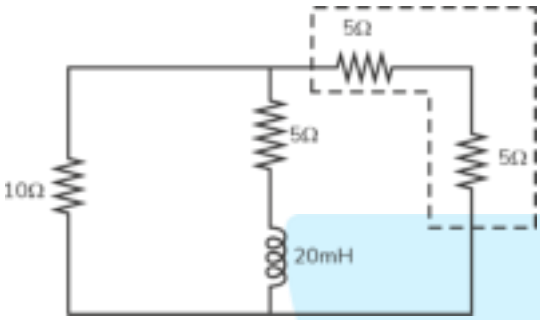
(A) The value of 'R' is 0.3Ω

(B) The value of R is 3.6Ω

(C) When the angular frequency is 1000rad / sec, the quality factor is 33.3.

(D) When the angular frequency is 1000rad / sec, the quality factor is 2.77.

Q.50 In the given R-L circuit, the time constant T = ___ ms.



Q.51 A series RLC circuit has a resonant frequency of 50 rad/sec and quality factor of 80. If the impedance at resonance is 10 Ω, then value of R, L and C is given by-

(A) 10 Ω, 2.5 mH, 160 mH

(B) 20 Ω, 32 H, 12.5 μF

(C) 10 Ω, 16 H, 25 μF

(D) 10 Ω, 62.5 mH, 6.4 mF

Q.52 For a parallel RLC circuit, which one of the following statement(s) is/are correct? MSQ

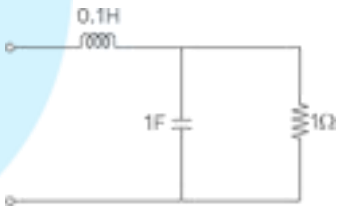
(A) The bandwidth of the circuit decreases if R is increased.

(B) The bandwidth of the circuit remains same if L is increased.

(C) At resonance, input impedance is a real quantity.

(D) At resonance, the magnitude of input impedance attains its minimum values.

Q.53 The resonance frequency for the given circuit will be-



(A) 3 rad/sec (B) 4 rad/sec

(C) 2 rad/sec (D) 1 rad/sec

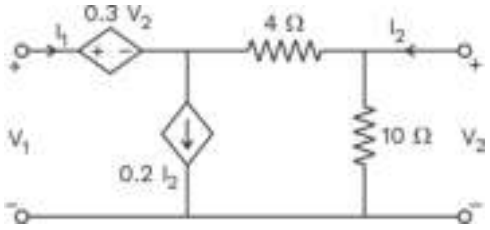
Q.54 For a series RLC circuit, the resonant frequency is 9 MHz and bandwidth is 81 KHz. If the value of capacitance is 159.15 pf, then the value of resistance is

(A) 1 Ω (B) 2 Ω

(C) 2.5 Ω (D) 5 Ω

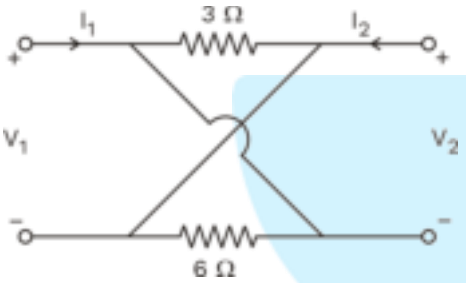
8 Two Port Network

Q.1 Find the value of |Z| for the given network –

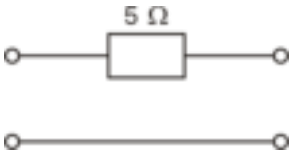


Q.2 Determine the value of Z_{11}?

_____(Ω)



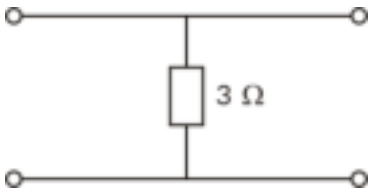
Q.3 Determine Y parameters for given network



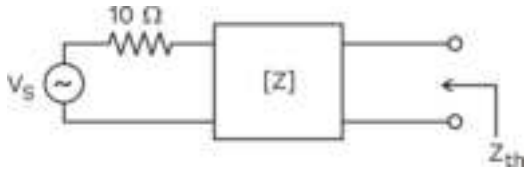
(A) $\begin{bmatrix} 0.2 & -0.2 \\ -0.2 & 0.2 \end{bmatrix}$ (B) $\begin{bmatrix} -0.2 & 0.2 \\ 0.2 & -0.2 \end{bmatrix}$

(C) $\begin{bmatrix} -0.2 & 0.3 \\ 0.3 & -0.1 \end{bmatrix}$ (D) $\begin{bmatrix} 0.2 & -0.3 \\ -0.3 & 0.1 \end{bmatrix}$

Q.4 Find the value of |Z| for the given network –



Q.5 Find the value of Z_{th} =?

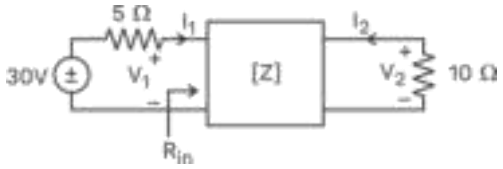


$$[Z] = \begin{bmatrix} 40 & 60 \\ 80 & 100 \end{bmatrix}$$

(A) 5 Ω (B) 4 Ω

(C) 2 Ω (D) 3 Ω

Q.6 Find the average power dissipated by R_2.

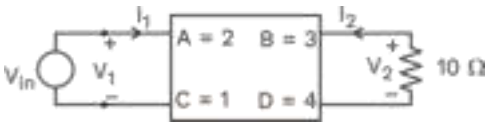


$$[Z] = \begin{bmatrix} 15 & 30 \\ 15 & 20 \end{bmatrix}$$

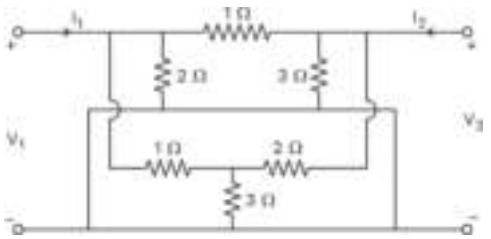
(A) 80 W (B) 70 W

(C) 75 W (D) 90 W

Q.7 Find the input resistance of the network $\frac{V_1}{I_1} = ?$

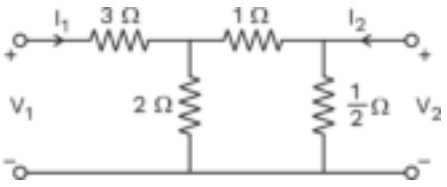


Q.8 A combination of two port network is shown in figure below–



For the combined network, the admittance parameter Y_{21} is _____ Ʊ.

Q.9 Which of the following statement is correct for the transmission parameter matrix of the entire network is –



(A) Symmetric but not reciprocal

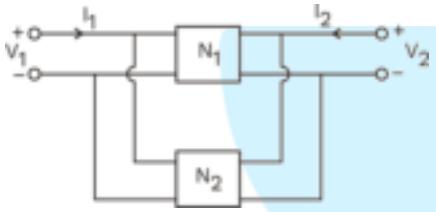
(B) Neither symmetric nor reciprocal

(C) Both symmetric and reciprocal

(D) Reciprocal but not symmetric

Q.10 If $[Y]_{N_1} = \begin{bmatrix} 3 & 2 \\ 1 & 3 \end{bmatrix}$ and $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = [Y]_{N_2}$

Then h-parameter matrix of the entire network is –



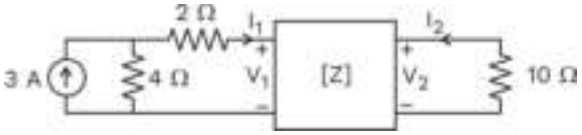
(A) $\begin{bmatrix} \frac{1}{5} & -\frac{3}{5} \\ \frac{2}{5} & \frac{19}{5} \end{bmatrix}$ (B) $\begin{bmatrix} \frac{19}{5} & \frac{3}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix}$

(C) $\begin{bmatrix} -\frac{1}{5} & \frac{3}{5} \\ \frac{2}{5} & -\frac{19}{5} \end{bmatrix}$ (D) $\begin{bmatrix} -\frac{19}{5} & \frac{3}{5} \\ \frac{2}{5} & -\frac{1}{5} \end{bmatrix}$

Q.11 For the circuit shown in figure let

$$[z] = \begin{bmatrix} 10 & -6 \\ -4 & 12 \end{bmatrix}$$

Find the value of V_1 and I_1? MSQ



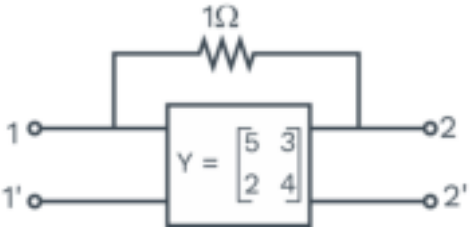
(A) I_1 is 0.1463 A (B) I_1 is 0.8049 A

(C) V_1 is 7.17 V (D) V_1 is –1.463 V

Q.12 The Y – parameter of two port network are –

$$[Y] = \begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix} s$$

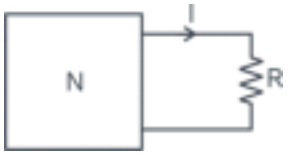
A single element of 1Ω is connected across as shown in the given figure. The new Y – parameter will be



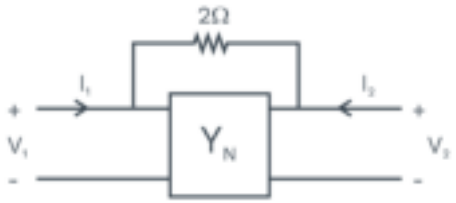
(A) $\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$ (D) $\begin{bmatrix} 6 & 2 \\ 1 & 5 \end{bmatrix}$

Q.13 Network N contains resistor and independent sources for R = 0 and 2Ω the value of current is I = 3 A and 1.5 A respectively find the value of I, where R = 3Ω.



Q.14 Find the Y-parameter of the given network



If $Y_N = \begin{bmatrix} 2 & -1 \\ 5 & 3 \end{bmatrix}$

(A) $\begin{bmatrix} \frac{5}{2} & \frac{-3}{2} \\ \frac{9}{2} & \frac{7}{2} \end{bmatrix}$ (B) $\begin{bmatrix} \frac{-5}{2} & \frac{3}{2} \\ \frac{-9}{2} & \frac{7}{2} \end{bmatrix}$

(C) $\begin{bmatrix} \frac{5}{4} & \frac{-3}{2} \\ \frac{9}{2} & \frac{7}{3} \end{bmatrix}$ (D) $\begin{bmatrix} \frac{-5}{4} & \frac{3}{2} \\ \frac{9}{2} & \frac{-7}{3} \end{bmatrix}$

Q.15 For a two port network, the transmission parameter 'C' can be given by –

(A) $\frac{1}{Z_{21}}$ (B) $\frac{-h_{22}}{h_{21}}$

(C) $\frac{Y_{12}Y_{21}}{Y_{22}} - Y_{11}$ (D) $\frac{-1}{Z_{12}}$

Q.16 For a two port network $Z_{11} = \frac{1}{Y_{11}}$, it is given that , then which one of the following relation is true?

(A) AC – BD = 0 (B) AD – BC = 1

(C) AD – BC = 0 (D) A = D

Q.17 Consider the network shown below having transmission parameter matrix – $T = \begin{bmatrix} 10 & -20 \\ 30 & -40 \end{bmatrix}$



The value of voltage gain $A_V = \frac{V_2}{V_1} = ?$

Q.18 The condition AD – BC = 1 for a two port network implies that the network is a -

(A) Reciprocal network

(B) Lumped element network

(C) Loss less network

(D) Unilateral element network

Q.19 For a two port network the condition for being symmetrical network in terms of Z-parameters is-

(A) $Z_{11} = Z_{22}$ (B) $Z_{12} = Z_{21}$

(C) $Z_{11} = Z_{12}$ (D) $Z_{21} = Z_{22}$

Q.20 Match List I with List II and select the correct answer using the codes given below for each list:

List I (Two port parameter)		List II (Also known as)	
A	g-parameters	1	Transmission parameters
B	y-parameters	2	Hybrid parameters
C	h-parameters	3	Admittance parameters
D	ABCD-parameters	4	Inverse Hybrid parameters

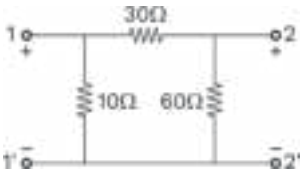
(A) A-4, B-3, C-2, D-1

(B) A-1, B-2, C-3, D-4

(C) A-4, B-3, C-1, D-2

(D) A-2, B-1, C-3, D-4

Q.21 For a two-port network shown in figure, Find the value of Z_{11} and Z_{22}:

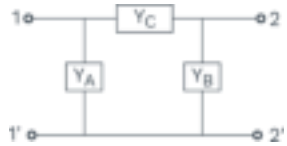


MSQ

(A) 9 Ω (B) 15 Ω

(C) 20 Ω (D) 24 Ω

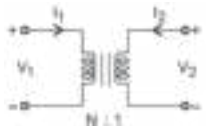
Q.22 For the two-port network shown in figure Y_{12} is-



(A) $Y_A + Y_B$ (B) $Y_C + \left(\frac{Y_A Y_B}{Y_A + Y_B}\right)$

(C) $-Y_C$ (D) Y_C

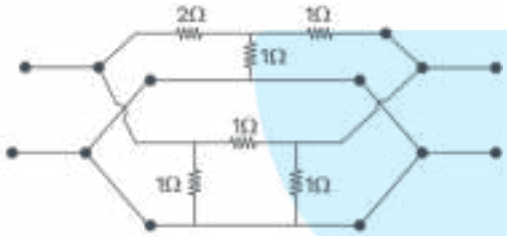
Q.23 Find the ABCD parameters of an ideal N : 1 transform shown in figure.



(A) $\begin{bmatrix} N & 0 \\ 0 & \frac{1}{N} \end{bmatrix}$ (B) $\begin{bmatrix} N & 0 \\ 0 & -\frac{1}{N} \end{bmatrix}$

(C) $\begin{bmatrix} N & 0 \\ 0 & N \end{bmatrix}$ (D) None of these

Q.24 Find the overall [z].



(A) $\begin{bmatrix} 0.54 & 0.25 \\ 0.25 & 0.5 \end{bmatrix}$ (B) $\begin{bmatrix} 0.5 & 0.25 \\ 0.25 & 0.5 \end{bmatrix}$

(C) $\begin{bmatrix} 0.5 & -0.25 \\ -0.25 & 0.5 \end{bmatrix}$ (D) $\begin{bmatrix} -0.54 & 0.25 \\ 0.25 & 0.5 \end{bmatrix}$

Q.25 Find the overall [T].



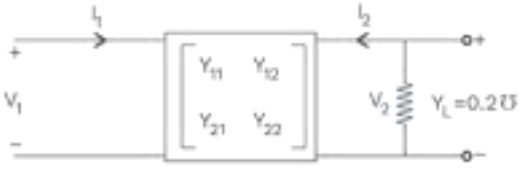
(A) $\begin{bmatrix} -3 & 5 \\ 1 & 2 \end{bmatrix}$ (B) $\begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ (D) $\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$

Q.26 Consider the following networks and Y-parameters-

$Y_{11} = 2\,℧$, $Y_{12} = 0.5\,℧$

$Y_{21} = 4.8\,℧$, $Y_{22} = -0.2\,℧$



Find V_2 for V_1 = 10V.

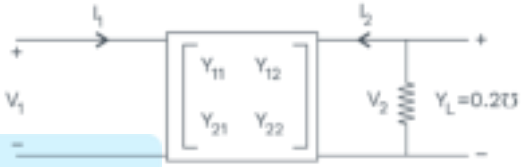
(A) 1 V (B) 0 V

(C) 10 V (D) $-\infty$

Q.27 Consider the following networks and Y-parameters

$Y_{11} = 2\,℧$, $Y_{12} = 0.5\,℧$

$Y_{21} = 4.8\,℧$, $Y_{22} = -0.2\,℧$



Find h_{11} in terms of Y-parameters-

(A) 0.5 (B) 0.25

(C) 0.208 (D) 5

Q.28 Given Y-parameter of two port network as-

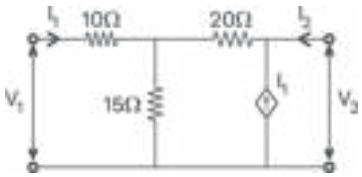
$$[Y] = \begin{bmatrix} 0.3 & -0.2 \\ -0.2 & 0.3 \end{bmatrix}$$

The Z-parameter of the network Z_{22} is-

(A) 5 Ω (B) 6 Ω

(C) 4 Ω (D) 1.5 Ω

Q.29 The parameter Z_{11} of the below circuit is-



(A) 50 Ω (B) 15 Ω

(C) 35 Ω (D) 40 Ω

Q.30 The h parameters h_{11} and h_{22} are related to Z and Y parameters as-

(A) $h_{11} = Z_{11}$ and $h_{22} = \frac{1}{Z_{22}}$

(B) $h_{11} = Z_{11}$ and $h_{22} = Y_{22}$

(C) $h_{11} = \frac{1}{Y_{11}}$ and $h_{22} = \frac{1}{Z_{22}}$

(D) $h_{11} = \frac{1}{Y_{11}}$ and $h_{22} = Y_{22}$

Q.31 Asymmetrical two-port network have:

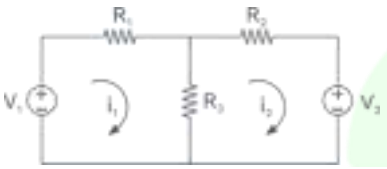
(A) $Z_{SC_1} = Z_{OC_1}$

(B) $Z_{SC_1} = Z_{SC_2}$

(C) $Z_{SC_1} = Z_{OC_2} \times Z_{OC_1}$

(D) $Z_{SC_1} \neq Z_{SC_2}$ and $Z_{OC_1} \neq Z_{OC_2}$

Q.32 If the below circuit can be represented in the form of $\begin{bmatrix} A & B \\ C & D \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix}$. Then the values of the matrix are-



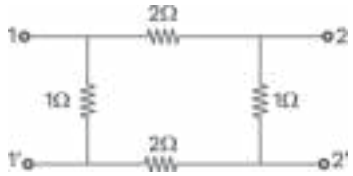
(A) $\begin{bmatrix} R_1 + R_3 & R_3 \\ R_3 & R_1 + R_3 \end{bmatrix}$

(B) $\begin{bmatrix} R_1 + R_2 & R_3 \\ R_3 & R_1 + R_2 \end{bmatrix}$

(C) $\begin{bmatrix} R_1 + R_2 & -R_3 \\ -R_3 & R_2 + R_3 \end{bmatrix}$

(D) $\begin{bmatrix} R_1 + R_2 & -R_3 \\ -R_3 & R_1 + R_2 \end{bmatrix}$

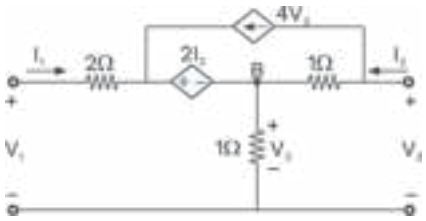
Q.33 What is the value of Z_{11} and Z_{22} for the circuit shown below respectively?



(A) $\frac{5}{6}\Omega, \frac{5}{3}\Omega$ (B) $\frac{5}{6}\Omega, \frac{5}{6}\Omega$

(C) $\frac{5}{3}\Omega, \frac{5}{6}\Omega$ (D) $\frac{5}{3}\Omega, \frac{5}{3}\Omega$

Q.34 Find the Z-parameters of the circuit shown below.



(A) $\begin{bmatrix} 2 & 3 \\ 3 & 3 \end{bmatrix}$ (B) $\begin{bmatrix} -2 & -2 \\ 3 & 3 \end{bmatrix}$

(C) $\begin{bmatrix} 3 & 3 \\ 5 & 6 \end{bmatrix}$ (D) $\begin{bmatrix} 3 & 3 \\ -3 & -2 \end{bmatrix}$

Q.35 Z-parameters of a two port network are $Z_{11} = 10\Omega$, $Z_{22} = 20\Omega$, $Z_{12} = Z_{21} = 5\Omega$. It's ABCD constant will be- MSQ

(A) A = 2

(B) B = 35 Ω

(C) C = 0.2 Ω, D = 4

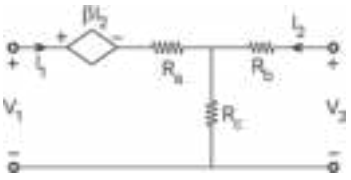
(D) None of these

Q.36 For the two-port network shown below the Z-parameters are $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$



If the Z-parameter terminated with a load Z_L = 1 Ω then the input impedance is _____ Ω.

Q.37 For the two-port network given below, the value of Z_{12} is-

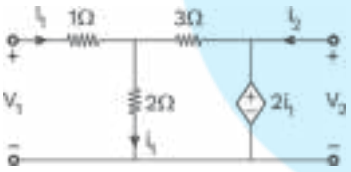


(A) $R_a + R_c$ (B) $\beta + R_a + R_c$

(C) $\beta + R_c$ (D) $\beta + R_b$

Q.38 For a two-port network Z-matrix is given as $Z = \begin{bmatrix} 14 & 8 \\ 6 & 4 \end{bmatrix}$. The value of h_{11} is _____. (in ohms).

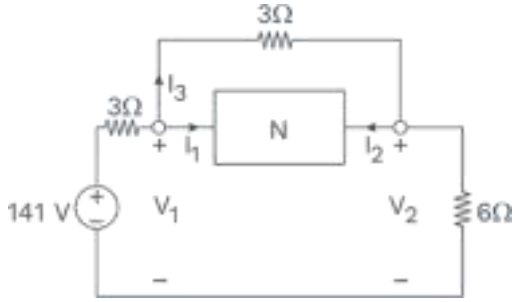
Q.39 What is the short circuit parameter matrix for the network shown in figure below?



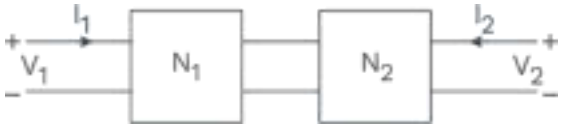
(A) $\begin{bmatrix} 0 & -1 \\ -3 & 2 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & -2 \\ -2 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 3 & -\frac{1}{2} \\ -\frac{1}{2} & 2 \end{bmatrix}$ (D) None of these

Q.40 For the two-port network N shown below in figure. Z-parameter matrix is $\begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$. The value of I_1 in Ampere is _____ .



Q.41 Two identical networks are connected in cascaded as shown in figure. Transmission matrix for one network is $\begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$. The value of Z_{12} is _____ . in (ohms)



Q.42 Consider the two port network given below. Which of the following parameters are possible for above network.



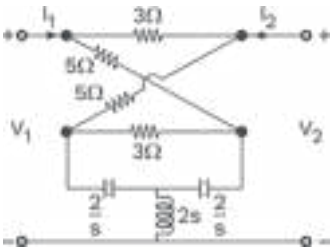
(A) Z-parameters

(B) Y-parameters

(C) h-parameters

(D) None of these above

Q.43 What is the Z-parameter matrix for the following circuit?



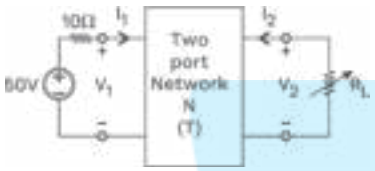
(A) $\begin{bmatrix} 4+2s+\frac{2}{s} & 4+2s \\ 4+2s & 4+2s+\frac{2}{s} \end{bmatrix}$

(B) $\begin{bmatrix} 4+2s+\frac{2}{s} & 2s+1 \\ 2s+1 & 4+2s+\frac{2}{s} \end{bmatrix}$

(C) $\begin{bmatrix} 2+s+\frac{2}{s} & 2s+1 \\ 2s+1 & 2+s+\frac{2}{s} \end{bmatrix}$

(D) $\begin{bmatrix} 2+4s+\frac{2}{s} & 2s+1 \\ 2s+1 & 2+4s+\frac{2}{s} \end{bmatrix}$

Q.44 The transmission matrix of the two-port network N are [T] = $\begin{bmatrix} 4 & 20 \\ 0.1 & 2 \end{bmatrix}$. The output port is connected with variable resistor R_L. So that power transferred to the R_C is maximum. The value of R_L is _____Ω.



Q.45 For cascade or series connections of any number of two-port networks the transmission parameter matrices have to be-

(A) Multiplied

(B) Added

(C) Inverted and then multiplied

(D) Inverted and then added

Q.46 Two port networks are connected in parallel. The combination is to be represented as a single two-network the parameters of this network that are obtained by addition of the individual parameters are-

(A) Z-parameters

(B) h-parameters

(C) Y-parameters

(D) ABCD-parameters

Q.47 Two port networks are connected in parallel. The combination is to be represented as a single two-network the parameters of this network that are obtained by addition of the individual parameters are-

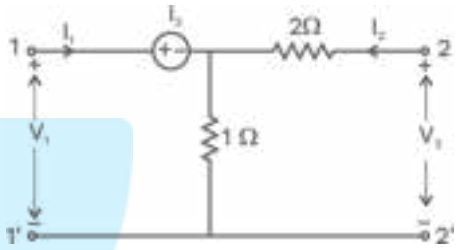
(A) Z-parameters

(B) h-parameters

(C) Y-parameters

(D) ABCD-parameters

Q.48 Consider the 2-port network shown in figure.

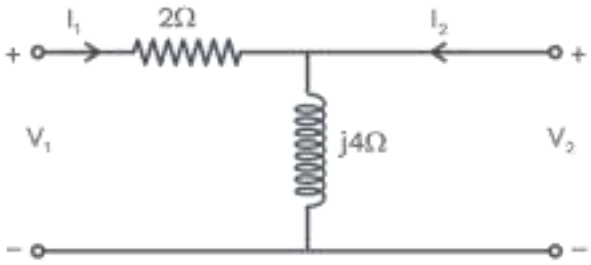


The Y-parameters of the network is:

(A) $\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} -1 & 1 \\ 3 & -2 \end{bmatrix}$

Q.49 Consider the given circuit shown below figure.



The h-parameter of the given two port network is,

(A) $\begin{bmatrix} 2 & 1 \\ -1 & -j3 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 2 \\ -2 & -j4 \end{bmatrix}$

(C) $\begin{bmatrix} 3 & 1 \\ -2 & -j1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 3 \\ -1 & -j4 \end{bmatrix}$

Q.50 If short circuit the port 1 of a two-port network, $I_1 = 4I_2$ and $V_2 = 0.136I_2$ then the value of Y_{12} and Y_{22} is respectively,

(A) 29.41℧, 7.35℧ (B) 11.23℧, 3.43℧

(C) 14.32℧, 6.45℧ (D) 16.41℧, 5.35℧

Q.51 Consider the given two port equation –

$$V_1 = AV_2 + B(-I_2)$$

$$I_1 = 1CV_2 + D(-I_2)$$

Let, A = 3, B = 4, C = 8, D = 6

Then, Z – Parameter will be

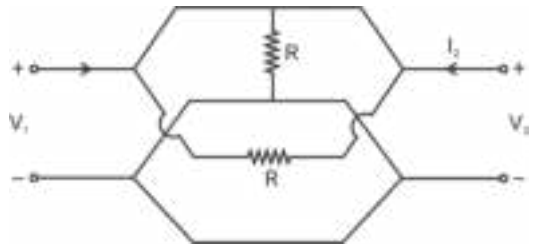
(A) $\begin{bmatrix} 0.38 & -1.75 \\ 0.13 & 0.75 \end{bmatrix}\Omega$

(B) $\begin{bmatrix} 1.33 & 0.53 \\ -1.33 & 1.23 \end{bmatrix}\Omega$

(C) $\begin{bmatrix} 0.15 & -1.33 \\ 1.83 & 0.75 \end{bmatrix}\Omega$

(D) $\begin{bmatrix} 0.23 & 0.13 \\ 1.33 & -4.5 \end{bmatrix}\Omega$

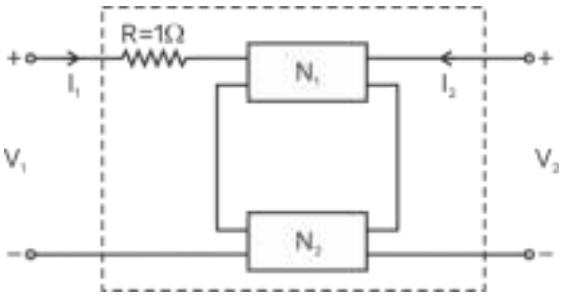
Q.52 The y-parameter matrix of the circuit shown below is



(A) $\begin{bmatrix} 2R & 2R \\ 2R & 2R \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} \frac{1}{2R} & \frac{1}{2R} \\ \frac{1}{2R} & \frac{1}{2R} \end{bmatrix}$ (D) Doesn't Exist

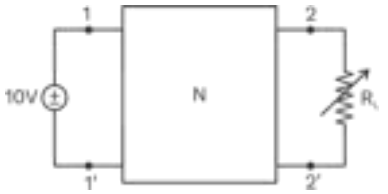
Q.53 If $[Y]_{N_1} = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$, $[Y]_{N_2} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, then the ABCD or transmission parameter matrix of the entire network is



(A) $\begin{bmatrix} +4 & -1.5 \\ 2 & -2 \end{bmatrix}$ (B) $\begin{bmatrix} -4 & -3.5 \\ -2 & -2 \end{bmatrix}$

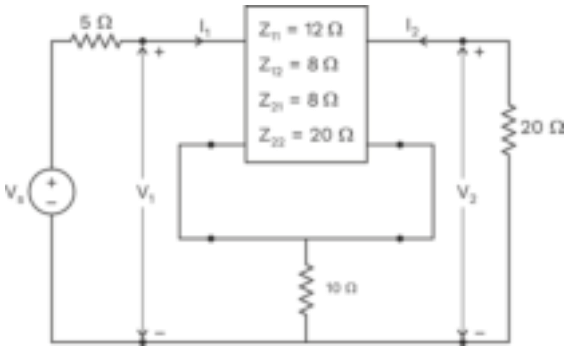
(C) $\begin{bmatrix} -4 & -0.5 \\ 2 & -2 \end{bmatrix}$ (D) $\begin{bmatrix} 4 & 3.5 \\ -2 & 2 \end{bmatrix}$

Q.54 Consider the network shown below :



If the hybrid parameter matrix of the network 'N' is $[h] = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$, then the maximum power that can be transferred to R_L is _____ W.

Q.55 For the following network $\frac{V_2}{V_s}$ is _____.



Q.56 The terminal volt-ampere conditions of a linear reciprocal network N are shown in the figure (i) shown below. The current I correspond to terminal conditions shown in figure (ii) is,

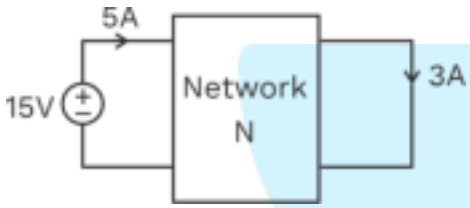


Figure (i)

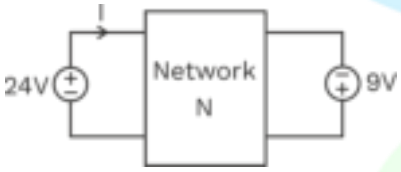
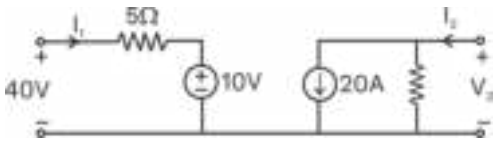


Figure (ii)

(A) 9.8 A (B) 5.6 A

(C) 7.2 A (D) 10.4 A

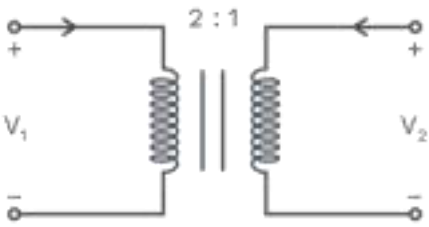
Q.57 The circuit which is shown below demonstrates h-parameter model of some electrical network. Find the value of h_{21} parameter.



(A) 5 (B) 5/3

(C) 20/3 (D) 10/3

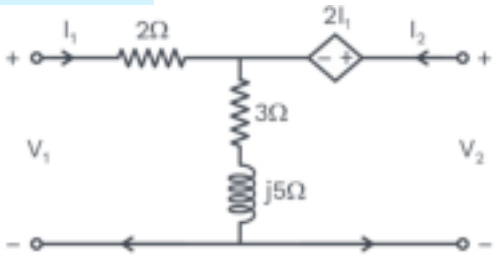
Q.58 Find the g-parameters for the given circuit below.



(A) $\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}$

Q.59 Find the Z-parameter for the shown circuit.



(A) $\begin{bmatrix} 7+j3 & 5+j3 \\ 5+j3 & 8+j3 \end{bmatrix}$

(B) $\begin{bmatrix} 5+j5 & 3+j5 \\ 5+j5 & 3+j5 \end{bmatrix}$

(C) $\begin{bmatrix} 5+j5 & 3+j5 \\ 3+j5 & 5+j5 \end{bmatrix}$

(D) $\begin{bmatrix} 8+i3 & 5+j3 \\ 5+j3 & 7+j3 \end{bmatrix}$

Q.60 Which of the following relation is/are correct for a two-port network?

1. $Z_{11} = \frac{A}{C}$

2. $Z_{22} = \frac{1}{h_{22}}$

3. $Z_{21} = \frac{-1}{C}$

4. $Z_{12} = \frac{h_{12}}{h_{22}}$

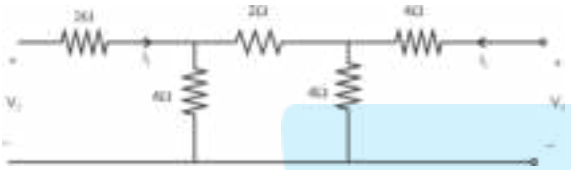
(A) 1 and 2 only

(B) 1, 2 and 3 only

(C) 2 and 3 only

(D) 1, 2 and 4 only

Q.61 Determine the value of the h_{21} for the two port network shown in figure.



(A) -0.125 (B) 0.125

(C) 0.25 (D) -0.25

Q.62 In a two port network the admittance parameters are given as.

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

If the output terminal is terminated by a load resistance R_L then the input admittance in input port will be

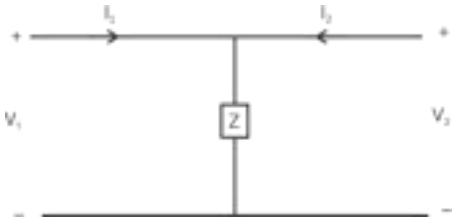
(A) $\frac{(Y_{11}Y_{22} - Y_{12}Y_{21})R_L}{1 + Y_{22}R_L}$

(B) $\frac{Y_{11} + (Y_{11}Y_{22} - Y_{12}Y_{21})R_L}{1 + Y_{22}R_L}$

(C) $\frac{Y_{11} + (Y_{11}Y_{22} - Y_{12}Y_{21})R_L}{1 + Y_{22}R_L}$

(D) $\frac{Y_{11}Y_{22} - Y_{12}Y_{21}}{1 + Y_{22}R_L}$

Q.63 The ABCD parameters of the given two port network are



(A) $\begin{bmatrix} \frac{1}{z} & 0 \\ 1 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 1 \\ 0 & \frac{1}{z} \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 0 \\ \frac{1}{z} & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & z \\ 0 & 1 \end{bmatrix}$

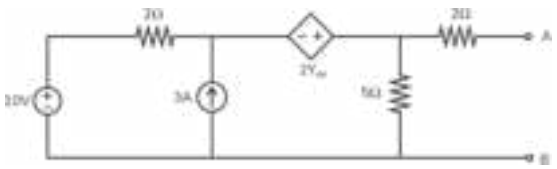
Q.64 The ABCD parameters of the given two port network are.



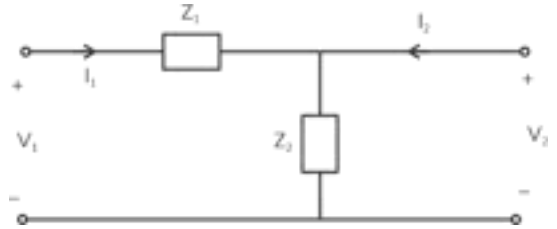
(A) $\begin{bmatrix} 1 & z \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 1 \\ z & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & z \\ 1 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & \frac{1}{z} \\ 0 & 1 \end{bmatrix}$

Q.65 In the given circuit below, the short circuit current across the terminal AB will be _____



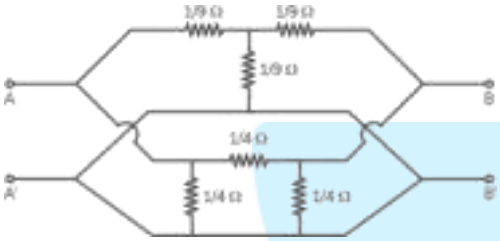
Q.66 The T-Parameter of the given two port network shown in figure will be



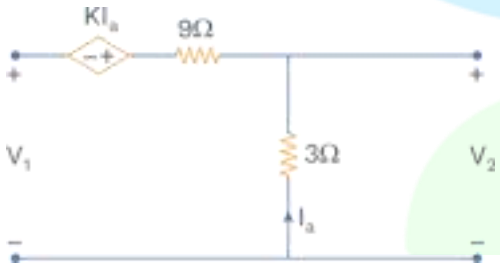
(A) $\begin{bmatrix} 1+\frac{z_1}{z_2} & 1 \\ z_2 & z_1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & \frac{z_1}{z_2} \\ z_2 & \frac{1}{z_1} \end{bmatrix}$

(C) $\begin{bmatrix} 1+\frac{z_1}{z_2} & z_1 \\ \frac{1}{z_2} & 1 \end{bmatrix}$ (D) $\begin{bmatrix} \frac{z_1}{z_2} & z_1 \\ z_2 & 1 \end{bmatrix}$

Q.67 In the network shown in figure, the Y_{21} parameter of the network is_____

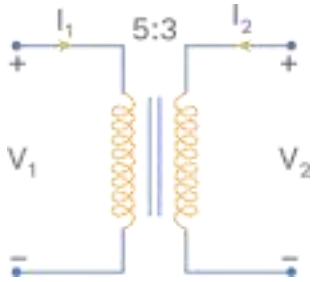


Q.68 A two-port network is shown below, whose admittance parameter Y_{22} is given as 3S.



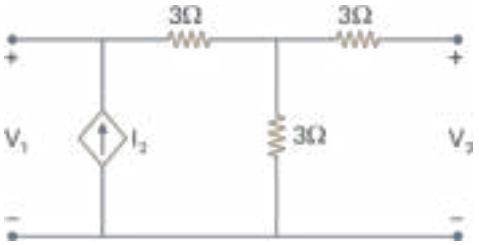
The value of K is _____. (Rounded off to the nearest integer)

Q.69 Consider the network shown below.



The value of $g_{21} \times g_{12}$ is _____.(Rounded off to the nearest integer)

Q.70 For the circuit shown below, which of the statement(s) given in the options is/are TRUE? MSQ



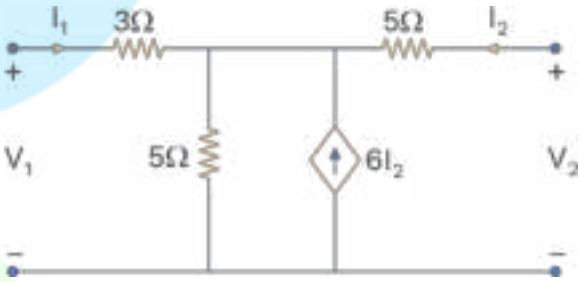
(A) The value of Y_{21} is $\frac{-2}{9}\Omega$.

(B) The value of $\frac{Y_{11}}{Y_{21}}$ is -3.

(C) The value of Y_{11} is $\frac{9}{2}$ ℧

(D) The value of $Z_{12}Y_{21}$ is -1

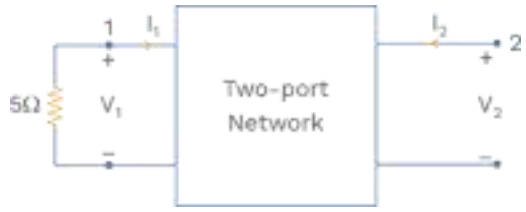
Q.71 For the two-port network shown in the figure, the Z-parametersare-



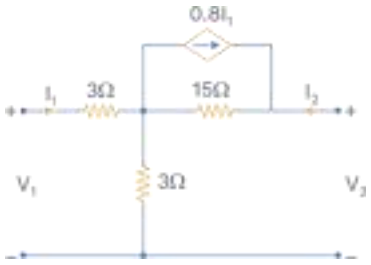
(A) $\begin{bmatrix} 4 & 15 \\ 5 & 20 \end{bmatrix}$ (B) $\begin{bmatrix} 5 & 16 \\ 16 & 5 \end{bmatrix}$

(C) $\begin{bmatrix} 8 & 5 \\ 30 & 15 \end{bmatrix}$ (D) $\begin{bmatrix} 8 & 35 \\ 5 & 40 \end{bmatrix}$

Q.72 TheY-parameters of the two-port network are given as: $Y_{11} = 8\vartheta$, $Y_{12} = Y_{21} = 4\vartheta$ and $Y_{22}=2\vartheta$. If a resistor of 5Ω is connected to port-1 of the two-port network, then the output impedance is _____ Ω. (Upto one decimal place)



Q.73 A two-port network is shown in the figure below. Which of the following satisfy the given circuit?



MSQ

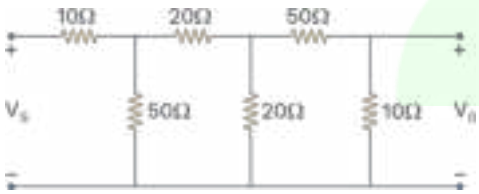
(A) $\frac{Y_{22}}{Y_{12}} = -18$

(B) $Y_{21} \times Z_{21} = -3.57$

(C) $\frac{Y_{22}}{Y_{11}} = 63$

(D) $Z_{22} \times Y_{22} = 1.71$

Q.74 The ABCD parameters of the network shown in the figure below are-



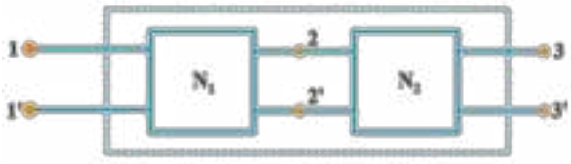
(A) $\begin{bmatrix} 15.23 & 163.2 \\ 0.68 & 5.9 \end{bmatrix}$ (B) $\begin{bmatrix} 15.23 & 163.2 \\ 2.32 & 7.9 \end{bmatrix}$

(C) $\begin{bmatrix} 20.8 & 179 \\ 0.68 & 5.9 \end{bmatrix}$ (D) $\begin{bmatrix} 20.8 & 179 \\ 2.32 & 7.9 \end{bmatrix}$

Q.75 Two two-port networks are cascaded as shown in the figure below. The transmission parameters of the two two-port networksare shown below.

$\begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$ and

$\begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{5} & 1 \end{bmatrix}$

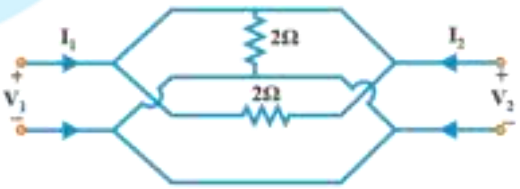


The z-parameters of resultant two-port networkare-

(A) $\begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix}$ (B) $\begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix}$

(C) $\begin{bmatrix} 5 & 1 \\ \frac{1}{5} & \frac{1}{5} \end{bmatrix}$ (D) $\begin{bmatrix} 2 & 5 \\ \frac{1}{5} & 1 \end{bmatrix}$

Q.76 The Y Parameter of the circuit shown in the figure is –



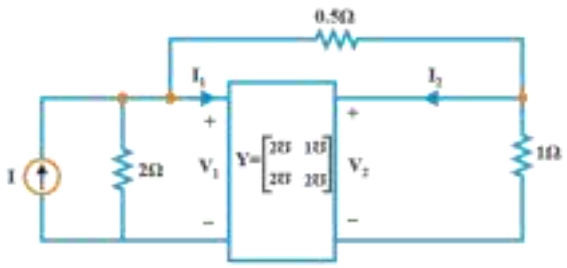
(A) $\begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix}$

(D)does not exit

Q.77 For the two-port network shown below, current I = 1A.



Which of the following statement(s) is/are true?

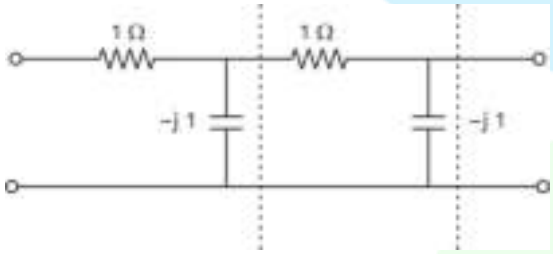
(A) The input voltage to the two-port network = 0V.

(B) The input voltage to the two-port network = 0.22V

(C) The output voltage to the two-port network =0V

(D) The output voltage to the two-port network =0.22V

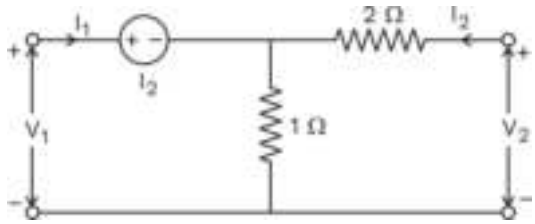
Q.78 Find the overall transmission parameter of the given network –



(A) $\begin{bmatrix} 3j & j+2 \\ 2j-1 & j+1 \end{bmatrix}$ (B) $\begin{bmatrix} -3j & j+2 \\ 2j-1 & j-1 \end{bmatrix}$

(C) $\begin{bmatrix} 3j & 2-j \\ 2j+1 & j+1 \end{bmatrix}$ (D) $\begin{bmatrix} 3j & 2-j \\ 2j-1 & j+1 \end{bmatrix}$

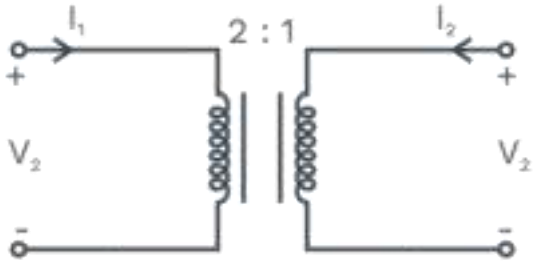
Q.79 For the given network, find the Y-parameter



(A) $\begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} -3 & 2 \\ -1 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} -3 & -2 \\ 1 & 1 \end{bmatrix}$

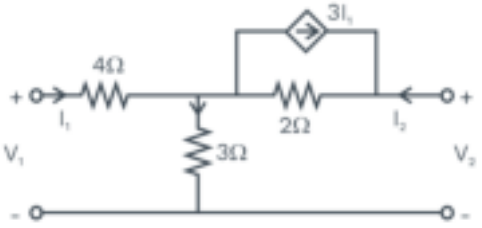
Q.80 In the circuit shown in figure, T – parameter are –



(A) $\begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ (B) $\begin{bmatrix} 2 & 0 \\ 0 & -\frac{1}{2} \end{bmatrix}$

(C) $\begin{bmatrix} 0 & -2 \\ \frac{1}{2} & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & 2 \\ -\frac{1}{2} & 0 \end{bmatrix}$

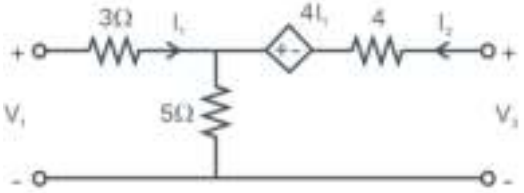
Q.81 Find Z – Parameter for the given network –



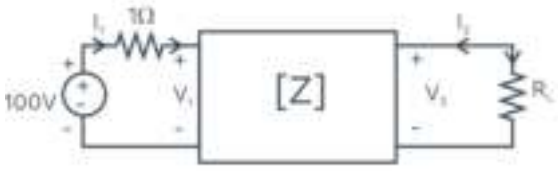
(A) $\begin{bmatrix} 4 & 3 \\ 3 & 5 \end{bmatrix}$ (B) $\begin{bmatrix} 7 & 3 \\ 9 & 5 \end{bmatrix}$

(C) $\begin{bmatrix} 4 & 7 \\ 9 & 4 \end{bmatrix}$ (D) $\begin{bmatrix} 7 & 3 \\ 3 & 5 \end{bmatrix}$

Q.82 For the given circuit find the value of D for a T-parameter _____.



Q.83 Find the maximum power transferred to R_L in watt _____.

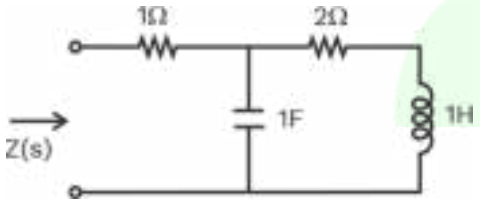


$$[Z] = \begin{bmatrix} 4 & 3 \\ 4 & 5 \end{bmatrix}$$

Q.84 The Z – parameter matrix of a particular network 'N' shown below is $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$. If the power dissipated in the load resistor of 4Ω, 50W. Then the supply voltage V_s will be _ ___ volt?



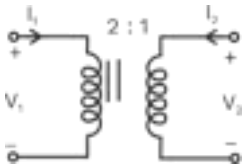
Q.85 Find the driving point impedance of the network shown in figure.



(A) $\frac{s^2+3s+3}{s^2+2s+1}$ (B) $\frac{s^2+2s+3}{s^2+2s+2}$

(C) $\frac{s^2+3s+4}{s^2+2s+1}$ (D) $\frac{3s^2+3s+2}{s^2+s+1}$

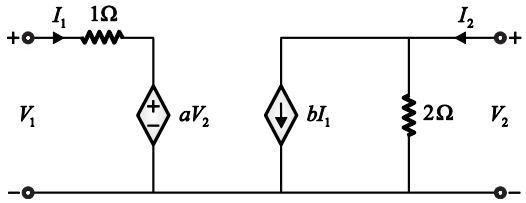
Q.86 In the circuit shown in figure, the g-parameters are



(A) $\begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$ (B) $\begin{pmatrix} 0 & \frac{1}{2} \\ \frac{-1}{2} & 0 \end{pmatrix}$

(C) $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ (D) $\begin{pmatrix} 0 & \frac{-1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}$

Q.87 Consider a two- port network shown in figure below.

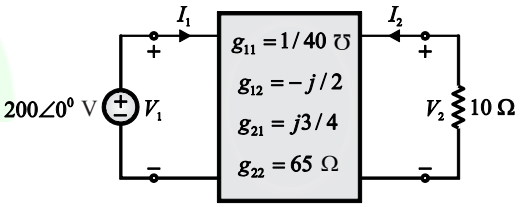


If the value of transmission parameter A is 20, then the relationship between a and b is-

(A) $ab - 20b = 1$ (B) $ab + 20b = 1$

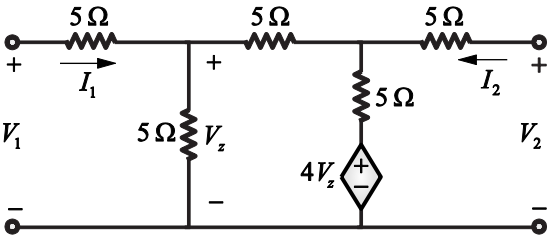
(C) $ab - 20b = 0.5$ (D) $ab + 20b = 0.5$

Q.88 Consider the two-port network described by inverse hybrid parameters as shown below.

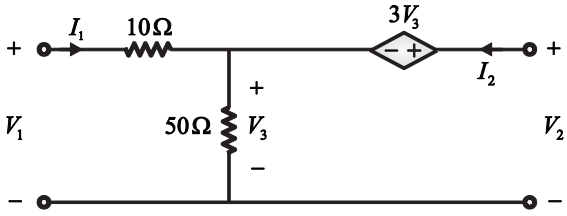


The magnitude of current I_1 is _____ A. (in integer)

Q.89 Consider the circuit shown in figure. The impedance (in Ω) seen at the port-1 with port-2 open circuited is _____. (in integer)



Q.90 Consider the two port network shown in figure below



Which of the following statements is/are correct? MSQ

(A) The value of hybrid parameter $h_{11} = 10\,\Omega$

(B) The value of z-parameter $z_{11} = 60\,\Omega$

(C) The value of hybrid parameter $h_{21} = -1$

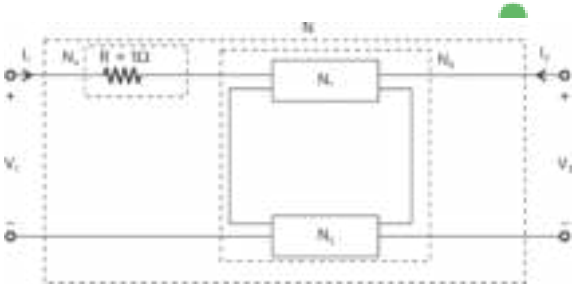
(D) The value of z-parameter $z_{12} = 70\,\Omega$

Q.91 For the circuit shown below, the network N has Z parameter matrix as $\begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix}$. The transformer used in the circuit is ideal. Power delivered to R_L is_____ W. (Rounded off to nearest integer).



Q.92 The Y-parameter of the circuit given below is- MSQ

$$Y_{N_1} = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \qquad Y_{N_2} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$



Which of the following statements is/are true?

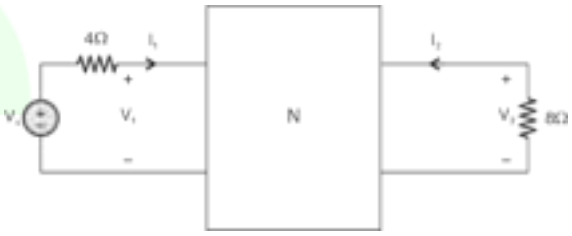
(A) T parameter of overall network is $\begin{bmatrix} -4 & \frac{-7}{2} \\ -2 & -2 \end{bmatrix}$

(B) T parameter, $[T]_{N_A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

(C) Z parameter, $[Z]_{N_B} = \begin{bmatrix} 1 & \frac{-1}{2} \\ \frac{-1}{2} & 1 \end{bmatrix}$

(D) Z parameter, $[Z]_{N_2} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{bmatrix}$

Q.93 A network is shown below

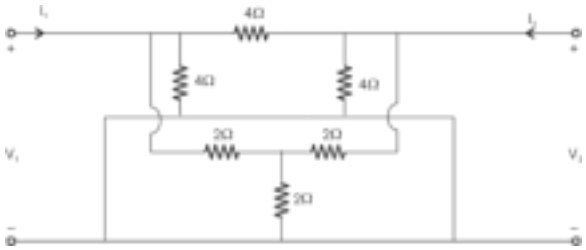


The h-parameter of the network is

$$[h] = \begin{bmatrix} 4\Omega & 8 \\ -10 & 4\Omega \end{bmatrix}$$

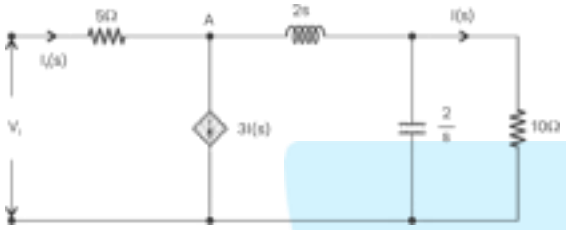
The power dissipated in load resistor of 8Ω is 50W, the supply voltage, V_s is _____V. (Rounded off to nearest integer).

Q.94 A two-port network is shown below.



The admittance parameter Y_{21} of the network = _____ ℧. (up to 3 decimal places)

Q.95 Find the driving point impedance for the given circuit as shown in the figure below.



(A) $\frac{10s^2+27s+30}{5s+4}$

(B) $\frac{10s^2+7s+14}{5s+4}$

(C) $\frac{10s^2+27s+30}{5s+1}$

(D) $\frac{10s^2+7s+14}{5s+1}$

Q.96 The condition AD – BC = 1 for two-port network implies that the network is a-

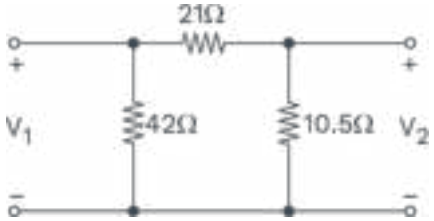
(A) Reciprocal network

(B) Lumped element network

(C) Losses less network

(D) Unilateral element network

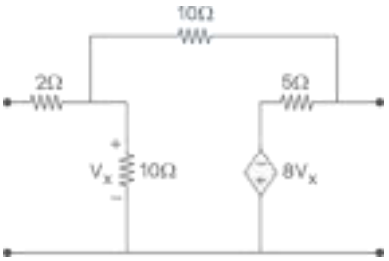
Q.97 Determine the Z-parameter of two-port network shown in figure.



(A) $Z=\begin{bmatrix}6 & 9\\ 9 & 18\end{bmatrix}$ (B) $Z=\begin{bmatrix}9 & 6\\ 6 & 9\end{bmatrix}$

(C) $Z=\begin{bmatrix}18 & 6\\ 6 & 9\end{bmatrix}$ (D) $Z=\begin{bmatrix}18 & 9\\ 9 & 6\end{bmatrix}$

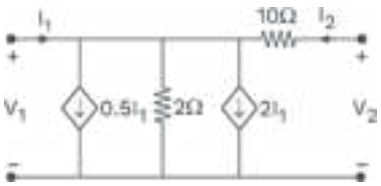
Q.98 For the two-port network shown below, the hybrid parameters are-



(A) $\begin{bmatrix}0.6\,\Omega & 3\\ 0.5 & 7\,S\end{bmatrix}$ (B) $\begin{bmatrix}7\Omega & 0.5\\ 7.5 & 1.05\,S\end{bmatrix}$

(C) $\begin{bmatrix}5S & 0.5\\ 6 & 30\Omega\end{bmatrix}$ (D) $\begin{bmatrix}7\,\Omega & 0.5\\ 3 & 0.6\,S\end{bmatrix}$

Q.99 Determine the admittance parameter for the given two-port network.



(A) $\begin{bmatrix}0.4 & 0.06\\ 0.1 & 0.21\end{bmatrix}$ (B) $\begin{bmatrix}-0.1 & -0.21\\ -0.4 & -0.06\end{bmatrix}$

(C) $\begin{bmatrix}-0.4 & 0.06\\ -0.1 & 0.09\end{bmatrix}$ (D) $\begin{bmatrix}0.6 & -0.1\\ -0.1 & -0.21\end{bmatrix}$

Q.100 A two - port network is defined by two voltage and current equations given as follows :

$V_1 = 4V_2 - I_2$ and $I_1 = 2V_2 - I_2$

If two of such two-port network are cascaded then the transmission parameter of the overall network will be

(A) $\begin{bmatrix} 8 & 2 \\ 4 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 18 & 5 \\ 10 & 3 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 2 \\ 4 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} 17 & 9 \\ 9 & 5 \end{bmatrix}$

9 Magnetic Coupled Circuits

Q.1 The resonant frequency of the given circuit is-



(A) $\frac{1}{2\pi\sqrt{3}}$ Hz (B) $\frac{1}{4\pi\sqrt{3}}$ Hz

(C) $\frac{1}{4\pi\sqrt{2}}$ Hz (D) $\frac{1}{2\pi\sqrt{2}}$ Hz

Q.2 Two coils having equal resistances but different inductances are connected in series. The time constant of the series combination is _____ .

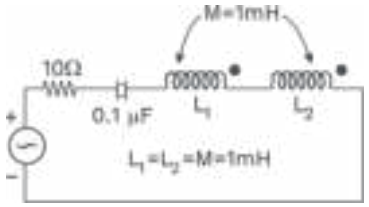
(A) Sum of the time constants of the individual coils.

(B) Average of the time constants of the individual coils.

(C) Geometric mean of the time constants of the individual's coils.

(D) Products of the time constant of the individual coils.

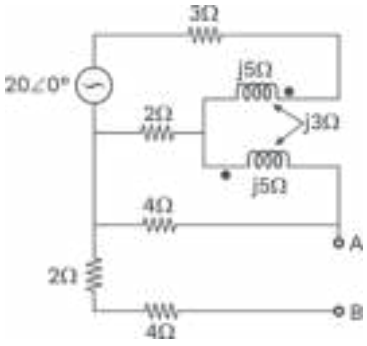
Q.3 The resonant frequency of the circuit is-



(A) $\frac{50}{2\pi}$ k rad / sec (B) $\frac{50}{2\pi}$ k Hz

(C) $\frac{100}{2\pi}$ k Hz (D) $\frac{100}{4\pi}$ k Hz

Q.4 Consider the circuit given below, the value of Thevenin voltage across the terminal A – B is-



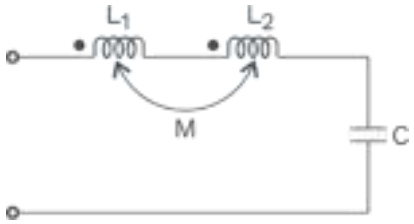
(A) $1.064\angle -137.8°$

(B) $4.256\angle -137.8°$

(C) $2.30\angle -41.7°$

(D) $9.21\angle -41.7°$

Q.5 Two mutually coupled inductors connected in series with the capacitor are operating at resonance. If number of turns in both the coils are doubled. Then-



(A) Resonating frequency will be doubled.

(B) Resonating frequency will become half.

(C) Resonating frequency will become $\sqrt{2}$ times.

(D) Resonating frequency will become $\frac{1}{\sqrt{2}}$ times.

Q.6 Two coils with negligible resistances and self-inductances of 0.2 H and 0.1 H respectively are connected in parallel. If their mutual inductance is 0.1 H then the possible effective inductance are-

MSQ

(A) 0.02 H (B) 0.5 H

(C) 0.002 H (D) 0.1 H

Q.7 Two inductors of 5H and 4H have mutual inductance of 2.5 H between them. The coupling coefficient is-

(A) 1.78 (B) 0.125

(C) 0.56 (D) 4.47

Q.8 Consider the following statements-

MSQ

(A) Mutual inductance describes the voltage induced at the ends of a wire due to the magnetic field generated by a second coil.

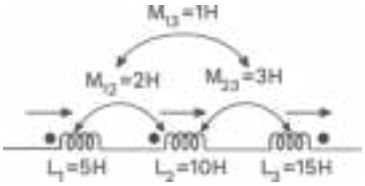
(B) The dot convention allows a sign to be assigned to the voltage induced due to mutual inductance term.

(C) The coupling coefficient is given by $k = \frac{M}{\sqrt{L_1 + L_2}}$

(D) None of these

Which of the above statement(s) are correct?

Q.9 Find the total inductance of the series coil shown below-

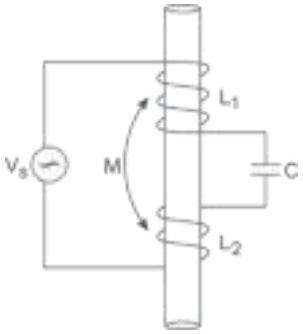


(A) 42 H (B) 26 H

(C) 36 H (D) 18 H

Q.10 What is the resonant frequency of circuit shown below?

Where R = 10Ω, $L_1 = L_2$ = 20 mH, Mutual inductance M = 5 mH, C = 0.05 μF.



(A) 4.18 kHz (B) 2.88 kHz

(C) 5.18 kHz (D) 3.16 kHz

Q.11 The combined inductance of two coils connected in series is 0.6 H or 0.1 H depending upon the relative directions of the currents in the coils. How much is the mutual inductance between the coils?

(A) 0.25 H (B) 0.3 H

(C) 0.05 H (D) 0.125 H

Q.12 Given two coupled inductors L_1 and L_2 their mutual inductance M satisfies-

(A) $M = \sqrt{L_1^2 + L_2^2}$ (B) $M > \frac{L_1 + L_2}{2}$

(C) $M > \sqrt{L_1 L_2}$ (D) $M \leq \sqrt{L_1 L_2}$

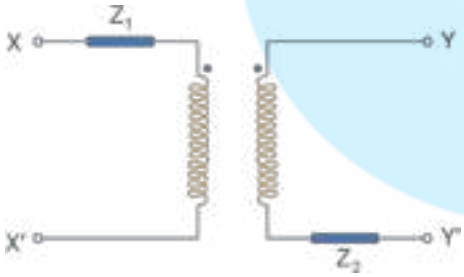
Q.13 Consider two coils which are identically coupled with each other such that the maximum equivalent inductance of coil is 50 mH and minimum equivalent inductance is 30 mH. Calculate the value of coefficient of coupling between them.

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) $\frac{1}{5}$

Q.14 Two magnetically coupled coils, having same parameters, are in series with a 240V, 50Hz voltage supply. The circuit draws a current of 12A. Now the terminals of one coil are inverted such that the new circuit so formed draws a current of 8A. The value of inductor is _____ H. (upto 3 decimal places)

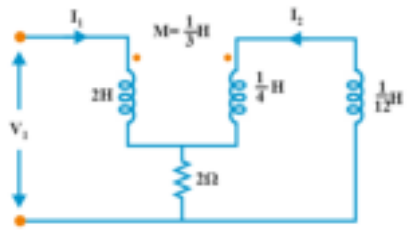
Q.15 An ideal transformer is shown below.



The primary winding of the transformer has j4Ω reactance and the secondary winding has j8Ω reactance. The mutual inductance is _____ mH. (Up to three decimal places)

Consider ω = 1200 rad/sec

Q.16 For the circuit shown in the figure below, the ratio $\frac{V_1}{I_1}$ is –

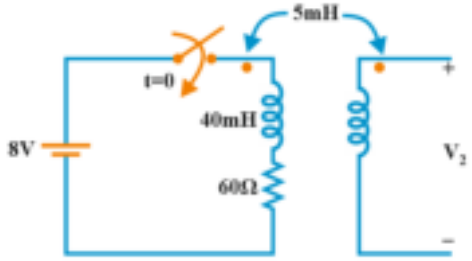


(A) 1/3s (B) 3s

(C) 5/3s (D) 7/3

Q.17 For the circuit shown in the figure below, the magnitude of V_2 at t=0 is ___V (Consider current in output loop = 0A).

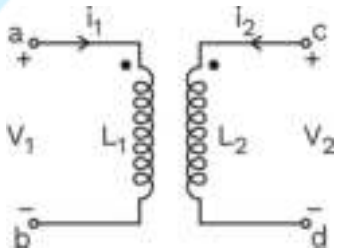
(Rounded off to the nearest integer)



Q.18 In given figure, L_1 = 2 H, L_2 = 8 H and K = 0.6, the total energy stored in the system at t = 0 is if,

$i_1(t) = 5 \cos(100t - 30°)$

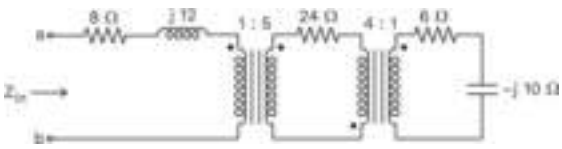
$i_2(t) = 3 \cos(100t - 30°)$



(A) 28.5 J (B) 28.92 J

(C) 19.34 J (D) 18.75 J

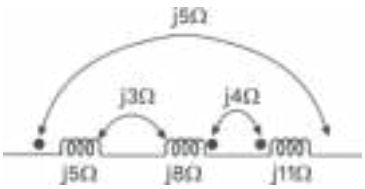
Q.19 Calculate the magnitude of input impedance (Z_{in}) for the given network –



(A) 15 Ω (B) 13.97 Ω

(C) 14 Ω (D) 15.43 Ω

Q.20 Find the equivalent impedance of the following configuration.



(A) 16 Ω (B) 20 Ω

(C) 10 Ω (D) 18 Ω

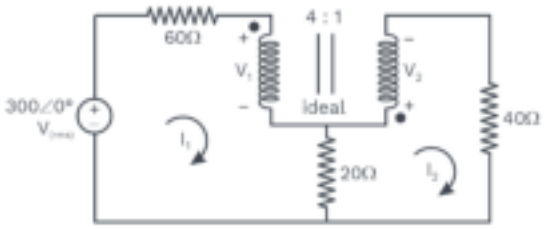
Q.21 A load $Z_L = (4 + j3)\Omega$ is to be matched using an ideal transformer to a generator of initial impedance, $Z_S = (12 + j6)\Omega$. If the secondary winding of transformer has 40 turns then number of turns in primary winding are _____ .

Q.22 There are two coupled coils which are connected in series having equivalent inductance of 20mH or 12 mH depending upon the type of connection. Find the value of mutual inductance.

(A) 4 mH (B) 2 mH

(C) 8 mH (D) 6 mH

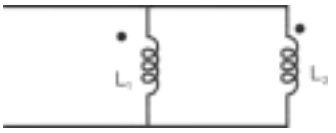
Q.23 Find the power consumed by 20 Ω resistor.



(A) 31.25 Ω (B) 40.25 Ω

(C) 36.25 Ω (D) 20.25 Ω

Q.24 The equivalent inductance of two coils connected in parallel as shown in figure are:



(A) $\frac{L_1L_2 - M^2}{L_1 + L_2 + 2M}$ (B) $\frac{L_1L_2 - M^2}{L_1 + L_2 - 2M}$

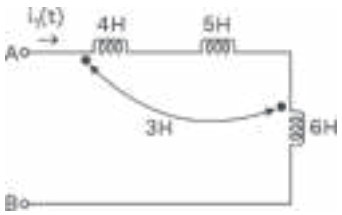
(C) $\frac{L_1L_2 + M^2}{L_1 + L_2 - 2M}$ (D) $\frac{L_1L_2 + M^2}{L_1 + L_2 - 2M}$

Q.25 Two inductors of 6 H and 4 H have mutual inductance of 3H between them. The coupling coefficient is-

(A) 1.78 (B) 0.125

(C) 0.61 (D) 0.47

Q.26 In the following network if $i_1(t) = 4 \cos 100t$ A then what is the value of maximum energy stored in the (Joule) network?



Q.27 Two coils are mutually coupled with L_1 = 16 mH, L_2 = 25 mH and K = 0.6. If the two coils are connected in series, then what is the value of maximum possible equivalent inductance?

(A) 57 mH (B) 58 mH

(C) 64 mH (D) 65 mH

10 Graph Theory

Q.1 If a complete graph consists of 12 branches and 5 nodes, then which of the following statements is/are true? MSQ

(A) The number of links is 5

(B) The order of tie-set matrix is 8×12

(C) The number of tie-setsis 8

(D) The order of the reduced incidence matrix is 4×12

Q.2 The number of undirected graphs (not necessarily connected), that can be constructed for the given 5 vertices set: $X = \{X_1, X_2, X_3, X_4, X_5\}$ is _____

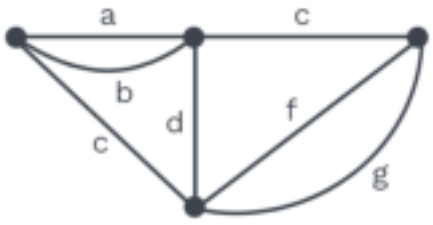
(Rounded off to the nearest integer)

Q.3 For a complete graph consisting of 10 fundamental cut sets. Find the number of tie sets of the corresponding graph.

(A) 12 (B) 26

(C) 45 (D) 35

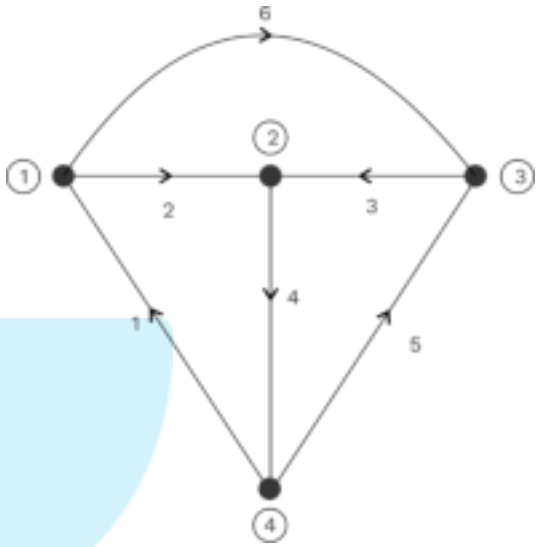
Q.4 For the given network graph shown in figure, which of the following is not a cut set ?



(A) abc (B) abdfg

(C) cdfg (D) cdfe

Q.5 For the graph shown in the figure below,



The loop incidence matrix for the graph is-

(A) $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 1 & 1 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$

❖❖❖

Answer Keys

CH-1 Basic of Network									
1.	300	2.	C	3.	C	4.	B	5.	A
6.	A	7.	A, D	8.	A	9.	D	10.	C
11.	C	12.	C	13.	C	14.	A	15.	D
16.	C	17.	6.396	18.	0.4	19.	C	20.	300
21.	C	22.	D	23.	A	24.	B	25.	C
26.	10	27.	2	28.	D	29.	B	30.	3000
31.	138.88	32.	A	33.	300	34.	C	35.	D
36.	7	37.	D	38.	C	39.	C	40.	D
41.	69.56	42.	A	43.	B	44.	B	45.	A

Answer Keys

CH-2 Circuit Analysis									
1.	B	2.	C	3.	C	4.	C	5.	D
6.	D	7.	C	8.	B	9.	A	10.	D
11.	D	12.	B	13.	A	14.	A	15.	A
16.	D	17.	C	18.	3	19.	A	20.	C
21.	C	22.	C	23.	B, C	24.	B	25.	C
26.	D	27.	B	28.	C	29.	4.5	30.	A
31.	C	32.	C	33.	72.1	34.	A, B, C, D	35.	C
36.	40	37.	0.407	38.	0	39.	248	40.	C
41.	4	42.	A, C, D	43.	–36	44.	4	45.	3
46.	C	47.	B	48.	C	49.	1.732	50.	84
51.	-19.75	52.	D	53.	–0.416	54.	C	55.	0
56.	0.66	57.	105.94	58.	0.4	59.	D	60.	D
61.	A	62.	D	63.	D	64.	A	65.	B
66.	2.714	67.	86.6	68.	B, C	69.	D	70.	B
71.	2.464Ω	72.	D	73.	B	74.	C	75.	C
76.	D	77.	B	78.	A	79.	18	80.	A
81.	–5V	82.	B, D	83.	B, D	84.	0.707 A	85.	B, C
86.	12.5 watt	87.	4.44	88.	A	89.	D	90.	33.33
91.	C	92.	A	93.	100	94.	D	95.	D
96.	C	97.	0	98.	B	99.	C	100.	3.06

101.	2.5	102.	B	103.	A, D	104.	C	105.	0
106.	A	107.	4Ω	108.	C	109.	36	110.	3.236Ω
111.	D	112.	B	113.	C	114.	A, D	115.	D
116.	0.92	117.	5	118.	750	119.	1.25	120.	B
121.	187.73	122.	D	123.	A	124.	B	125.	D
126.	C	127.	50	128.	B	129.	0	130.	–10V
131.	C	132.	D	133.	C	134.	C	135.	C
136.	B	137.	B	138.	A	139.	D	140.	C
141.	B	142.	B	143.	-4	144.	6.48	145.	B
146.	A	147.	B, D	148.	B	149.	A	150.	0
151.	1	152.	D	153.	13.2	154.	D	155.	0.4
156.	D	157.	5	158.	57.09	159.	8.4	160.	16

Answer Keys

CH-3 Network Theorem									
1.	C	2.	B, C	3.	1	4.	B	5.	18W
6.	4mw	7.	A	8.	D	9.	A	10.	A, B
11.	5.303∠-45°	12.	20Ω	13.	A	14.	54	15.	4.48
16.	D	17.	A	18.	84.05	19.	0	20.	C
21.	C	22.	1	23.	B	24.	D	25.	28
26.	2	27.	A	28.	B	29.	48	30.	B
31.	A	32.	D	33.	1	34.	A	35.	B
36.	D	37.	A, B	38.	A, C	39.	B	40.	A, B
41.	C	42.	A	43.	C	44.	B	45.	B
46.	C	47.	D	48.	A	49.	25V	50.	2
51.	–6A	52.	A	53.	1.6	54.	D	55.	C
56.	B	57.	A	58.	B	59.	6	60.	B
61.	A	62.	D	63.	100	64.	D	65.	D
66.	C	67.	A, D	68.	B, C	69.	0.33	70.	A
71.	16	72.	B	73.	0.33	74.	4	75.	D
76.	A	77.	A	78.	D	79.	A	80.	C
81.	A	82.	A, B, C, D	83.	C	84.	4.27	85.	12.50
86.	D	87.	A	88.	18	89.	B	90.	A
91.	B, C	92.	A, D	93.	1.53	94.	49	95.	20
96.	B	97.	A	98.	7	99.	A	100.	15.8
101.	D	102.	-8	103.	B	104.	D	105.	D

106.	A, D	107.	A, C	108.	D	109.	D	110.	(0)
111.	4.645	112.	A	113.	4				

Answer Keys

CH-4 Phasor Analysis and Three Phase Circuits									
1.	B, C	2.	306.87	3.	1	4.	B	5.	B
6.	A	7.	C	8.	12	9.	27	10.	B
11.	D	12.	D	13.	A, C	14.	50Ω	15.	114
16.	C	17.	0.125F	18.	D	19.	0.576 leading	20.	0.25
21.	306.87	22.	5	23.	B	24.	141.2V	25.	(200Ω)
26.	D	27.	A, C	28.	C	29.	D	30.	A
31.	B	32.	A	33.	0.97	34.	B	35.	A
36.	30	37.	45	38.	D	39.	A	40.	B
41.	C	42.	B	43.	D	44.	8.195	45.	C
46.	3.459	47.	287.04	48.	D	49.	B	50.	B
51.	B	52.	5.567	53.	C	54.	2.59	55.	A
56.	6.104	57.	C	58.	B	59.	B	60.	A
61.	B	62.	D	63.	9.68Ω	64.	33%	65.	B
66.	2.4	67.	B	68.	930 μf	69.	D	70.	B
71.	A	72.	D	73.	C				

Answer Keys

CH-5 Transient Analysis									
1.	A, B, D	2.	5A	3.	D	4.	50	5.	A
6.	16V	7.	C	8.	B	9.	6.99	10.	B
11.	A	12.	3.809	13.	C	14.	A	15.	24.40
16.	C	17.	A, B, D	18.	5	19.	C	20.	D
21.	50	22.	A	23.	4A	24.	B	25.	B
26.	C	27.	0	28.	–96	29.	–0.5	30.	B
31.	1	32.	11.24	33.	A	34.	B	35.	C
36.	B	37.	D	38.	1	39.	B	40.	B
41.	48	42.	4	43.	C	44.	D	45.	B
46.	A	47.	B	48.	D	49.	10	50.	100
51.	4	52.	30	53.	1.58	54.	A, B	55.	B
56.	B	57.	0.5	58.	B	59.	A	60.	0.50
61.	62.643	62.	C	63.	3.43	64.	3.37	65.	C
66.	A, C	67.	D	68.	B, D	69.	A	70.	A, B
71.	600	72.	D	73.	C	74.	B	75.	C
76.	A	77.	(3)	78.	B	79.	2.653	80.	A

81.	A	82.	C	83.	6.99	84.	C	85.	D
86.	C	87.	D	88.	5.056	89.	800	90.	A, C
91.	B	92.	A	93.	D	94.	C	95.	D
96.	A, B, D	97.	2.4	98.	B	99.	C	100.	C
101.	C	102.	80	103.	C	104.	A	105.	A
106.	D	107.	A	108.	C	109.	A	110.	A
111.	D	112.	–10	113.	15	114.	A	115.	B
116.	B	117.	1.61	118.	2.49	119.	B	120.	D
121.	C	122.	C	123.	C	124.	C	125.	C
126.	B	127.	C	128.	A	129.	D	130.	0.375
131.	C	132.	(12)	133.	D	134.	D	135.	0.167
136.	5	137.	7.8	138.	D	139.	-2.22	140.	-0.04
141.	D	142.	600	143.	B, C, D	144.	C	145.	1.386
146.	D	147.	B	148.	D	149.	B	150.	C

Answer Keys

CH-6 Steady State Analysis									
1.	A	2.	C	3.	C	4.	6	5.	B
6.	B	7.	C	8.	D	9.	D	10.	2.57
11.	0.051	12.	0.10	13.	96.02	14.	A	15.	D
16.	0.43	17.	A, D	18.	A	19.	B	20.	C
21.	C	22.	D	23.	A	24.	A	25.	A
26.	C	27.	A	28.	2	29.	A	30.	6
31.	51.9	32.	B	33.	B	34.	A	35.	C
36.	939	37.	A, C	38.	9.522	39.	2.403	40.	3.46
41.	A	42.	1732.05	43.	A	44.	178.8	45.	B

Answer Keys

CH-7 Resonance									
1.	B	2.	0.707	3.	C	4.	B	5.	C
6.	A, C	7.	1.642Ω	8.	D	9.	C	10.	D
11.	B	12.	0.5	13.	33.42	14.	30	15.	C
16.	63.66	17.	D	18.	68.11	19.	D	20.	11.32
21.	C, D	22.	3.98	23.	D	24.	B	25.	C
26.	B, C	27.	A	28.	C	29.	D	30.	B
31.	B	32.	C	33.	5291	34.	D	35.	B
36.	D	37.	C	38.	B	39.	D	40.	C
41.	A	42.	B	43.	400	44.	1.919	45.	0.29
46.	D	47.	3	48.	C	49.	B, D	50.	2

51.	C	52.	A, B, C	53.	A	54.	A		

Answer Keys

CH-8 Two Port Network									
1.	40	2.	2	3.	A	4.	0	5.	B
6.	D	7.	1.64	8.	–1.27	9.	D	10.	A
11.	A, C	12.	D	13.	1.2	14.	A	15.	B
16.	C	17.	0.1	18.	A	19.	A	20.	A
21.	A, D	22.	C	23.	A	24.	A	25.	D
26.	D	27.	A	28.	B	29.	D	30.	C
31.	C	32.	C	33.	B	34.	D	35.	A, B, C
36.	1	37.	C	38.	2	39.	D	40.	24
41.	11.11	42.	D	43.	B	44.	(8)	45.	A
46.	C	47.	C	48.	C	49.	A	50.	A
51.	D	52.	D	53.	B	54.	8.33	55.	0.35
56.	A	57.	D	58.	D	59.	B	60.	A
61.	D	62.	C	63.	C	64.	A	65.	3.33
66.	C	67.	7	68.	69	69.	-0.36	70.	B, D
71.	D	72.	20.5	73.	B, C, D	74.	C	75.	A
76.	D	77.	B, C	78.	A	79.	B	80.	A
81.	B	82.	1	83.	320	84.	55	85.	A
86.	D	87.	C	88.	4	89.	5	90.	A, B, C
91.	9	92.	A, B, C, D	93.	226	94.	0.416	95.	A
96.	A	97.	C	98.	B	99.	C	100.	B

Answer Keys

CH-9 Magnetic Coupled Circuits									
1.	B	2.	B	3.	B	4.	B	5.	B
6.	A, D	7.	C	8.	A, B	9.	B	10.	D
11.	D	12.	D	13.	C	14.	0.039	15.	4.714
16.	C	17.	1500	18.	D	19.	B	20.	B
21.	80	22.	B	23.	A	24.	B	25.	C
26.	168	27.	D						

Answer Keys

CH-10 Graph Theory									
1.	B, C, D	2.	1024	3.	C	4.	D	5.	B